CHAPTER 11

Cranium, Facial Bones, and Paranasal Sinuses

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RADIOGRAPHIC ANATOMY

Skull

As with other body parts, radiography of the skull requires a good understanding of all related anatomy. The anatomy of the skull is very complex, and specific attention to detail is required of the technologist.

The **skull,** or bony skeleton of the head, rests on the superior end of the vertebral column and is divided into two main sets of bones—**8 cranial bones** and **14 facial bones** (Fig. 11.1). Anatomy and positioning for the cranial and facial bones are described in this chapter.

CRANIAL BONES

The eight bones of the cranium are divided into the calvarium (skullcap) and the floor. Each of these two areas primarily consists of four bones.

Calvarium (Skullcap)

1. Frontal
2. Right parietal *(pah-ri′-e-tal)*
3. Left parietal
4. Occipital *(ok-sip′-i-tal)*

Floor

1. Right temporal
2. Left temporal
3. Sphenoid *(sfe′-noid)*
4. Ethmoid *(eth′-moid)*

The eight bones that make up the calvarium (skullcap) and the floor or base of the cranium are demonstrated on these frontal, lateral, and superior cutaway view drawings (Figs. 11.2, 11.3, and 11.4). These cranial bones are fused in an adult to form a protective enclosure for the brain. Each of these cranial bones is demonstrated and described individually in the pages that follow.

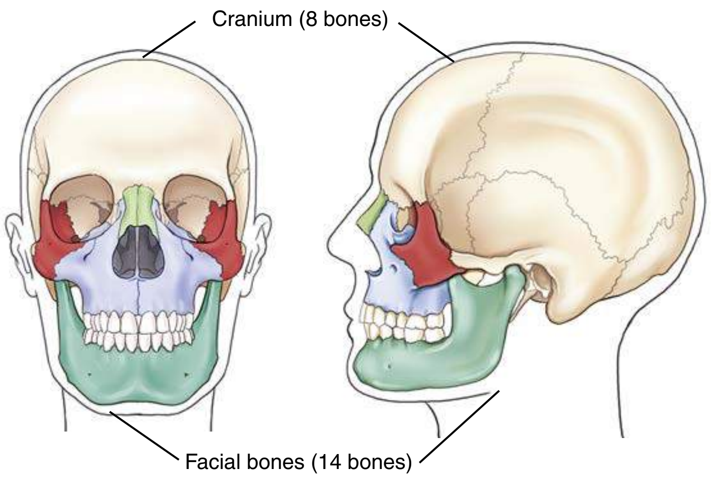


Fig. 11.1 Skull—bony skeleton of head (cranial and facial bones).

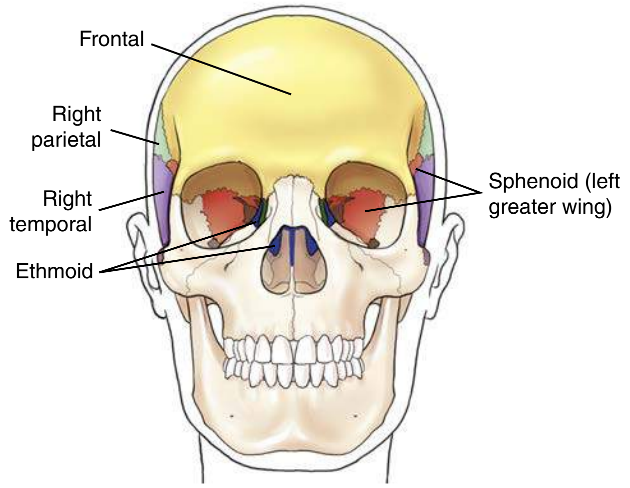


Fig. 11.2 Cranium—frontal view.

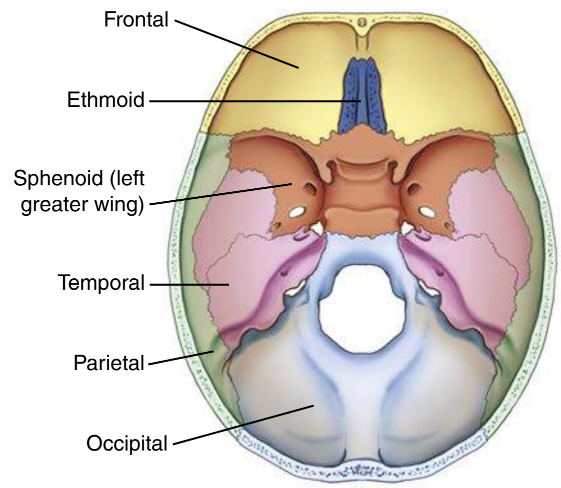


Fig. 11.3 Cranium—superior cutaway view.

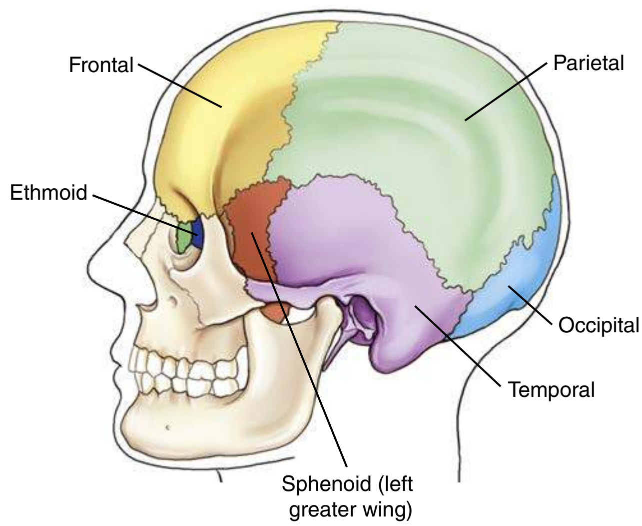


Fig. 11.4 Cranium—lateral view.

Frontal Bone As viewed from the front, the bone of the calvarium that is most readily visible is the **frontal bone.** This bone contributes to the formation of the forehead and the superior part of each orbit. It consists of two main parts: the **squamous** or **vertical portion,** which forms the forehead, and the **orbital** or **horizontal portion,** which forms the superior part of the orbit.

Squamous or vertical portion The **glabella** is the smooth, raised prominence between the eyebrows just above the bridge of the nose (Figs. 11.5 and 11.6).

The **supraorbital groove (SOG)** is the slight depression above each eyebrow. The SOG becomes an important landmark because it corresponds to the floor of the anterior fossa of the cranial vault, which is also at the level of the orbital plate or at the highest level of the facial bone mass (Fig. 11.7).

NOTE: You can locate the SOG on yourself by placing your finger against the length of your eyebrow and feeling the raised arch of bone, then allowing your finger to slide upward and drop slightly into the SOG.

The superior rim of each orbit is the **supraorbital margin (SOM).** The **supraorbital notch** (foramen) is a small hole or opening within the SOM slightly medial to its midpoint. The supraorbital nerve and artery pass through this small opening.

On each side of the squamous portion of the frontal bone above the SOG is a larger, rounded prominence termed the **frontal tuberosity** (eminence).

Orbital or horizontal portion As can be seen from the inferior aspect, the frontal bone shows primarily the horizontal or orbital portion (see Fig. 11.7), which consists of the **SOMs, superciliary ridges, glabella,** and **frontal tuberosities.**

The **orbital plate** on each side forms the superior part of each orbit. Below the orbital plates lie facial bones, and above the orbital plates is the anterior part of the floor of the brain case.

Each orbital plate is separated from the other by the **ethmoidal notch.** The ethmoid bone, one of the bones of the floor of the cranium, fits into this notch.

Articulations The frontal bone articulates with **four** cranial bones: right and left parietals, sphenoid, and ethmoid. These can be identified on frontal, lateral, and superior cutaway drawings in Figs. 11.3 and 11.4. (The frontal bone also articulates with eight facial bones.)

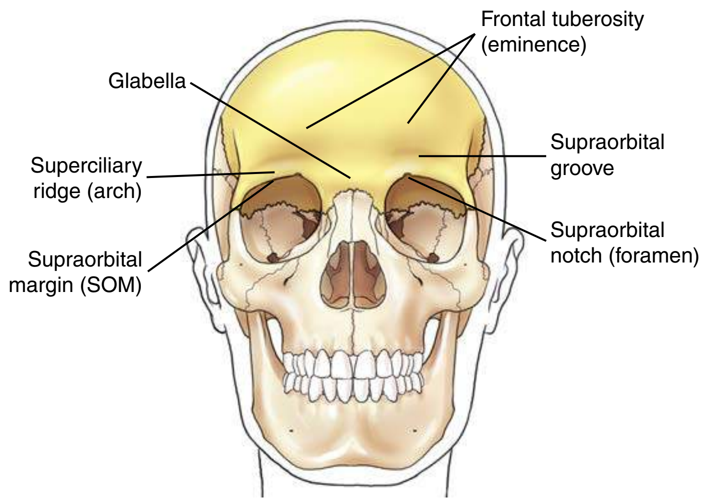


Fig. 11.5 Frontal bone—frontal view.

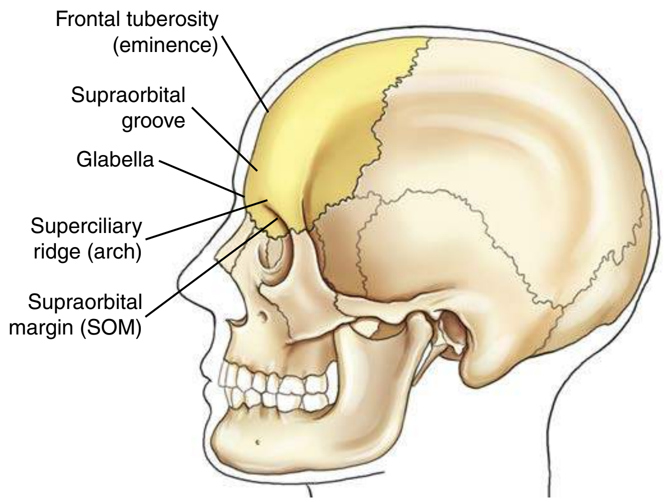


Fig. 11.6 Frontal bone—lateral view.

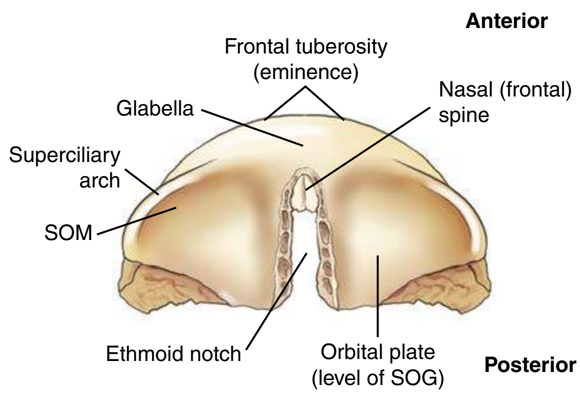


Fig. 11.7 Orbital portion of frontal bone—inferior view.

Parietal Bones The paired **right** and **left parietal bones** are well demonstrated on the lateral and superior view drawings of Figs. 11.8 and 11.9. The lateral walls of the cranium and part of the roof are formed by the two parietal bones. The parietal bones are roughly square and have a concave internal surface.

The widest portion of the entire skull is located between the **parietal tubercles (eminences)** of the two parietal bones. The frontal bone is primarily anterior to the parietals; the occipital bone is posterior; the temporal bones are inferior; and the greater wings of the sphenoid are inferior and anterior.

Articulations Each parietal bone articulates with five cranial bones: frontal, occipital, temporal, sphenoid, and opposite parietal bones.

Occipital Bone The inferoposterior portion of the calvarium (skull-cap) is formed by the single occipital bone. The external surface of the occipital bone presents a rounded part called the **squamous portion.** The squamous portion forms most of the back of the head and is the part of the occipital bone that is superior to the **external occipital protuberance,** or **inion,** which is the prominent bump or protuberance at the inferoposterior portion of the skull (Fig. 11.10).

The large opening at the base of the occipital bone through which the spinal cord passes as it leaves the brain is called the **foramen magnum** (literally meaning "great hole").

The two lateral **condylar portions (occipital condyles)** are oval processes with convex surfaces, with one on each side of the foramen magnum. These articulate with depressions on the first cervical vertebra, called the *atlas.* This two-part articulation between the skull and the cervical spine is called the **atlantoccipital joint.** They form a pair of ellipsoid joints that permits flexion, extension and a limited amount of lateral flexion and rotation.

Articulations The occipital bone articulates with **six** bones: two parietals, two temporals, sphenoid, and atlas (first cervical vertebra).

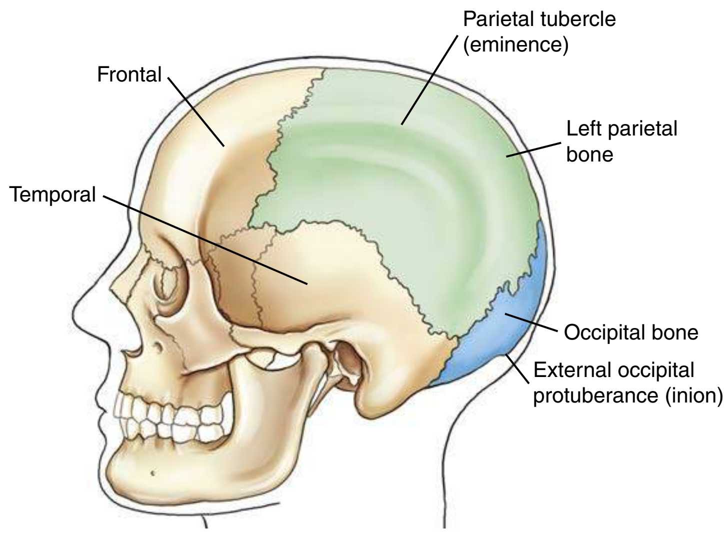


Fig. 11.8 Parietal and occipital bones—lateral view.

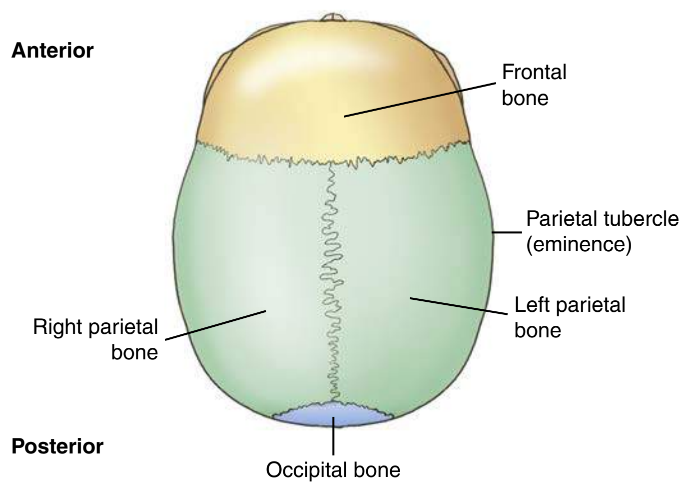


Fig. 11.9 Parietal and occipital bones—superior view.

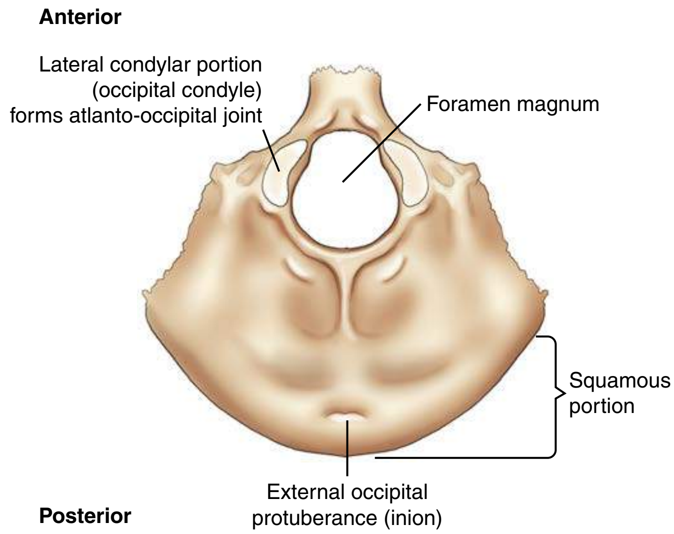


Fig. 11.10 Occipital bone—inferior view.

Temporal Bones

Lateral view The paired **right** and **left temporal bones** are complex structures that house the delicate organs of hearing and balance. As seen from this lateral view drawing (Fig. 11.11), the left temporal bone is situated between the greater wing of the sphenoid bone anteriorly and the occipital bone posteriorly.

Extending anteriorly from the squamous portion of the temporal bone is an arch of bone termed the **zygomatic** *(zi″-go-mat′-ik)* **process.** This process meets the temporal process of the zygomatic bone (one of the facial bones) to form the easily palpated **zygomatic arch.**

Inferior to the zygomatic process and just anterior to the **external acoustic (auditory) meatus (EAM)** is the **temporomandibular (TM) fossa,** into which the mandible fits to form the **temporomandibular joint (TMJ).**

Projecting inferior to the mandible and anterior to the EAM is a slender bony projection called the **styloid process.**

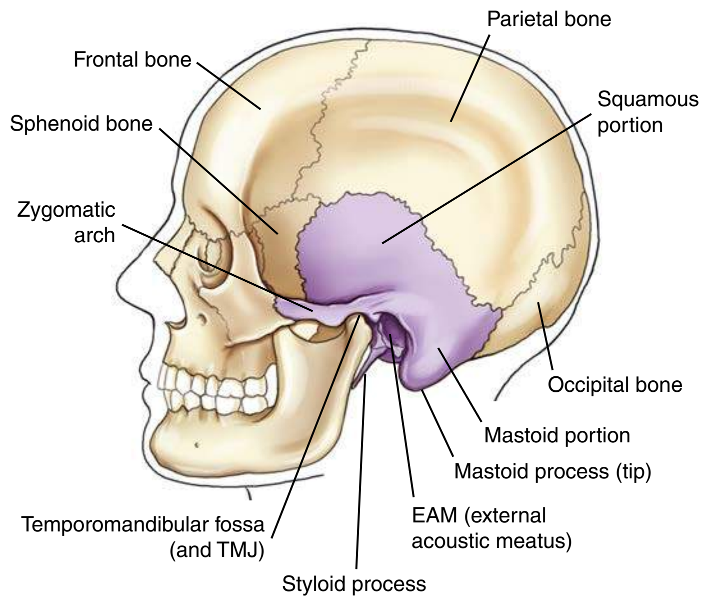


Fig. 11.11 Temporal bone—lateral view.

Frontal cutaway view Each temporal bone is divided into **three primary parts** (Fig. 11.12). First is the thin upper portion that forms part of the wall of the skull, the **squamous portion.** This part of the skull is quite thin and is the most vulnerable portion of the entire skull to fracture.

The second portion is the area posterior to the EAM, the **mastoid portion,** with a prominent **mastoid process,** or **tip.** Many air cells are located within the mastoid process.

The third main portion is the dense **petrous** *(pet′-rus)* **portion,** which also is called the **petrous pyramid,** or **pars petrosa;** it houses the organs of hearing and equilibrium, including the mastoid air cells, as described later in this chapter. Sometimes this is also called the **petromastoid portion** of the temporal bone because internally it includes the mastoid portion. The upper border or ridge of the petrous pyramids is commonly called the **petrous ridge,** or petrous apex.

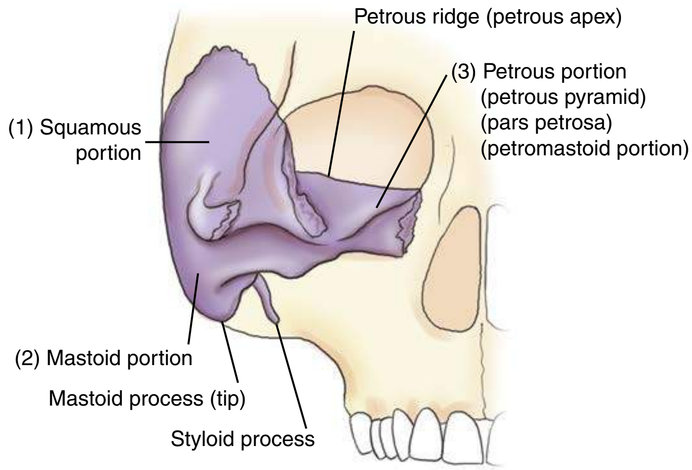


Fig. 11.12 Temporal bone, three primary parts—frontal cutaway view.

Superior view The floor of the cranium is well visualized in this drawing (Fig. 11.13). The single occipital bone resides between the paired temporal bones. The third main portion of each temporal bone, the **petrous portion,** again is shown in this superior view. This pyramid-shaped portion of the temporal bone is the thickest and densest bone in the cranium. The **petrous pyramids** project anteriorly and toward the midline from the area of the **EAM.**

The **petrous ridge** of these pyramids **corresponds to the level of an important external landmark,** the **TEA** (top of the ear attachment). Near the center of the petrous pyramid on the posterior surface just superior to the **jugular foramen** is an opening or orifice called the **internal acoustic meatus,** which serves to transmit the nerves of hearing and equilibrium. The bilateral jugular foramina are located in the base of the cranium and are where the internal jugular veins are formed and three cranial nerves (IX, X, and XI) pass.[1]

NOTE: The openings of the external and internal acoustic meatus cannot be visualized on this superior view drawing (Fig. 11.13) because they are located on the posteroinferior aspect of the petrous pyramid.

Articulations Each temporal bone articulates with **three** cranial bones: parietal, occipital, and sphenoid. (Each temporal bone also articulates with two facial bones.)

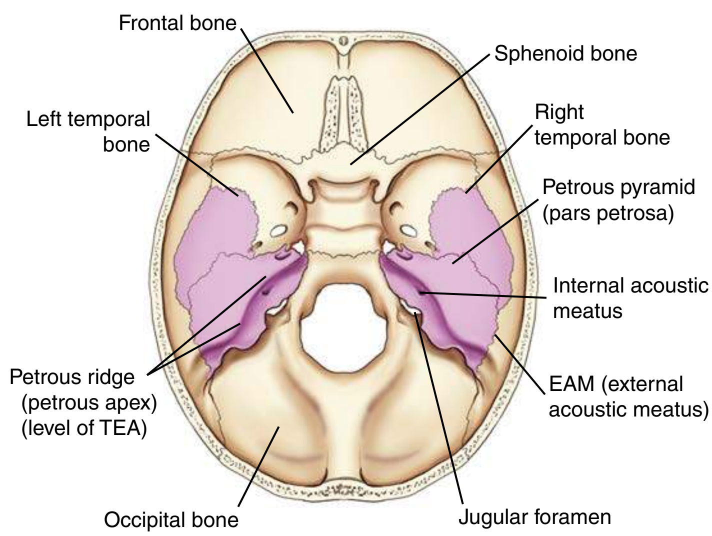


Fig. 11.13 Temporal bones—superior view.

Sphenoid Bone

Superior view The single centrally located **sphenoid bone** forms the anchor for the other seven cranial bones. The central portion of the sphenoid is the body, which lies in the midline of the floor of the cranium and contains the sphenoid sinus, as is best shown on a sagittal sectional drawing (see Fig. 11.18).

The central depression on the body is termed the **sella turcica** *(sel′-a tur′-si-ka).* This depression looks like a saddle from the side (see Fig. 11.16), and it derives its name from words meaning "Turkish saddle." The sella turcica partially surrounds and protects a major gland of the body, the **hypophysis cerebri,** or **pituitary gland.** Posterior to the sella turcica is the back of the saddle, the **dorsum sellae** *(dor′-sum sel′-e)* (also best seen in Fig. 11.16).

The **clivus** *(kli′-vus)* is a shallow depression that begins on the posteroinferior aspect of the dorsum sellae of the sphenoid bone and extends posteriorly to the foramen magnum at the base of the occipital bone (Fig. 11.14; also see Fig. 11.16). This slightly depressed area forms a base of support for the pons (a portion of the brainstem) and for the basilar artery.

Extending laterally from the body to either side are two pairs of wings. The smaller pair, termed the **lesser wings,** are triangular and are nearly horizontal, ending medially in the two **anterior clinoid processes.** They project laterally from the superoanterior portion of the body and extend to the middle of each orbit. The **greater wings** extend laterally from the sides of the body and form a portion of the floor of the cranium and a portion of the sides of the cranium.

Three pairs of small openings or foramina exist in the greater wings for passage of certain cranial nerves (see Fig. 11.14). Lesions that can cause erosion of these foramina can be detected radiographically. The **foramen rotundum** *(ro-tun′-dum)* and the **foramen ovale** *(o-va′-le)* are seen as small openings on superior and oblique view drawings (Figs. 11.14 and 11.15). The small rounded **foramen spinosum** *(spi-no′-sum)* (one of a pair) is also labeled on the superior view drawing (see Fig. 11.14).

Oblique view An oblique drawing of the sphenoid bone demonstrates the complexity of this bone. The shape of the sphenoid has been compared with a bat with its wings and legs extended as in flight. The centrally located depression, the **sella turcica,** again is seen on this view (see Fig. 11.15).

Arising from the posterior aspect of the **lesser wings** are two bony projections termed **anterior clinoid processes.** The anterior clinoid processes are larger and are spread farther apart than the **posterior clinoid processes** that extend superiorly from the **dorsum sellae,** which is best seen on the lateral drawing (Fig. 11.16).

Between the anterior body and the lesser wings on each side are groovelike canals through which the optic nerve and certain arteries pass into the orbital cavity. These canals begin in the center as the **chiasmatic** *(ki-az-mat′-ik)* or **optic groove,** which leads on each side to an **optic canal,** which ends at the **optic foramen,** or the opening into the orbit. The optic foramina can be demonstrated radiographically with the parieto-orbital oblique projection (Rhese method) described later in this chapter. Slightly lateral and posterior to the optic foramina on each side are irregularly shaped openings, which are seen best on this oblique view, called **superior orbital fissures.** These openings provide additional communication with the orbits for numerous cranial nerves and blood vessels. The foramen rotundum and the foramen ovale are seen again on this oblique view (Fig. 11.15).

Projecting downward from the inferior surface of the body are four processes that correspond to the legs of the imaginary bat. The more lateral, flat extensions are called the **lateral pterygoid** *(ter′-i-goyd)* **processes,** which sometimes are called *plates.* Directly medial to these are two **medial pterygoid processes** or plates, which end inferiorly in small hooklike processes, called the **pterygoid hamuli.** The pterygoid processes or plates form part of the lateral walls of the nasal cavities.

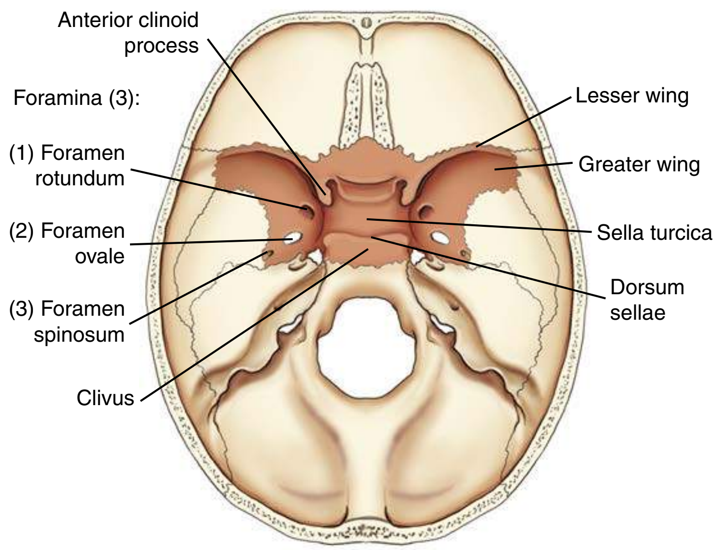


Fig. 11.14 Sphenoid bone—superior view.

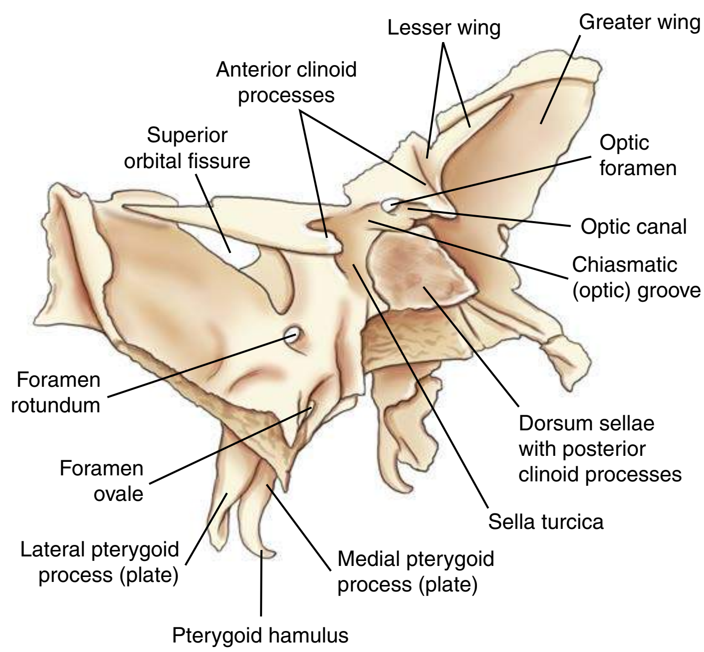


Fig. 11.15 Sphenoid bone—oblique view.

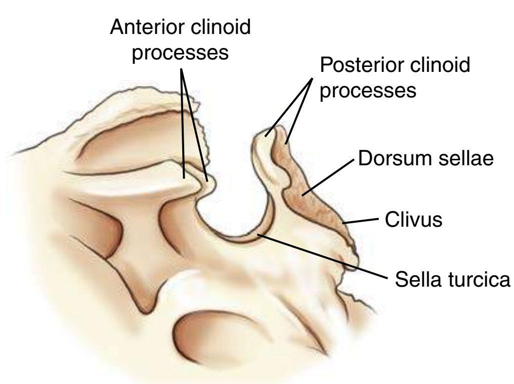


Fig. 11.16 Sella turcica of sphenoid bone—lateral view.

Sella turcica—lateral view A true lateral view of the sella turcica would look similar to the image in Fig. 11.16. Deformity of the sella turcica is often an indication that a lesion exists intracranially as seen radiographically. Computed tomography (CT) and magnetic resonance imaging (MRI) of the **sella turcica** may be performed to detect such deformities. The **sella turcica** and the **dorsum sellae** are also demonstrated best on a lateral projection of the cranium.

Articulations Because of its central location, the sphenoid articulates with the other **seven** cranial bones. The sphenoid also articulates with five facial bones.

Ethmoid Bone

The eighth and last cranial bone to be studied is the **ethmoid bone.** The single ethmoid bone lies primarily below the floor of the cranium. Only the top of the ethmoid is shown on a superior view situated in the ethmoidal notch of the frontal bone (Fig. 11.17).

A magnified coronal view of the entire ethmoid is shown on the right in Fig. 11.17. The small upper horizontal portion of the bone, termed the **cribriform plate,** contains many small openings or foramina through which segmental branches of the olfactory nerves (or the nerves of smell) pass. Projecting superiorly from the cribriform plate is the **crista galli** *(kris′-ta gal′-le),* which is derived from words meaning "rooster's comb."

The major portion of the ethmoid bone lies beneath the floor of the cranium. Projecting downward in the midline is the **perpendicular plate,** which helps to form the bony nasal septum. The two **lateral labyrinths** (masses) are suspended from the undersurface of the cribriform plate on each side of the perpendicular plate. The lateral masses contain the ethmoid air cells or sinuses and help to form the medial walls of the orbits and the lateral walls of the nasal cavity. Extending medially and downward from the medial wall of each labyrinth are thin, scroll-shaped projections of bone. These projections are termed the **superior** and **middle nasal conchae** *(kong′-ha)* or **turbinates;** they are best shown on facial bone drawings in Figs. 11.46 and 11.47.

Articulations The ethmoid articulates with **two** cranial bones: frontal and sphenoid. The ethmoid bone also articulates with 11 facial bones.

Cranium—Sagittal View

Fig. 11.18 represents the right half of the skull, which is sectioned near the midsagittal plane (MSP). The centrally located **sphenoid** and **ethmoid** bones are well demonstrated, showing their relationship to each other and to the other cranial bones.

The **ethmoid bone** is located anterior to the sphenoid bone. The smaller **crista galli** and **cribriform plate** project superiorly, and the larger **perpendicular plate** extends inferiorly. The perpendicular plate forms the upper portion of the bony nasal septum.

The **sphenoid bone,** which contains the saddle-shaped sella turcica, is located directly posterior to the ethmoid bone. Shown again is one of the two long, slender **pterygoid processes** or plates extending down and forward and ending with the small pointed process called the **pterygoid hamulus.** Inferior and slightly anterior to the sella turcica of the sphenoid bone in this sagittal view is a hollow-like body area of the sphenoid, which houses the **sphenoid sinus.**

The larger **frontal bone** also demonstrates a cavity directly posterior to the glabella that contains the **frontal sinus.** The vomer (a facial bone) is shown as a midline structure between parts of the sphenoid and parts of the ethmoid, as is seen in Fig. 11.18.

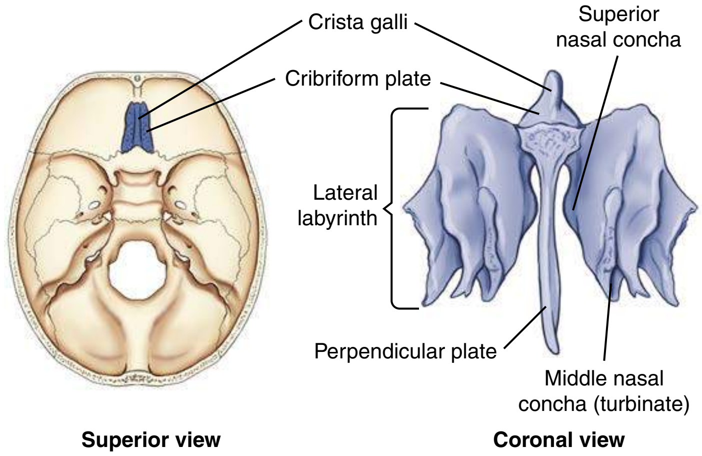


Fig. 11.17 Ethmoid bone.

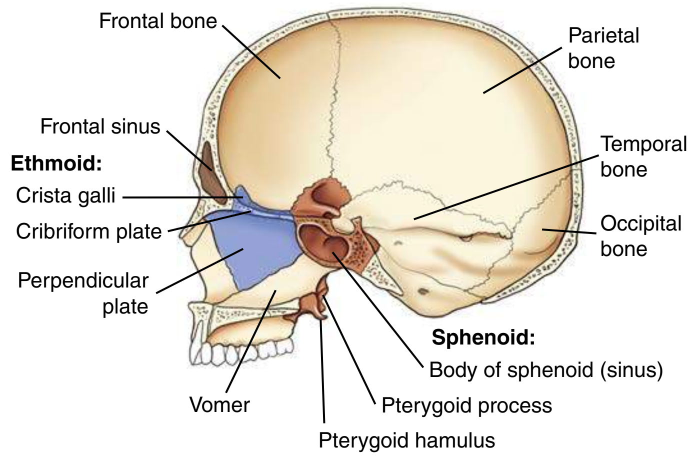


Fig. 11.18 Cranium—midsagittal view of sphenoid and ethmoid bones.

JOINTS OF THE CRANIUM—SUTURES

Adult Cranium

The articulations or joints of the cranium are called **sutures** and are classified as **fibrous joints.** In an adult, these are immovable and therefore are **synarthrodial-type joints.** They are demonstrated in Fig. 11.19 in lateral, superior oblique, and posterior views.

The **coronal** *(ko-ro′-nal)* **suture** separates the frontal bone from the two parietal bones. Separating the two parietal bones in the midline is the **sagittal suture.** Posteriorly, the **lambdoidal** *(lam′-doy-dal)* **suture** separates the two parietal bones from the occipital bone. The **squamosal** *(skwa-mo′-sal)* **sutures** are formed by the inferior junctions of the two parietal bones with their respective temporal bones.

Each end of the sagittal suture is identified as a point or area with a specific name as labeled. The anterior end of the sagittal suture is termed the **bregma** *(breg′-mah)*, and the posterior end is called the **lambda** *(lam′-dah).* The right and left **pterions** *(ter′-re-ons)* are points at the junction of the frontal, parietals, temporals, and the greater wings of the sphenoid. (The pterions are at the **posterior** end of the sphenoparietal suture.[1])

The right and left **asterions** *(as-te′-re-ons)* are points posterior to the ear where the squamosal and lambdoidal sutures meet. These six recognizable bony points are used in surgery or other cases in which specific reference points for cranial measurements are necessary.

Infant Cranium

The calvarium (skullcap) on an infant is very large in proportion to the rest of the body, but the facial bones are quite small, as can be seen on these drawings (Fig. 11.20). Ossification of the individual cranial bones is incomplete at birth, and the sutures are membrane-covered spaces that fill in soon after birth. However, certain regions where sutures join are slower in their ossification, and these are called **fontanels** *(fon″-tah-nels′)*. The cranial sutures themselves generally do not ossify completely until the mid-to-late 20s, and some may not completely close until the fifth decade of life.[2]

Fontanels Early in life, the bregma and the lambda are not bony but are membrane-covered openings or "soft spots." These soft spots are termed the **anterior** and **posterior fontanels** in an infant. The anterior fontanel is the largest and at birth measures approximately 1 inch (2.5 cm) wide and 1½ inches (4 cm) long. It does not completely close until about 18 months of age.

Two smaller lateral fontanels that close soon after birth are the **sphenoid** (pterion in an adult) and **mastoid** (asterion in an adult) **fontanels,** which are located at the sphenoid and mastoid angles of the parietal bones on each side of the head. **Six fontanels** occur in an infant as follows:

INFANT	ADULT
Anterior fontanel	Bregma
Posterior fontanel	Lambda
Right sphenoid fontanel	Right pterion
Left sphenoid fontanel	Left pterion
Right mastoid fontanel	Right asterion
Left mastoid fontanel	Left asterion

Sutural, or Wormian, Bones

Certain small, irregular bones called *sutural,* or *wormian, bones* sometimes develop in adult skull sutures. These isolated bones most often are found in the lambdoidal suture but occasionally also are found in the region of the fontanels, especially the posterior fontanel. In the adult skull, these are completely ossified and are visible only by the sutural lines around their borders.

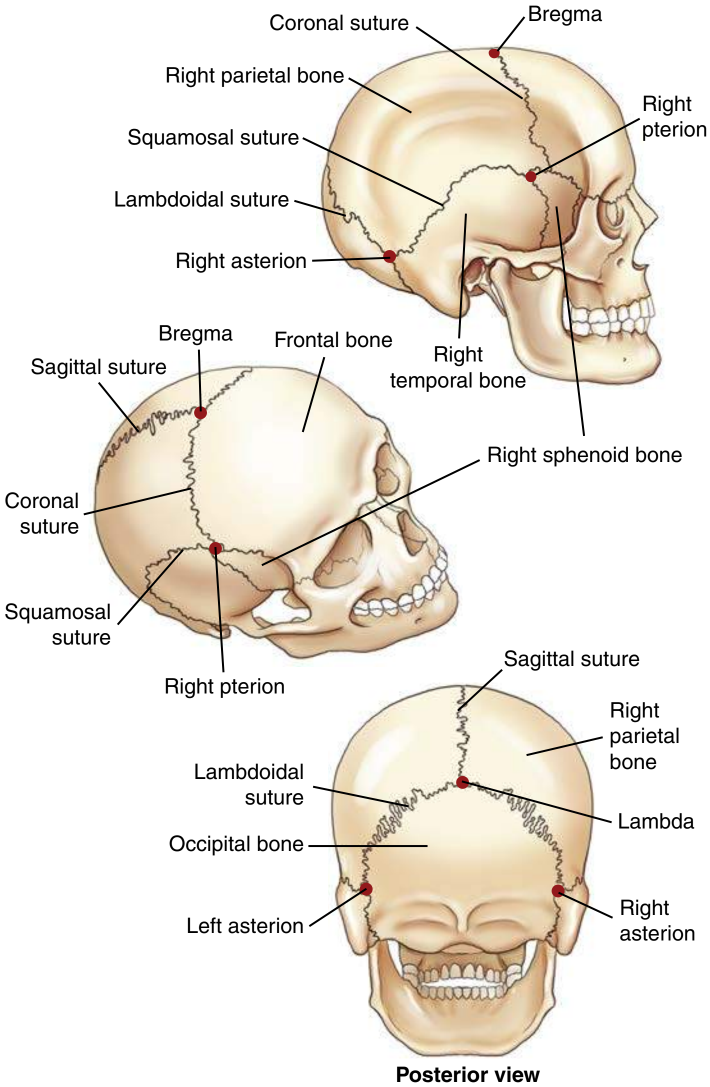


Fig. 11.19 Adult cranial sutures—**fibrous joints, synarthrodial** (immovable).

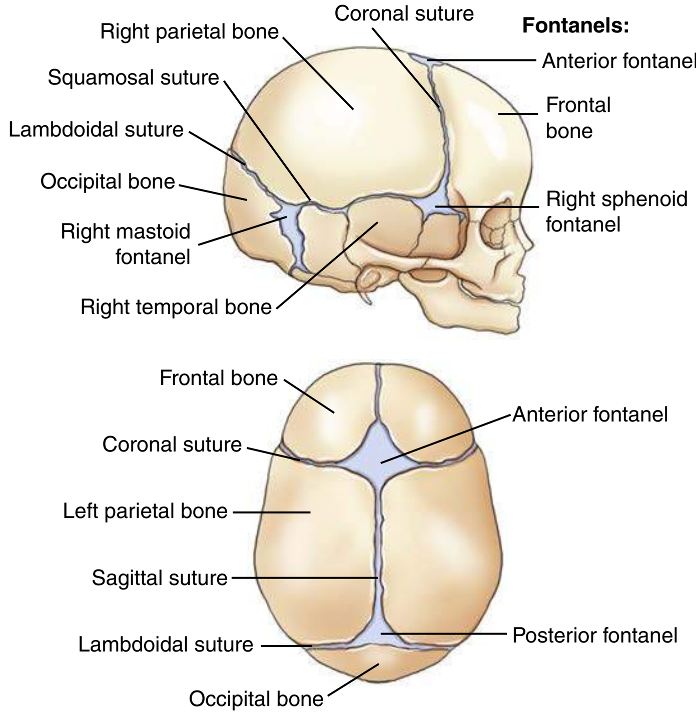


Fig. 11.20 Infant cranial sutures and **fontanels.**

ANATOMY REVIEW WITH RADIOGRAPHS

The following review exercises focus on the anatomy of the eight cranial bones as labeled on the radiographs on the right. A recommended method of review and reinforcement is to cover the answers and first try to identify each of the labeled parts from memory. Specific anatomic parts may be more difficult to recognize on radiographs compared with drawings, but knowing locations and relationships to surrounding structures and bones should aid in identifying these parts.

Cranial Bones—PA Axial Caldwell Projection, 15° caudad (Fig. 11.21)

A. Supraorbital margin of right orbit
B. Crista galli of ethmoid
C. Sagittal suture (posterior skull)
D. Lambdoidal suture (posterior skull)
E. Petrous ridge

Cranial Bones—AP Axial Projection (Fig. 11.22)

A. Dorsum sellae of sphenoid
B. Posterior clinoid processes
C. Petrous ridge or petrous pyramid
D. Parietal bone
E. Occipital bone
F. Foramen magnum

Cranial Bones—Lateral Projection (Fig. 11.23)

A. EAM
B. Mastoid portion of temporal bone
C. Occipital bone
D. Lambdoidal suture
E. Clivus
F. Dorsum sellae
G. Posterior clinoid processes
H. Anterior clinoid processes
I. Vertex of cranium
J. Coronal suture
K. Frontal bone
L. Orbital plates
M. Cribriform plate
N. Sella turcica
O. Body of sphenoid (sphenoid sinus)
P. Petrous portion of temporal bone

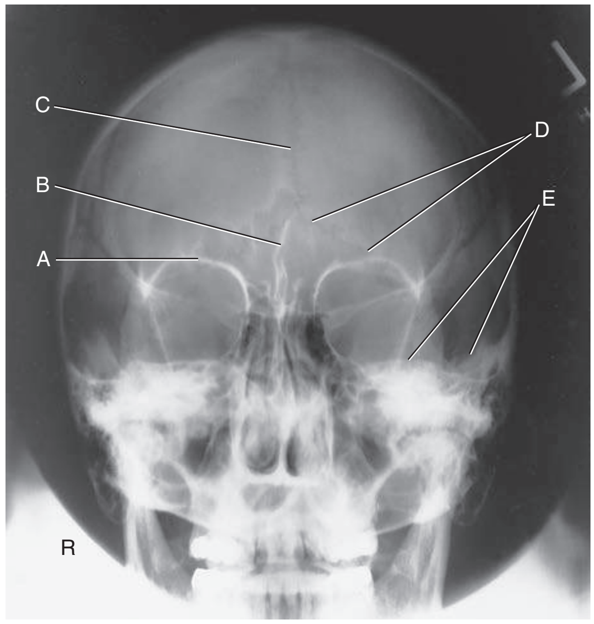


Fig. 11.21 PA Caldwell—15° caudad projection.

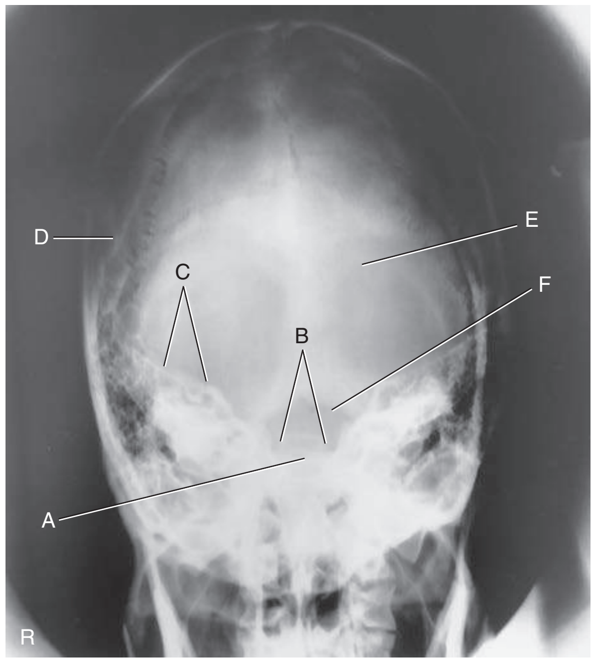


Fig. 11.22 AP axial projection.

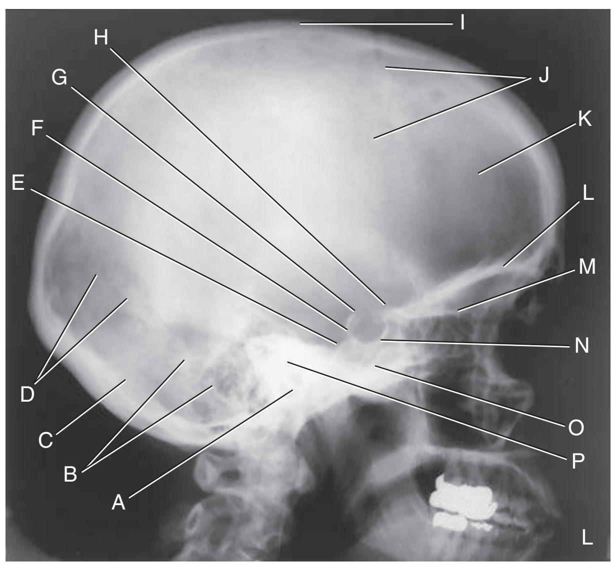


Fig. 11.23 Lateral projection.

11

Anatomy of Organs of Hearing and Equilibrium

Because of the density (brightness) and relative location of the temporal bones, the mastoids and petrous portions are difficult to visualize with conventional radiography. CT and MRI have largely replaced conventional radiography for imaging of these regions. However, a knowledge of temporal bone anatomy is critical whether performing conventional radiography, CT, or MRI.

The organs of hearing and equilibrium are the main structures found within the petrous portion of the temporal bones. The three divisions of the ear—**external, middle,** and **internal portions**—are illustrated in Fig. 11.24.

EXTERNAL EAR

The **external ear** begins with the **auricle** or **pinna** on each side of the head. The **tragus** is part of this external structure. It is the small liplike structure located anterior to the EAM that acts as a partial shield to the ear opening.

The **EAM** is the opening or canal of the external ear. The external acoustic canal or meatus is approximately 1 inch (2.5 cm) long; half is bony in structure, and half is cartilaginous (Fig. 11.25).

The **mastoid process** and **mastoid tip** of the temporal bone are posterior and inferior to the EAM, whereas the **styloid process** is inferior and slightly anterior. The meatus narrows as it meets the **tympanic membrane** or **eardrum.** The eardrum is situated at an oblique angle, forming a depression, or well, at the lower medial end of the meatus.

MIDDLE EAR

The **middle ear** is an irregularly shaped, air-containing cavity located between the external and internal ear portions. The three main parts of the middle ear are the **tympanic membrane,** the three small bones called **auditory ossicles,** and the **tympanic cavity** (Fig. 11.26). The tympanic membrane is considered part of the middle ear even though it serves as a partition between the external and middle ears.

The tympanic cavity is divided further into two parts. The larger cavity opposite the eardrum is called the **tympanic cavity proper.** The area above the level of the EAM and the eardrum is called the **attic,** or **epitympanic recess.** The **drum crest,** or **spur,** is a structure that is important radiographically. The tympanic membrane is attached to this sharp, bony projection. The drum crest or spur separates the EAM from the epitympanic recess.

The tympanic cavity communicates anteriorly with the nasopharynx by way of the **eustachian tube,** or **auditory tube.**

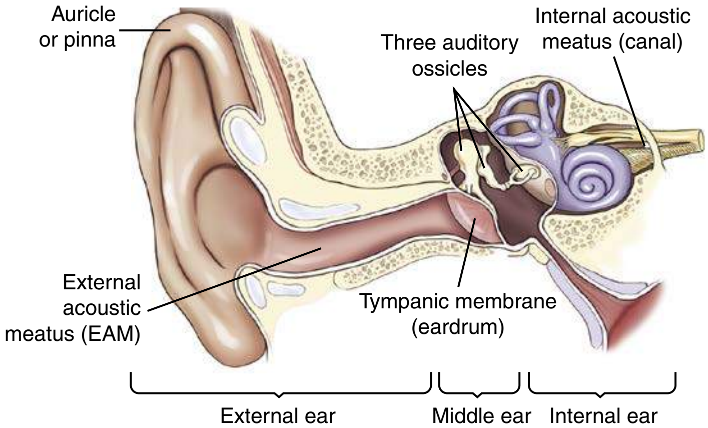


Fig. 11.24 Ear.

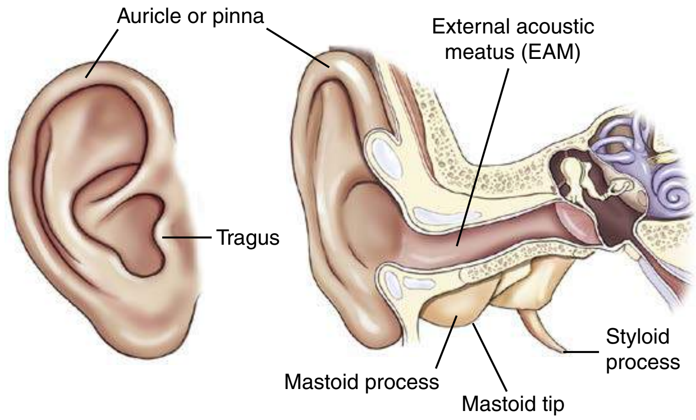


Fig. 11.25 External ear.

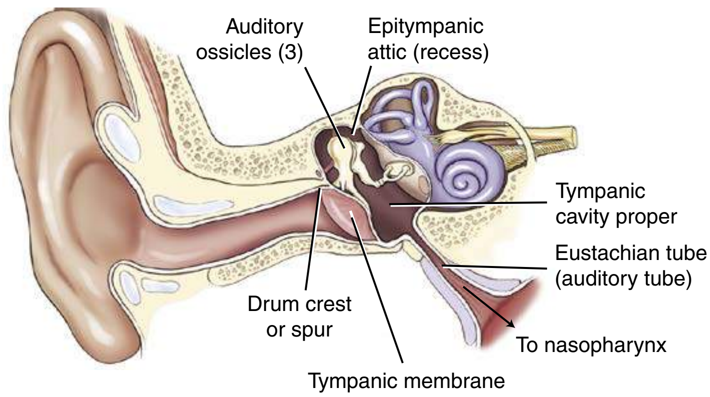


Fig. 11.26 Middle ear.

COMPUTED TOMOGRAPHY OF THE TEMPORAL BONE

Fig. 11.27 demonstrates select organs of hearing and equilibrium of the middle and inner ear as seen on this CT of the temporal bone.

Eustachian Tube

The **eustachian tube** is the passageway between the middle ear and the nasopharynx. This tube is approximately 1½ inches (4 cm) long and serves to equalize the pressure within the middle ear to the outside atmospheric air pressure through the nasopharynx (Fig. 11.28). The sensation of one's ears popping is caused by pressure being adjusted internally in the middle ear to prevent damage to the eardrum.

A problem associated with this direct communication between the middle ear and the nasopharynx is that disease organisms have a direct passageway from the throat to the middle ear. Therefore, ear infections often accompany sore throats, especially in children whose immune system is still developing.

Internal Acoustic Meatus

Fig. 11.29 illustrates the ear structures as they would appear in a **modified posteroanterior (PA) projection.** A 5° to 10° central ray (CR) caudad angle to the orbitomeatal line projects the petrous ridges to the **midorbital level,** as is shown in this drawing. This results in a special transorbital view, which may be taken to demonstrate the **internal acoustic meatus.** The opening to the internal acoustic meatus is an oblique aperture that is smaller in diameter than the opening to the EAM and is very difficult to demonstrate clearly on any conventional radiographic projection. It is best demonstrated with CT (for bony erosion) and MRI (for demonstration of acoustic neuromas).

In the drawing of a PA axial projection (see Fig. 11.29), the internal acoustic meatus is projected into the orbital shadow slightly below the petrous ridge, allowing it to be visualized on radiographs taken in this position. The lateral portions of the petrous ridges are at approximately the level of the **TEA.**

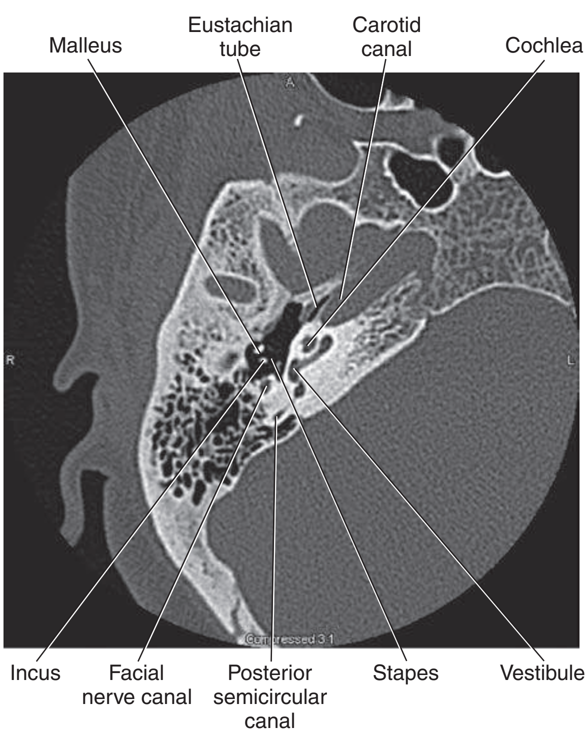


Fig. 11.27 CT of temporal bone.

Mastoids

A second direct communication into the middle ear occurs posteriorly to the **mastoid air cells.** The schematic drawing in Fig. 11.30 is a sagittal section that shows the relationships of the mastoid air cells to the **attic,** or **epitympanic recess,** and the **tympanic cavity proper.** The **aditus** is the opening between the epitympanic recess and the mastoid portion of the temporal bone.

The aditus connects directly to a large chamber within the mastoid portion termed the **antrum.** The antrum connects to the various **mastoid air cells.** This communication allows infection in the middle ear, which may have originated in the throat, to pass into the mastoid area. Infection within the mastoid area is separated from brain tissue only by thin bone. Before effective antibiotics were commonly used, this was often a pathway for **encephalitis,** a serious infection of the brain. The thin plate of bone that forms the roof of the antrum, aditus, and attic area of the tympanic cavity is called the **tegmen tympani.**

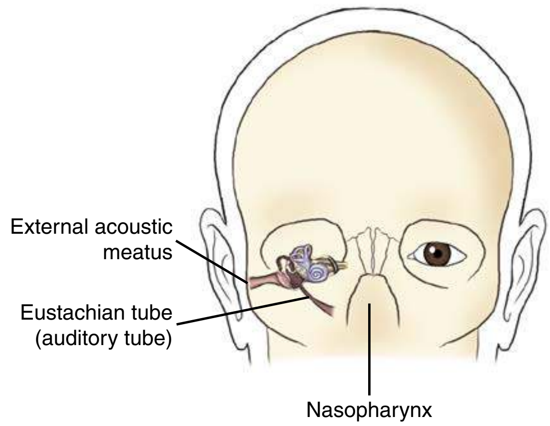


Fig. 11.28 Middle ear.

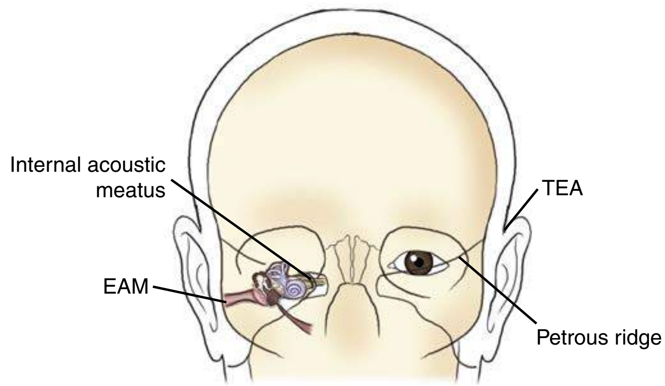


Fig. 11.29 Modified PA Caldwell projection (CR 5° to 10° caudad).

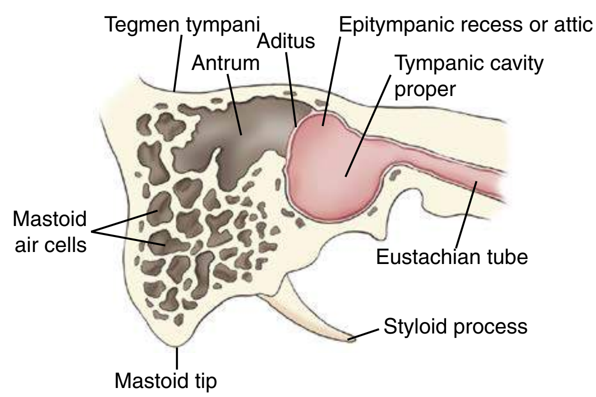


Fig. 11.30 Mastoid connection.

Auditory Ossicles

The **auditory ossicles** are three small bones that are prominent structures within the middle ear. Figs. 11.31 and 11.32 show that these three small bones are articulated to permit vibratory motion. The three auditory ossicles are located partly in the attic, or epitympanic recess, and partly in the tympanic cavity proper. These delicate bones bridge the middle ear cavity to transmit sound vibrations from the tympanic membrane to the oval window of the internal ear.

Vibrations are first picked up by the **malleus,** meaning "hammer," which is attached directly to the inside surface of the tympanic membrane. The head of the malleus articulates with the central ossicle, the **incus.** The incus receives its name from a supposed resemblance to an anvil, but it actually looks more like a premolar tooth with a body and two roots. The incus connects to the stirrup-shaped **stapes,** which is the smallest of the three auditory ossicles. The footplate of the stapes is attached to another membrane called the **oval window,** which leads into the inner ear.

Auditory Ossicles—Frontal and Lateral View

Fig. 11.32 illustrates the relationship of the auditory ossicles to one another in a close-up frontal view and a lateral view. As can be seen from the front, the most lateral of the three bones is the **malleus,** whereas the most medial of the three bones is the **stapes.** The lateral view drawing demonstrates how the ossicles would appear if one looked through the **EAM** to see the bony ossicles of the middle ear. The malleus, with its attachment to the eardrum, is located slightly anterior to the other two bones.

The resemblance of the **incus** to a premolar tooth with a body and two roots is well visualized in the lateral drawing. The longer root of the incus connects to the stapes, which connects to the oval window of the cochlea, resulting in the sense of hearing.

INTERNAL EAR

The complex **internal ear** contains the essential sensory apparatus of both **hearing** and **equilibrium.** Lying within the densest portion of the petrous pyramid, it can be divided into two main parts—the **osseous,** or **bony, labyrinth,** which is important radiographically, and the **membranous labyrinth.** The osseous labyrinth is a bony chamber that houses the membranous labyrinth, a series of intercommunicating ducts and sacs. One such duct is the **endolymphatic duct,** a blind pouch or closed duct contained in a small, canal-like, bony structure. The canal of the endolymphatic duct arises from the medial wall of the vestibule and extends to the posterior wall of the petrous pyramid, located both posterior and lateral to the **internal acoustic meatus.**

Osseous (Bony) Labyrinth

The osseous, or bony, labyrinth is divided into **three** distinctly shaped parts: the **cochlea** (meaning "snail shell"), the **vestibule,** and the **semicircular canals.** The osseous labyrinth completely surrounds and encloses the ducts and sacs of the membranous labyrinth. As is illustrated on the frontal cutaway view in Fig. 11.33, the snail-shaped, bony cochlea houses a long, coiled, tubelike duct of the membranous labyrinth.

The **cochlea** is the most anterior of the three parts of the osseous labyrinth. This is best shown on the lateral view of the osseous labyrinth in Fig. 11.34. The **round window,** sometimes called the **cochlear window,** is shown to be at the base of the cochlea.

The **vestibule,** the central portion of the bony labyrinth, contains the **oval window,** sometimes called the **vestibular window.**

Semicircular Canals

The **three semicircular canals** are located posterior to the other inner ear structures and are named according to their position: **superior, posterior,** and **lateral semicircular canals.** Each is located at a right angle to the other two, allowing a sense of equilibrium in addition to a sense of direction. The **semicircular canals relate to the sense of direction or equilibrium,** and the **cochlea relates to the sense of hearing** because of its connection to the stapes through the oval window.

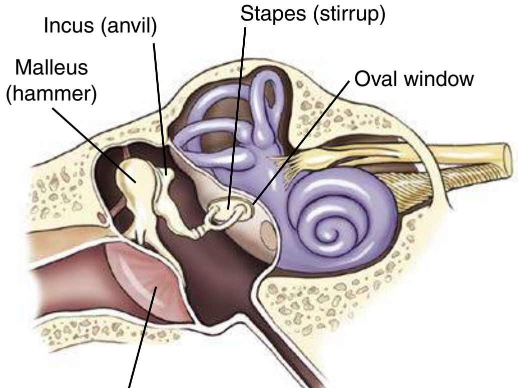


Fig. 11.31 Auditory ossicles—malleus, incus, and stapes.

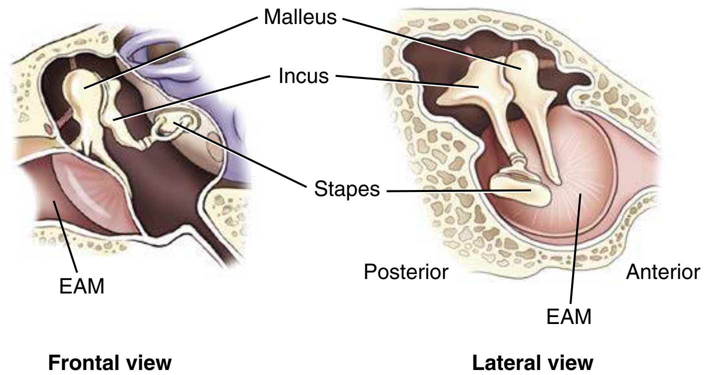


Fig. 11.32 Auditory ossicles.

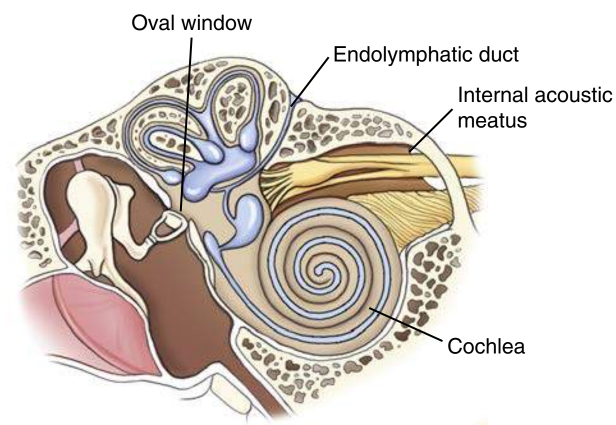


Fig. 11.33 Internal ear, osseous labyrinth—frontal view.

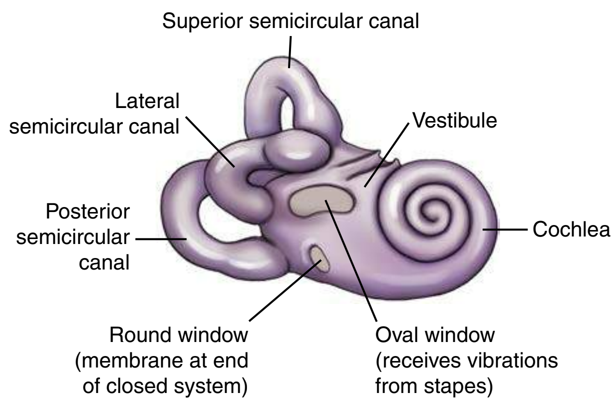


Fig. 11.34 Osseous (bony) labyrinth—lateral view.

"Windows" of the Internal Ear

The two openings into the internal ear are covered by membranes (see Fig. 11.34). The **oval,** or **vestibular, window** receives vibrations from the external ear through the distal aspect of the stapes of the middle ear and transmits these vibrations into the **vestibule** of the internal ear. The vestibule is the structure that houses the semicircular canals. The **round,** or **cochlear, window** is located at the base of the first coil of the cochlea. The round window is a membrane that allows movement of fluid within the closed duct system of the membranous labyrinth. As the oval window moves slightly inward with a vibration, the round window moves outward because this is a closed system and fluid does not compress. Vibrations and associated slight fluid movements within the cochlea produce impulses that are transmitted to the auditory nerve within the internal acoustic meatus, creating the sense of hearing.

ANATOMY REVIEW WITH RADIOGRAPHS

Specific anatomy of the temporal bone is difficult to recognize on conventional radiographs. Conventional positioning for mastoids is rarely performed today, but these two projections are provided to review the anatomy of the inner ear and mastoids.

Axiolateral Projection (Fig. 11.35)

A. EAM
B. Mastoid antrum
C. Mastoid air cells
D. Downside mandibular condyle (just anterior to EAM)
E. Upside (magnified) mandibular condyle

Posterior Profile Position (Fig. 11.36)

A. Petrous ridge
B. Bony (osseous) labyrinth (semicircular canals)
C. EAM
D. Region of internal acoustic canal

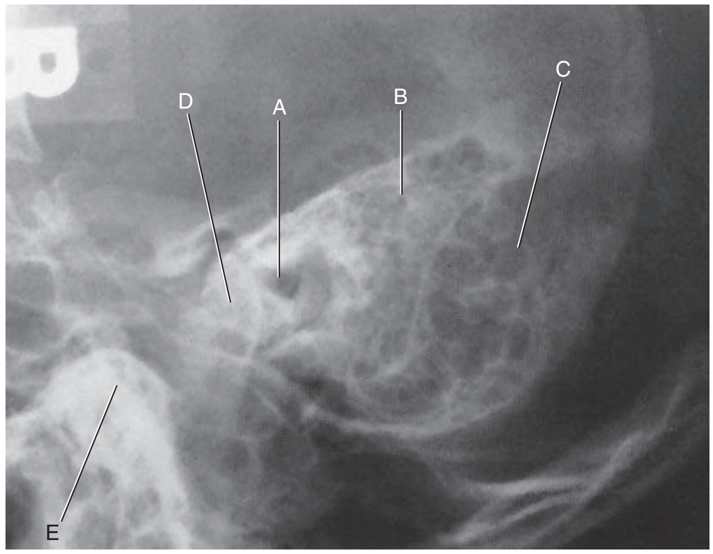


Fig. 11.35 Axiolateral projection for mastoids.

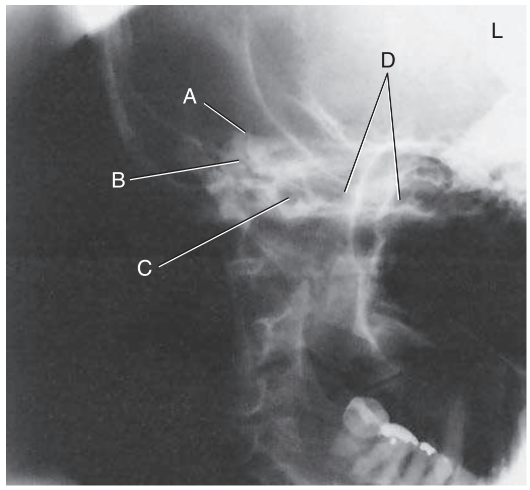


Fig. 11.36 Posterior profile projection for mastoids.

Facial Bones

Each of the facial bones can be identified on frontal and lateral drawings (Figs. 11.37 and 11.38) except for the two palatine bones and the vomer, both of which are located internally and are not visible on a dry skeleton from the exterior. These bones are identified on sectional drawings later in this chapter.

The 14 facial bones contribute to the shape and form of a person's face. In addition, the cavities of the orbits, nose, and mouth are largely constructed from the bones of the face. Of the 14 bones that make up the facial skeleton, only 2 bones are unpaired, the vomer and mandible. The remaining 12 consist of six pairs of bones, with similar bones on each side of the face.

2	Maxillae *(mak-sil′-e)* (upper jaw), or maxillary bones
2	Zygomatic *(zi″-go-mat′-ik)* bones
2	Lacrimal *(lak′-ri-mal)* bones
2	Nasal bones
2	Inferior nasal conchae *(kong′-ke)*
2	Palatine *(pal′-ah-tin)* bones
1	Vomer *(vo′-mer)*
1	Mandible (lower jaw)
14	*Total*

Each of the facial bones is studied individually or in pairs. After the description of each in the figures is a listing of the specific adjoining bones with which they articulate. Knowledge of these anatomic relationships aids in understanding the structure of the bony skeleton of the head.

RIGHT AND LEFT MAXILLARY BONES

The two maxillae, or maxillary bones, are the largest immovable bones of the face (Fig. 11.39). The only facial bone larger than the maxilla is the movable lower jaw, or mandible. All the other bones of the upper facial area are closely associated with the two maxillae; they are structurally the most important bones of the upper face. The right and left maxillary bones are solidly united at the midline below the nasal septum. Each maxilla assists in the formation of three cavities of the face: (1) the mouth, (2) the nasal cavity, and (3) one orbit.

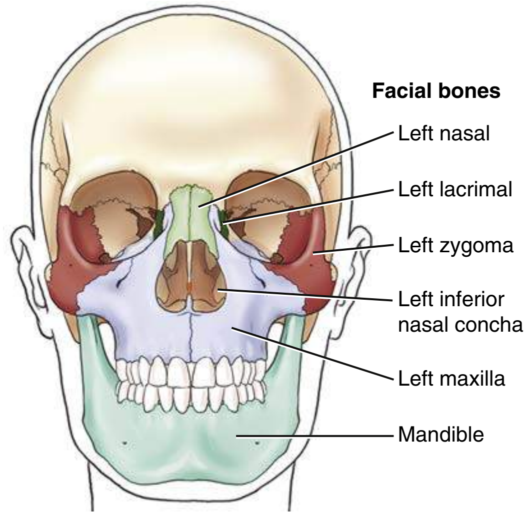


Fig. 11.37 Facial bones—frontal view.

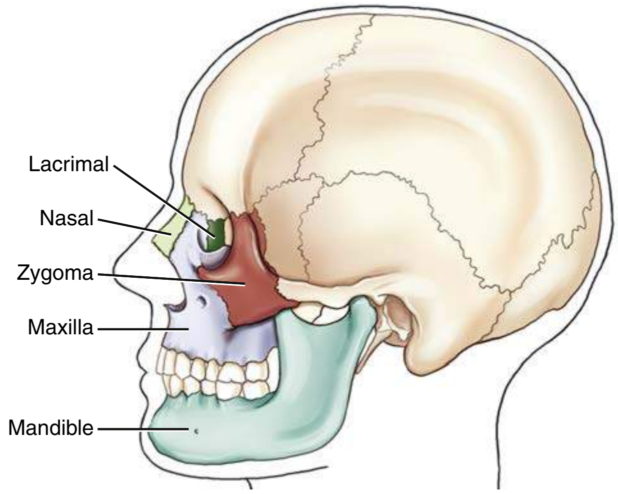


Fig. 11.38 Facial bones—lateral view.

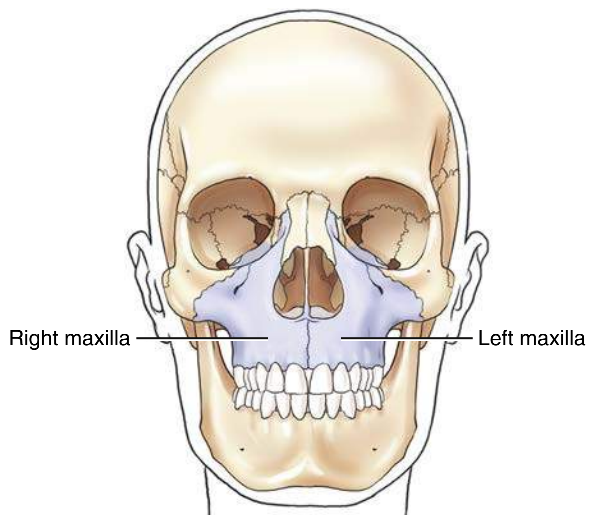


Fig. 11.39 Right and left maxillae.

Lateral View of Left Maxilla

Each maxilla consists of a centrally located **body** and **four processes** that project from that body. Three processes are more obvious and are visible on lateral and frontal drawings (Figs. 11.40 and 11.41). The fourth process, described later, is the palatine process, which is part of the hard palate.

The **body** of each maxilla is the centrally located portion that lies lateral to the nose. One of the three processes is the **frontal process,** which projects upward along the lateral border of the nose toward the frontal bone. The **zygomatic process** projects laterally to unite with the zygoma. The third process, the **alveolar process,** is the inferior aspect of the body of each maxilla. Eight upper teeth occur along the inferior margin of each alveolar process.

The two maxillae are solidly united in the midline anteriorly. At the upper part of this union is the **anterior nasal spine.** A blow to the nose sometimes results in separation of the nasal spine from the maxillae.

A point at the superior aspect of the anterior nasal spine is the **acanthion,** which is described later in this chapter as a surface landmark at the midline point where the nose and the upper lip meet.

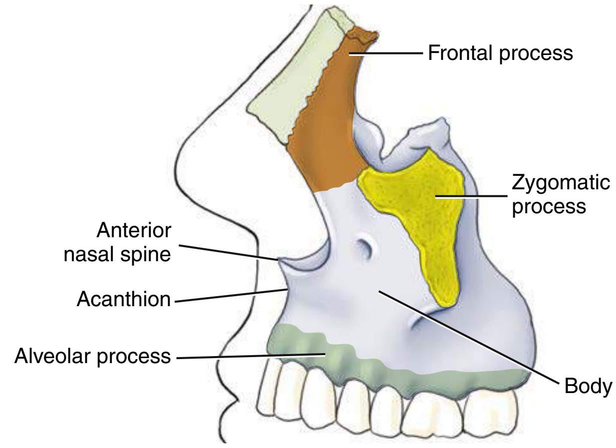


Fig. 11.40 Left maxilla—lateral view.

Frontal View

The relationship of the two maxillary bones to the remainder of the bones of the skull is well demonstrated in the frontal view (see Fig. 11.41). Note again **three processes,** as seen in the frontal view of the skull. Extending upward toward the frontal bone is the **frontal process.** Extending laterally toward the zygoma is the **zygomatic process,** and supporting the upper teeth is the **alveolar process.**

The body of each maxillary bone contains a large, air-filled cavity known as a **maxillary sinus.** Several of these air-filled cavities are found in certain bones of the skull. These sinuses communicate with the nasal cavity and are collectively termed *paranasal sinuses;* they are described further later on in this chapter.

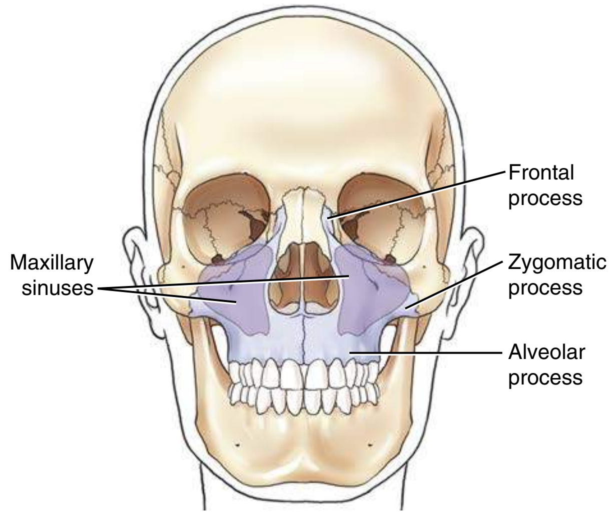


Fig. 11.41 Maxillae—frontal view.

Hard Palate (Inferior Surface)

The **fourth process** of each maxillary bone is the palatine process, which can be demonstrated only on an inferior view of the two maxillae (Fig. 11.42). The two palatine processes form the anterior portion of the roof of the mouth, called the *hard* or *bony palate.* The two palatine processes are solidly united at the midline to form a synarthrodial (immovable) joint. A common congenital defect called a *cleft palate* is an opening between the palatine processes that is caused by incomplete joining of the two bones.

The horizontal portion of two other facial bones, the **palatine bones,** forms the posterior part of the hard palate. Only the horizontal portions of the L-shaped palatine bones are visible on this view. The vertical portions are demonstrated later on a cutaway drawing (see Fig. 11.47).

Note the differences between the palatine process of the maxillary bone and the separate palatine facial bones.

The two small inferior portions of the sphenoid bone of the cranium also are shown on this inferior view of the hard palate. These two processes, the **pterygoid hamuli,** are similar to the feet of the outstretched legs of a bat, as described in an earlier drawing in the chapter (also see Fig. 11.15, p. 384).

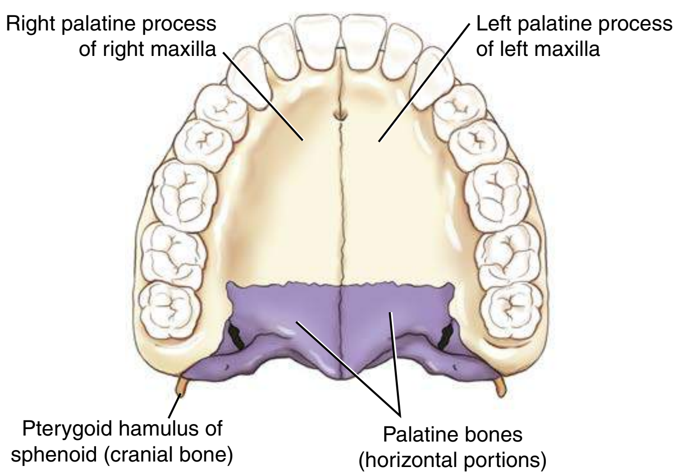


Fig. 11.42 Maxillae and palatine bones—hard palate (inferior surface).

Articulations

Each maxilla articulates with **two cranial bones** (frontal and ethmoid) and with **seven facial bones** (zygoma, lacrimal, nasal, palatine, inferior nasal concha, vomer, and adjacent maxilla).

RIGHT AND LEFT ZYGOMATIC BONES

One **zygoma** is located lateral to the zygomatic process of each maxilla. These bones (sometimes termed *malar bones*) form the prominence of the cheeks and make up the lower outer portion of the orbits.

Projecting posteriorly from the zygoma is a slender process that connects with the zygomatic process of the temporal bone to form the **zygomatic arch.** The zygomatic arch is a delicate structure that sometimes is fractured or "caved in" by a blow to the cheek. The anterior portion of the arch is formed by the zygoma, and the posterior portion is formed by the zygomatic process of the temporal bone. The **zygomatic prominence** is a positioning landmark, and the term refers to this prominent portion of the zygoma (Fig. 11.43).

Articulations

Each zygoma articulates with **three cranial bones** (frontal, sphenoid, and temporal) and with **one facial bone** (maxilla).

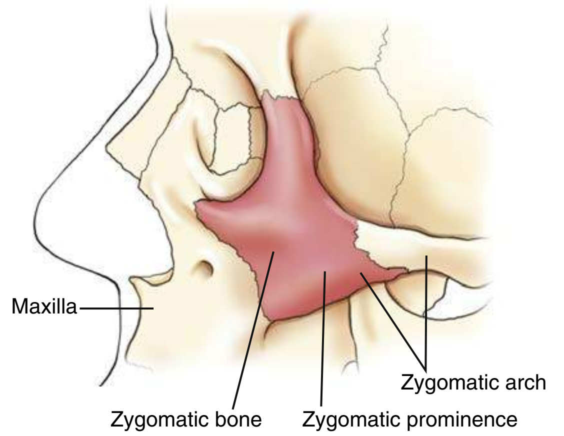


Fig. 11.43 Zygomatic bone—lateral view.

RIGHT AND LEFT NASAL AND LACRIMAL BONES

The lacrimal and nasal bones are the thinnest and most fragile bones in the entire body.

Lacrimal Bones

The two small and delicate lacrimal bones (about the size and shape of a fingernail) lie anteriorly on the medial side of each orbit just posterior to the frontal process of the maxilla (Fig. 11.44). *Lacrimal,* derived from a word meaning "tear," is an appropriate term because the lacrimal bones are closely associated with the tear ducts.

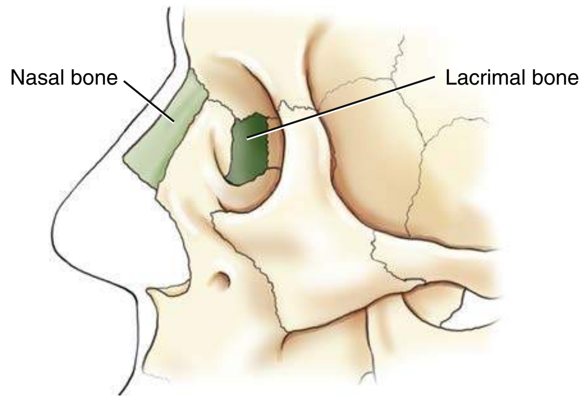


Fig. 11.44 Nasal and lacrimal bones—lateral view.

Nasal Bones

The two fused nasal bones form the bridge of the nose and are variable in size. Some people have very prominent nasal bones, whereas nasal bones are quite small in other people. Much of the nose is made up of cartilage, and only the two nasal bones form the bridge of the nose. The nasal bones lie anterior and superomedial to the frontal process of the maxillae and inferior to the frontal bone. The point of junction of the two nasal bones with the frontal bone is a surface landmark called the *nasion* (Fig. 11.45).

Articulations

Lacrimal Each lacrimal bone articulates with **two cranial bones** (frontal and ethmoid) and with **two facial bones** (maxilla and inferior nasal concha).

Nasal Each nasal bone articulates with **two cranial bones** (frontal and ethmoid) and with **two facial bones** (maxilla and adjacent nasal bone) (see Fig. 11.45).

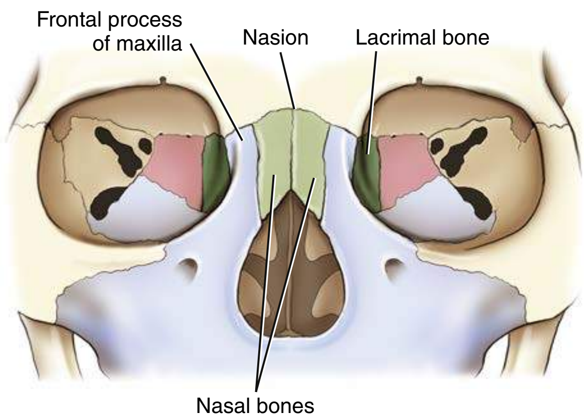


Fig. 11.45 Nasal and lacrimal bones—frontal view.

RIGHT AND LEFT INFERIOR NASAL CONCHAE

Within the nasal cavity are two platelike, curved (or scroll-shaped) facial bones called the **inferior nasal conchae** *(turbinates).* These two bones project from the lateral walls of the nasal cavity on each side and extend medially (Fig. 11.46).

There are three pairs of nasal conchae. The superior and middle pairs are parts of the ethmoid bone, and the inferior pair consists of separate facial bones.

The effect of the three pairs of nasal conchae is to divide the nasal cavities into various compartments. These irregular compartments tend to break up or mix the flow of air coming into the nasal cavities before it reaches the lungs. In this way, incoming air is warmed and cleaned as it comes in contact with the mucous membrane that covers the conchae.

Sectional Drawing

Inferior Nasal Conchae The relationship between the various nasal conchae and the lateral wall of one nasal cavity is illustrated in this sectional drawing (Fig. 11.47). The midline structures that make up the nasal septum have been removed so that the lateral portion of the right nasal cavity can be seen. The **superior and middle conchae** are part of the ethmoid bone, and the **inferior nasal conchae** are separate facial bones. The **cribriform plate** and the **crista galli** of the ethmoid bone help to separate the cranium from the facial bone mass. The **palatine process** of the maxilla is shown again.

RIGHT AND LEFT PALATINE BONES

The two **palatine bones** are difficult to visualize in the study of a dry skeleton because they are located internally and are not visible from the outside. Each palatine bone is roughly L-shaped (see Fig. 11.47). The vertical portion of the "L" extends upward between one maxilla and one pterygoid plate of the sphenoid bone. The horizontal portion of each "L" helps to make up the posterior portion of the hard palate, as shown in an earlier drawing (see Fig. 11.42). Additionally, the most superior small tip of the palatine can be seen in the posterior aspect of the orbit (see Fig. 11.71, p. 401).

Articulations

Inferior Nasal Conchae Each inferior nasal concha articulates with **one cranial bone** (ethmoid) and with **three facial bones** (maxilla, lacrimal, and palatine).

Palatine Each palatine articulates with **two cranial bones** (sphenoid and ethmoid) and **four facial bones** (maxilla, inferior nasal conchae, vomer, and adjacent palatine).

NASAL SEPTUM

The midline structures of the nasal cavity, including the **bony nasal septum,** are shown on this sagittal view drawing (Fig. 11.48). Two bones—the **ethmoid** and the **vomer**—form the bony nasal septum. Specifically, the septum is formed superiorly by the **perpendicular plate** of the ethmoid bone and inferiorly by the single vomer bone, which can be demonstrated radiographically. Anteriorly, the nasal septum is cartilaginous and is termed the **septal cartilage.**

Vomer

The single **vomer** (meaning "plowshare") bone is a thin, triangular bone that forms the inferoposterior part of the nasal septum. The surfaces of the vomer are marked by small, furrow-like depressions for blood vessels, a source of nosebleed with trauma to the nasal area. A deviated nasal septum describes the clinical condition wherein the nasal septum is deflected or displaced laterally from the midline of the nose. This deviation usually occurs at the site of junction between the septal cartilage and the vomer. A severe deviation can entirely block the nasal passageway, making breathing through the nose impossible.

Articulations The vomer articulates with **two cranial bones** (sphenoid and ethmoid) and with **four facial bones** (right and left palatine bones and right and left maxillae). The vomer also articulates with the septal cartilage.

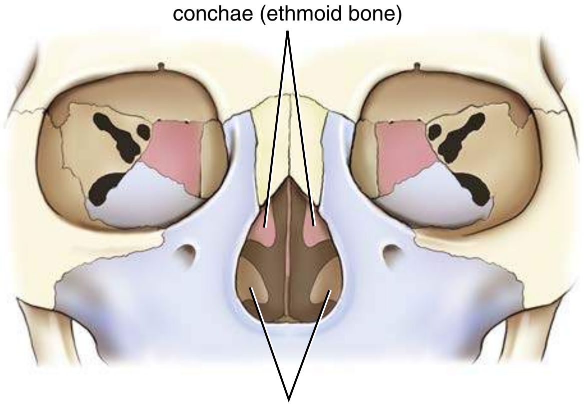


Fig. 11.46 Inferior nasal conchae.

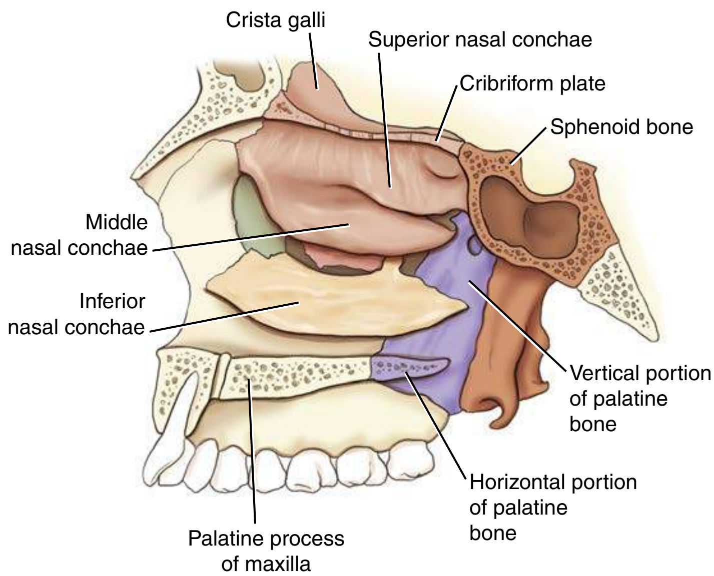


Fig. 11.47 Sectional view drawing—inferior nasal conchae and palatine bones.

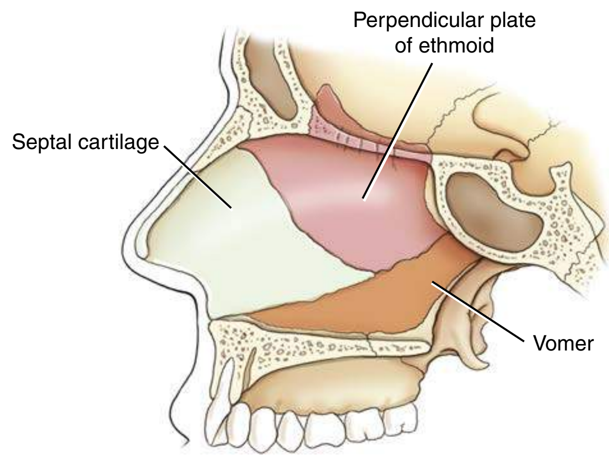


Fig. 11.48 Bony nasal septum and vomer.

11

MANDIBLE

The last and largest facial bone is the lower jaw, or **mandible.** It is the only movable bone in the adult skull. This large facial bone, which is a single bone in the adult, originates from two separate bones. The two bones in the infant join to become one at approximately 1 year of age.

Lateral View

The **angle** (gonion) of the mandible divides each half of the mandible into two main parts. That area anterior to the angle is termed the **body** of the mandible, whereas the area superior to each angle is termed the **ramus.** Because the mandible is a single bone, the body extends from the left angle around to the right angle (Fig. 11.49).

The lower teeth are rooted in the mandible. An **alveolar process,** or ridge, extends along the entire superior portion of the body of the mandible.

Frontal View

The anterior aspect of the adult mandible is best seen on a frontal view. The single body forms from each lateral half and unites at the anterior midline. This union is called the **symphysis** of the mandible, or the **symphysis menti.** The flat triangular area below the symphysis, marked by two knoblike protuberances that project forward, is called the **mental protuberance.** The center of the mental protuberance is described as the **mental point.** *Mentum* and *mental* are Latin words that refer to the general area known as the chin. The mental point is a specific point on the chin, whereas the mentum is the entire area.

Located on each half of the body of the mandible is a **mental foramen.** The mental foramina serve as passageways for the mental artery and vein and mental nerve (branch of inferior alveolar nerve) that innervates the lower lip and chin.

Ramus

The upper portion of each **ramus** terminates in a U-shaped notch termed the **mandibular notch.** At each end of the mandibular notch is a process. The process at the anterior end of the mandibular notch is termed the **coronoid process** (Fig. 11.50). This process does not articulate with another bone and cannot be palpated easily because it lies just inferior to the zygomatic arch and serves as a site for muscle attachment.

The **coronoid process** of the mandible must not be confused with the **coronoid process** of the proximal ulna of the forearm or the **coracoid process** of the scapula.

The posterior process of the upper ramus is termed the **condyloid process** and consists of two parts. The rounded end of the condyloid process is the **condyle** or **head,** whereas the constricted area directly below the condyle is the **neck.** The condyle of the condyloid process fits into the TM fossa of the temporal bone to form the **TMJ.**

Submentovertical Projection

The horseshoe shape of the mandible is well visualized on a **submentovertical** (SMV) projection (Fig. 11.51). The mandible is a thin structure, which explains why it is susceptible to fracture. The area of the **mentum** is well demonstrated, as are the **body, ramus,** and **gonion** of the mandible. The relative position of the upper ramus and its associated **coronoid process** and **condyle** are also demonstrated with this projection. The condyle projects medial and the coronoid process slightly lateral on this view.

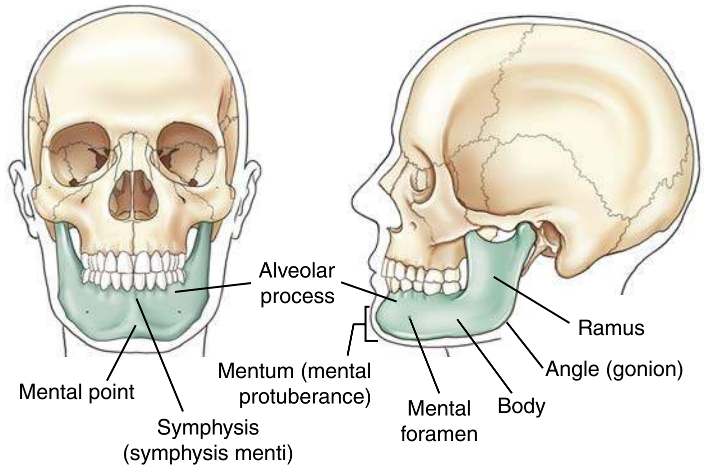


Fig. 11.49 Mandible—lateral and frontal views.

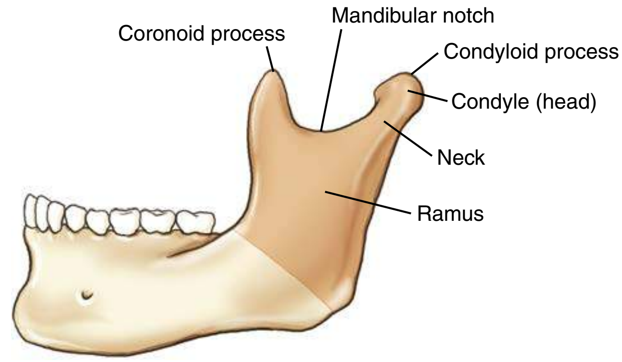


Fig. 11.50 Ramus of mandible—lateral view.

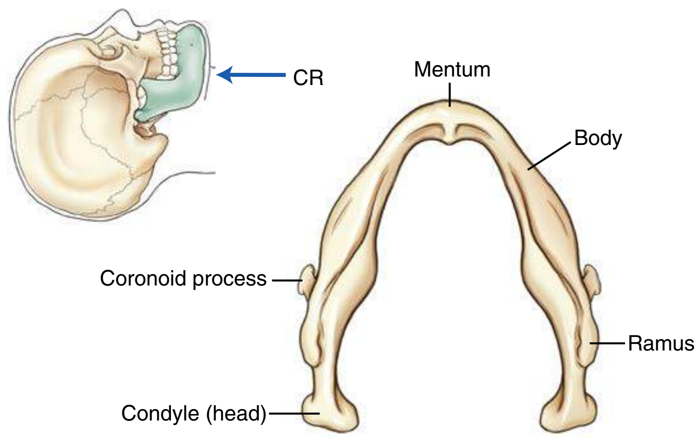


Fig. 11.51 SMV projection of mandible.

TEMPOROMANDIBULAR JOINT

The TMJ, the only movable joint in the skull, is shown on this lateral drawing (Fig. 11.52) and on the lateral view photograph of the skull (Fig. 11.53). The relationship of the mandible to the temporal bone of the cranium is well demonstrated. The TMJ is located just anterior and slightly superior to the **EAM.**

Joint Classifications (Mandible and Skull)

SYNOVIAL JOINTS (DIARTHRODIAL)

The complex TMJ is classified as a **synovial type** of joint divided into upper and lower synovial cavities by a single articular fibrous disk (Table 11.1). A series of strong ligaments join the condylar neck, ramus, and gonion of the mandible to the lower borders of the zygomatic process of the temporal bone.

This complete two-part synovial joint, along with its fibrous articular disk, allows for not only a hinge-type motion but also a gliding movement. The action of this type of joint is very complex. Two movements are predominant. When the mouth opens, the condyle and the fibrocartilage move forward, and at the same time, the condyle revolves around the fibrocartilage. The TMJ is classified as a bicondylar joint similar to the knee.[3]

FIBROUS JOINTS (SYNARTHRODIAL)

Two types of fibrous joints involve the skull, both of which are **synarthrodial,** or immovable. First are the **sutures** between cranial bones, as described earlier. Second is a unique type of fibrous joint involving the teeth with the mandible and maxillae. This is a **gomphosis** *(gom-fo′-sis)* subclass type of fibrous joint that is found between the roots of the teeth and the alveolar processes of both the maxillae and the mandible.

TMJ Motion

The drawings and radiographs illustrate the TMJ in both **open-mouth** and **closed-mouth** positions (Fig. 11.54). When the mouth is opened widely, the condyle moves forward to the front edge of the fossa. If the condyle slips too far anteriorly, the joint may dislocate. If the TMJ dislocates, either by force or by jaw motion, it may be difficult or impossible to close the mouth, which returns the condyle to its normal position.

Radiographs (Open and Closed Mouth)

Two axiolateral projections (Schuller method) of the TMJ are shown in closed-mouth and open-mouth positions (Figs. 11.55 and 11.56). The range of anterior movement of the condyle in relationship to the TM fossa is clearly demonstrated.

TABLE 11.1 **JOINTS OF MANDIBLE**

TEMPOROMANDIBULAR JOINT	ALVEOLI AND ROOTS OF TEETH
Classification Synovial (diarthrodial)	**Classification** Fibrous (synarthrodial)
Movement Types Bicondylar	**Subclass** Gomphosis
Plane (gliding)	

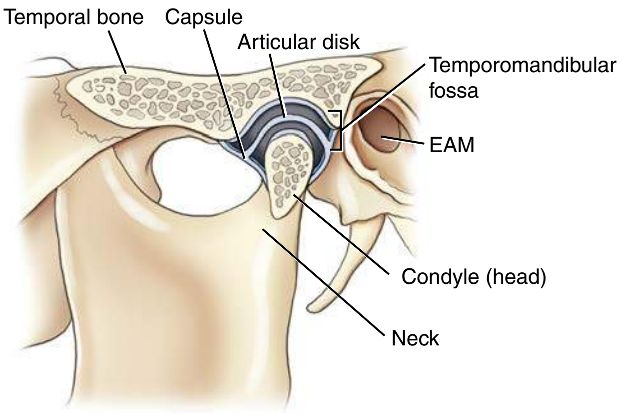


Fig. 11.52 Temporomandibular joint.

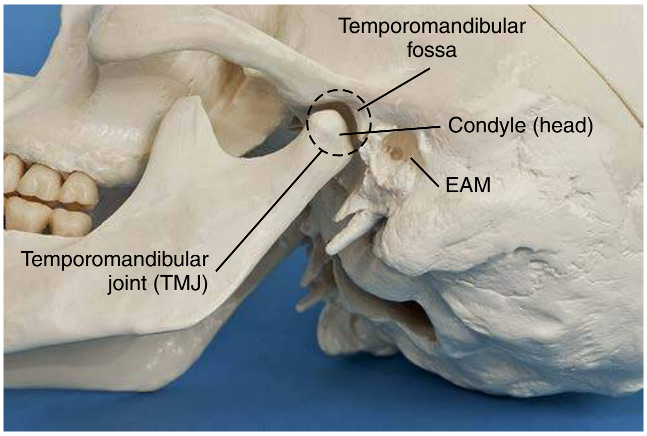


Fig. 11.53 Temporomandibular joint of mandible.

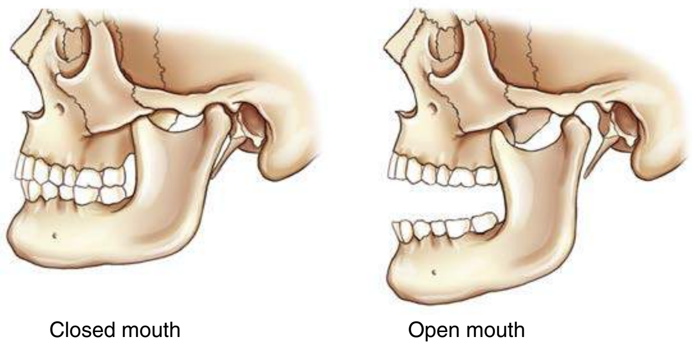


Fig. 11.54 TMJ movements.

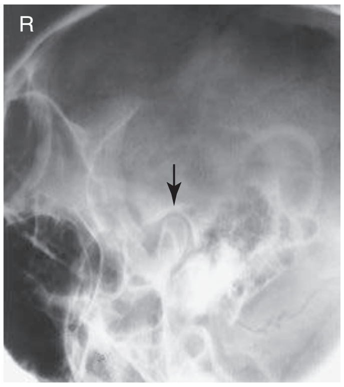


Fig. 11.55 Closed mouth.

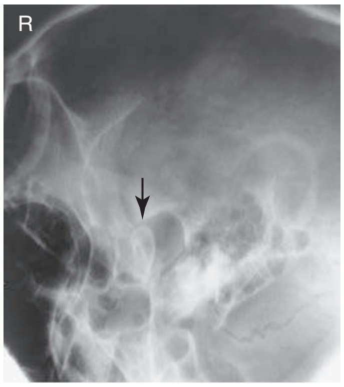


Fig. 11.56 Open mouth.

Paranasal Sinuses

The large, air-filled cavities of the **paranasal sinuses** are sometimes called the accessory nasal sinuses because they are lined with mucous membrane, which is continuous with the nasal cavity. These sinuses are divided into four groups, according to the bones that contain them:

Maxillary (2)	Maxillary (facial) bones
Frontal (usually 2)	Frontal (cranial) bones
Ethmoid (many)	Ethmoid (cranial) bones
Sphenoid (1 or 2)	Sphenoid (cranial) bone

Only the **maxillary sinuses** are part of the **facial bone** structure. The **frontal, ethmoid,** and **sphenoid** sinuses are contained within their respective **cranial bones** (Fig. 11.57).

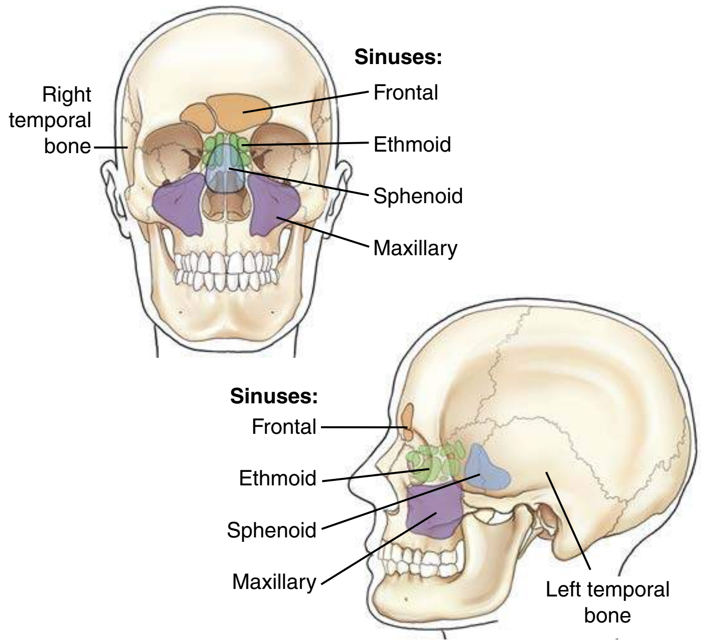


Fig. 11.57 Skull—paranasal sinuses and temporal bone.

PURPOSE

The purpose of the paranasal sinuses is subject to speculation. Various sources suggest they assist in vocal resonance, lighten the weight of the skull, and produce mucus to moisten the nasal passageways and air entering the nasal airway.

The paranasal sinuses begin to develop in the fetus, but only the maxillary sinuses exhibit a definite cavity at birth. The frontal and sphenoid sinuses begin to be visible on radiographs at age 6 or 7. The ethmoid sinuses develop last. All the paranasal sinuses generally are fully developed by the late teenage years.

Each of these groups of sinuses is studied, beginning with the largest, the maxillary sinuses.

MAXILLARY SINUSES

The large **maxillary sinuses** are paired structures, one of which is located within the body of each maxillary bone. An older term for maxillary sinus is **antrum,** an abbreviation for **antrum of Highmore**.

Each maxillary sinus is shaped like a pyramid on a frontal view. Laterally, the maxillary sinuses appear more cubic. The average total vertical dimension is 1 to 1½ inches (2.5 to 4 cm), and the other dimensions are approximately 1 inch (2.5 cm).

The bony walls of the maxillary sinuses are thin. The floor of each maxillary sinus is slightly below the level of the floor of each nasal fossa. The two maxillary sinuses vary in size from one person to another and sometimes from one side to the other. Projecting into the floor of each maxillary sinus are several conic elevations related to roots of the first and second upper molar teeth (Fig. 11.58). Occasionally, one or more of these roots can allow infection that originates in the teeth, particularly in the molars and premolars, to travel upward into the maxillary sinus.

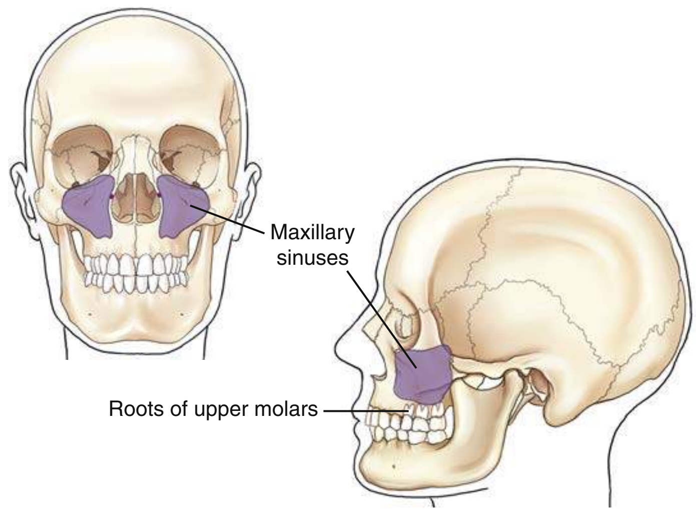


Fig. 11.58 Maxillary sinuses (2).

All the paranasal sinus cavities communicate with one another and with the **nasal cavity,** which is divided into two equal chambers, or **fossae.** In the case of the maxillary sinuses, this site of communication is the opening into the middle nasal meatus passageway located at the superior medial aspect of the sinus cavity itself, as demonstrated in Fig. 11.59. (The osteomeatal complex is illustrated in greater detail later in Figs. 11.63 and 11.64.) When a person is erect, any mucus or fluid trapped within the sinus tends to remain there and layer out, forming an air-fluid level. Therefore, radiographic positioning of the paranasal sinuses should be accomplished with the patient in the **erect position,** if possible, to delineate any possible air-fluid levels.

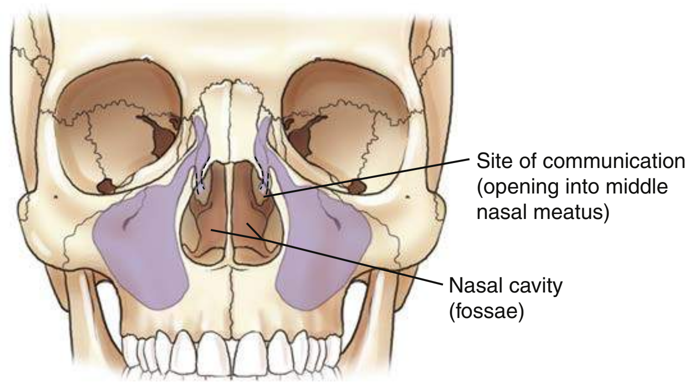


Fig. 11.59 Maxillary sinuses.

FRONTAL SINUSES

The **frontal sinuses** are located between the inner and outer tables of the skull, posterior to the glabella; they **rarely become aerated before age 6.** Whereas the maxillary sinuses are always paired and are usually fairly symmetric in size and shape, the frontal sinuses are rarely symmetric (Fig. 11.60). The frontal sinuses usually are separated by a septum, which deviates from one side to the other or may be absent entirely, resulting in a single cavity. However, two cavities generally exist, which vary in terms of size and shape. They generally are larger in men than in women. They may be singular on the right or the left side, they may be paired as shown, or they may be absent.

ETHMOID SINUSES

The **ethmoid sinuses** are contained within the lateral masses or labyrinths of the ethmoid bone. These air cells are grouped into **anterior, middle,** and **posterior collections,** but they all intercommunicate (Fig. 11.61).

When viewed from the side, the anterior ethmoid sinuses appear to fill the orbits. This occurs because portions of the ethmoid sinuses are contained in the lateral masses of the ethmoid bone, which helps to form the medial wall of each orbit.

SPHENOID SINUSES

The **sphenoid sinuses** lie in the body of the sphenoid bone directly below the sella turcica (Fig. 11.62). The body of the sphenoid that contains these sinuses is cubic and frequently is divided by a thin septum to form two cavities. This septum may be incomplete or absent entirely, resulting in only one cavity.

Because the sphenoid sinuses are so close to the base or floor of the cranium, sometimes pathologic processes make their presence known by their effect on these sinuses. An example is the demonstration of an air-fluid level within the sphenoid sinuses after skull trauma. This air-fluid level may provide evidence that the patient has a basal skull fracture and that either blood or cerebrospinal fluid is leaking through the fracture into the sphenoid sinuses, a condition referred to as **sphenoid effusion.**

Osteomeatal Complex

The drainage pathways of the frontal, maxillary, and ethmoid sinuses make up the **osteomeatal complex,** which can become obstructed, leading to infection of these sinuses, a condition termed **sinusitis.** The osteomeatal complex, sometimes called the osteomeatal unit (OMU), can be imaged with CT to evaluate for obstructions.

Figs. 11.63 and 11.64 illustrate **two key passageways** (infundibulum and middle nasal meatus) and their associated structures identified on coronal CT. The **large maxillary sinus** drains through the **infundibulum** passageway down through the **middle nasal meatus** into the **inferior nasal meatus.** The **uncinate process** of the ethmoid bone makes up the medial wall of the infundibulum passageway. The **ethmoid bulla** receives drainage from the frontal and ethmoid sinus cells, which drains down through the middle nasal meatus into the inferior nasal meatus, where it exits the body through the exterior nasal orifice.

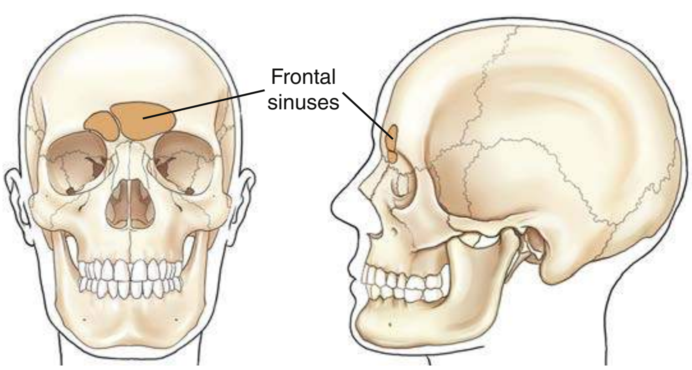


Fig. 11.60 Frontal sinuses.

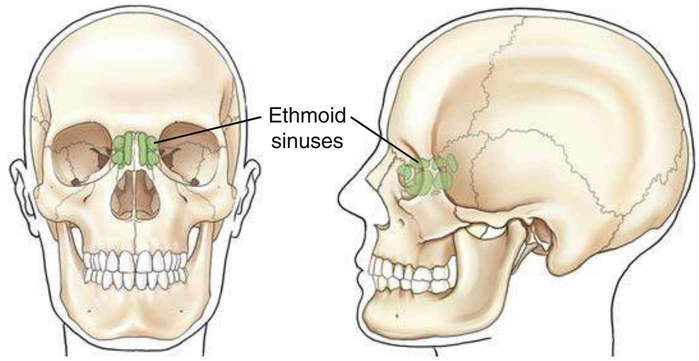


Fig. 11.61 Ethmoid sinuses.

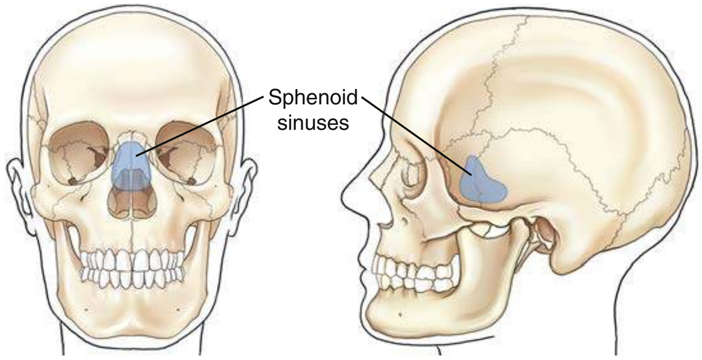


Fig. 11.62 Sphenoid sinuses.

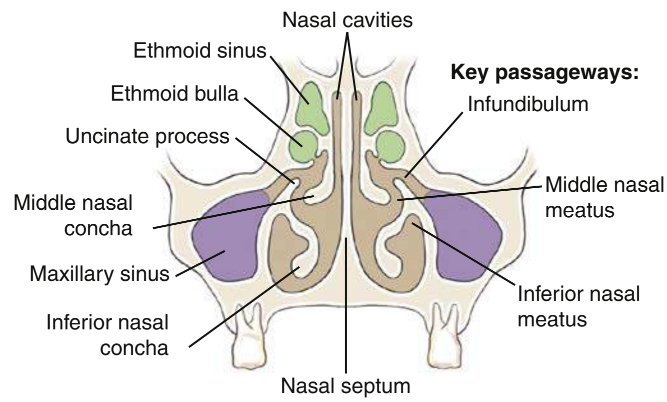


Fig. 11.63 Osteomeatal complex—coronal sectional view.

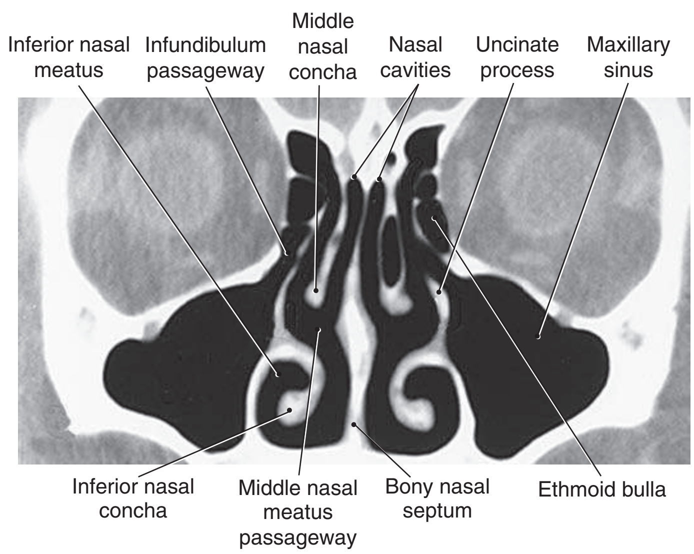


Fig. 11.64 Coronal CT, osteomeatal complex. (From Kelley L, Petersen C: *Sectional anatomy for imaging professionals,* ed 4, St. Louis, 2018, Elsevier.)

11

RADIOGRAPHS—PARANASAL SINUSES

Drawings of the sinuses on preceding pages revealed definite sizes and shapes of the sinuses with clear-cut borders. On actual radiographs, these borders are not nearly as defined because various sinuses overlap and superimpose each other, as can be seen on these radiographs of four common sinus projections. The labeled radiographs clearly demonstrate the relative locations and relationships of each of these sinuses. (Note the following abbreviations: **F**—frontal sinuses; **E**—ethmoid sinuses; **M**—maxillary sinuses; and **S**—sphenoid sinuses.)

Lateral Position

The frontal sinuses are clearly visualized between the inner and outer tables of the skull (Fig. 11.65).

The sphenoid sinuses appear to be continuous with the ethmoid sinuses anteriorly.

The large maxillary sinuses are clearly visualized. The roots of the molars and premolars of the upper teeth appear to extend up through the floor of the maxillary sinuses.

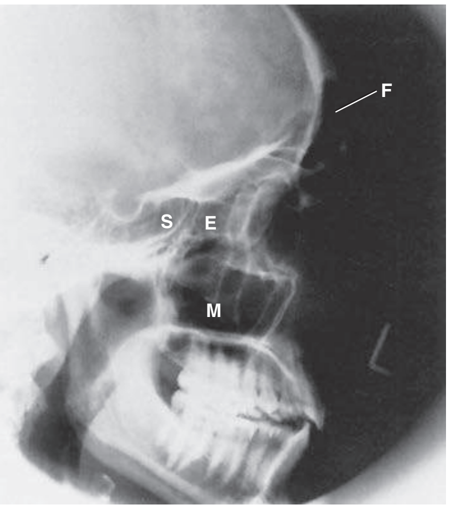


Fig. 11.65 Lateral sinuses projection.

PA (Caldwell) Projection

The frontal, ethmoid, and maxillary sinuses are clearly illustrated on this PA axial projection radiograph (Fig. 11.66). The sphenoid sinuses are not demonstrated specifically because they are located directly posterior to the ethmoid sinuses. This relationship is demonstrated on the lateral view (see Fig. 11.65) and the SMV projection (see Fig. 11.68).

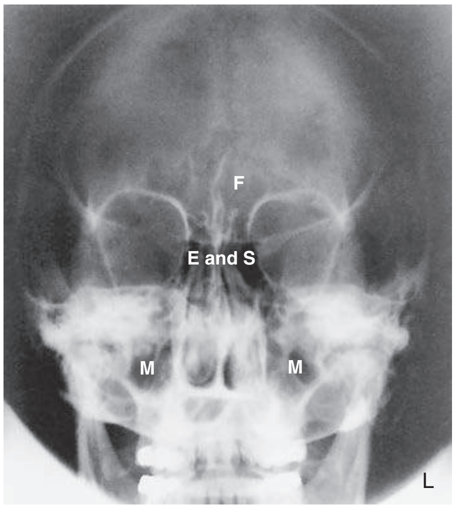


Fig. 11.66 PA axial—Caldwell projection.

Parietoacanthial Transoral Projection (Open-Mouth Waters)

All four groups of sinuses are clearly demonstrated on this projection taken with the mouth open and the head tipped back to separate and project the sphenoid sinuses inferior to the ethmoid sinuses (Fig. 11.67). The open mouth also removes the upper teeth from direct superimposition of the sphenoid sinuses. The pyramid-shaped maxillary sinuses are clearly seen.

SMV Projection

The SMV projection is obtained with the head tipped back so that the top of the head (vertex) is touching the table/upright imaging device surface and the CR is directed inferior to the chin (mentum) (Fig. 11.68).

The centrally located sphenoid sinuses are anterior to the large opening, the foramen magnum. The multiple clusters of ethmoid air cells extend to each side of the nasal septum. The mandible and teeth superimpose the maxillary sinuses. Portions of the maxillary sinuses are visualized laterally.

Fig. 11.68 demonstrates these air-filled mastoids (labeled A) and the dense petrous portions of the temporal bones (labeled B).

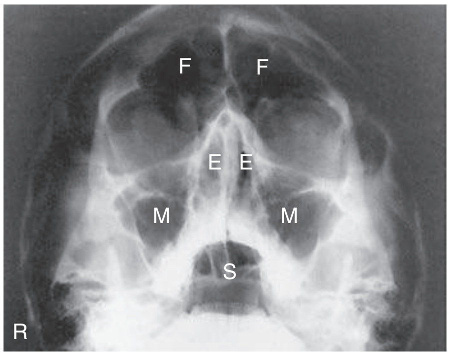


Fig. 11.67 Parietoacanthial transoral projection (open-mouth Waters method).

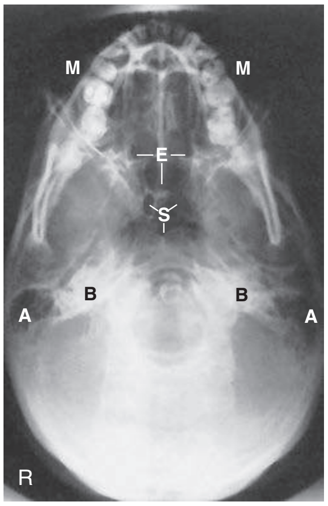


Fig. 11.68 SMV projection.

Orbits

The complex anatomy of the 14 facial bones helps to form several facial cavities. These cavities, which are formed in total or in part by the facial bones, include the mouth (oral cavity), the nasal cavities, and the orbits. The mouth and nasal cavities are primarily passageways and are rarely imaged. However, the orbits that contain the vital organs of sight and associated nerves and blood vessels are imaged more frequently. The structure and shape of the orbits are illustrated in this simplified drawing (Fig. 11.69). Each orbit is a **cone-shaped,** bony-walled structure, as is shown in the drawing.

The rim of the orbit, which corresponds to the outer circular portion of the cone, is called the **base.** However, the base of the orbit is not a true circle and may even look like a figure with four definite sides. The posterior portion of the cone, the **apex,** corresponds to the **optic foramen,** through which the optic nerve passes.

The long axis of the orbits projects both upward and toward the midline. With the head placed in an upright frontal or lateral position with the orbitomeatal line adjusted parallel to the floor, each orbit would project superiorly at an angle of **30°** and toward the MSP at an angle of **37°.** These two angles are important for radiographic positioning of the optic foramina. Each optic foramen is located at the apex of its respective orbit. To radiograph either optic foramen, it is necessary both to extend the patient's chin by 30° and to rotate the head 37°. The CR projects through the base of the orbit along the long axis of the cone-shaped orbit.

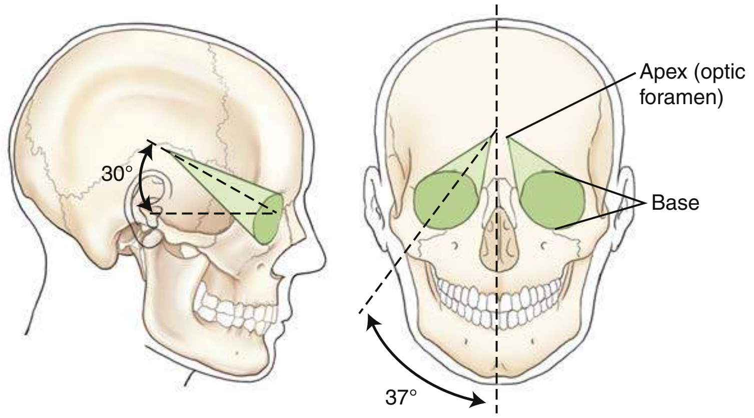


Fig. 11.69 Orbits (cone shaped).

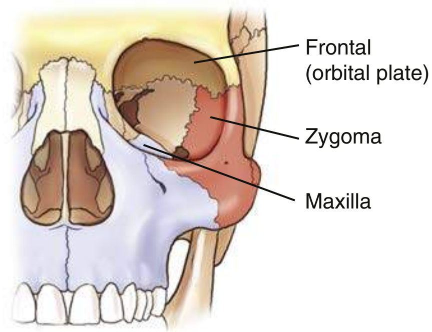


Fig. 11.70 Base of orbit—**three bones** (direct frontal view).

BONY COMPOSITION OF ORBITS

Each orbit is composed of parts of **seven bones.** The circumference or circular base of each orbit is composed of parts of **three** bones—the **frontal bone (orbital plate)** from the cranium and the **maxilla** and the **zygoma** from the facial bones (Fig. 11.70). A roof, a floor, and two walls, parts of which also are formed by these three bones, are found inside each orbital cavity. The orbital plate of the frontal bone forms most of the roof of the orbit. The zygoma forms much of the lateral wall and some of the floor of the orbit, whereas a portion of the maxilla helps to form the floor.

The slightly oblique frontal view in Fig. 11.71 demonstrates all seven bones that form each orbit. The **frontal bone, zygoma,** and **maxilla**, which form the base of the orbit, are shown again. A portion of the medial wall of the orbit is formed by the thin **lacrimal bone.** The **sphenoid** and **ethmoid** bones make up most of the posterior orbit, whereas only a small bit of the **palatine** bone contributes to the innermost posterior portion of the floor of each orbit.

The **seven** bones that make up each orbit include **three cranial bones** and **four facial bones,** as shown in Box 11.1.

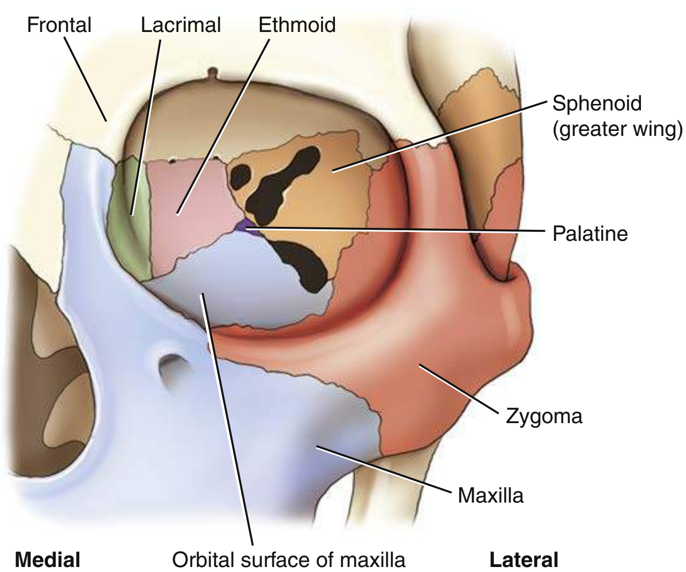


Fig. 11.71 Orbit—seven bones (slightly oblique frontal view).

BOX 11.1 **BONES OF ORBITS**

CRANIAL BONES	FACIAL BONES
Frontal	Maxilla
Sphenoid	Zygoma
Ethmoid	Lacrimal
	Palatine

OPENINGS IN POSTERIOR ORBIT

Each orbit also contains three holes or openings in the posterior portion, as shown in Fig. 11.72. These openings provide for passage of specific cranial nerves (**CN**). (The 12 pairs of cranial nerves are listed and described in the anatomy section of Chapter 18.)

The **optic foramen** is a small hole in the sphenoid bone that is located posteriorly at the apex of the cone-shaped orbit. The optic foramen allows for passage of the optic nerve (CN II), which is a continuation of the retina.

The **superior orbital fissure** is a cleft or opening between the greater and lesser wings of the sphenoid bone, located lateral to the optic foramen. It allows transmission of four primary cranial nerves (CN III to CN VI), which control movement of the eye and eyelid.

A third opening is the **inferior orbital fissure,** which is located between the maxilla, zygomatic bone, and greater wing of the sphenoid. It allows for transmission of the maxillary branch of CN V, which permits entry of sensory innervation for the cheek, nose, upper lip, and teeth.

The small root of bone that separates the superior orbital fissure from the optic canal is known as the **sphenoid strut.** The optic canal is a small canal into which the optic foramen opens. Any abnormal enlargement of the optic nerve could cause erosion of the sphenoid strut, which is actually a portion of the lateral wall of the optic canal.

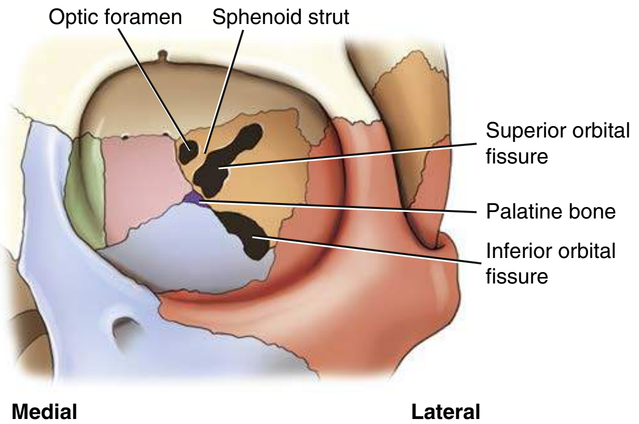


Fig. 11.72 Orbits—posterior openings (slightly oblique frontal view).

Anatomy Review

Review exercises for anatomy of the cranial and facial bones follow. Anatomy can be demonstrated on a dry skull or on radiographs. Some anatomic parts identified on the dry skull are not visualized on these radiographs. The parts that are identifiable are labeled as such. A good learning or review exercise is to study both the skull illustrations and the radiographs carefully and identify each part before looking at the answers listed next.

Seven Bones of Left Orbit (Fig. 11.73)

A. Frontal bone (orbital plate)
B. Sphenoid bone
C. Small portion of palatine bone
D. Zygomatic bone
E. Maxillary bone
F. Ethmoid bone
G. Lacrimal bone

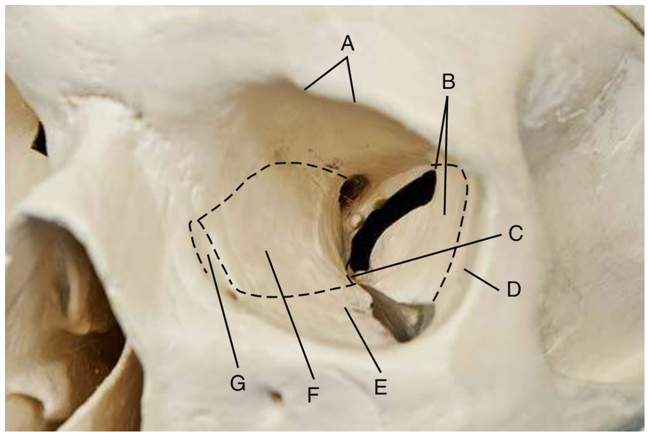


Fig. 11.73 Seven bones of left orbit.

Openings and Structures of Left Orbit (Fig. 11.74)

A. Optic foramen
B. Sphenoid strut
C. Superior orbital fissure
D. Inferior orbital fissure

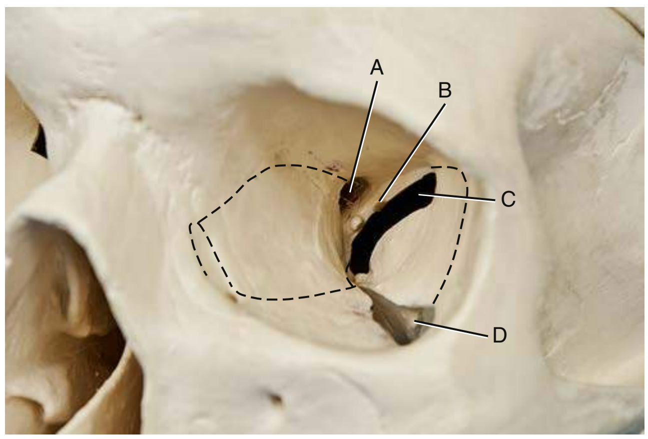


Fig. 11.74 Openings and structures of left orbit.

Parieto-Orbital Oblique (Rhese method) Projection of Orbits (Fig. 11.75)

A. Orbital plate of frontal bone
B. Sphenoid bone
C. Optic foramen and canal
D. Superior orbital fissure
E. Infraorbital margin (IOM)
F. Sphenoid strut (part of inferior and lateral wall of optic canal)
G. Lateral orbital margin
H. Supraorbital margin

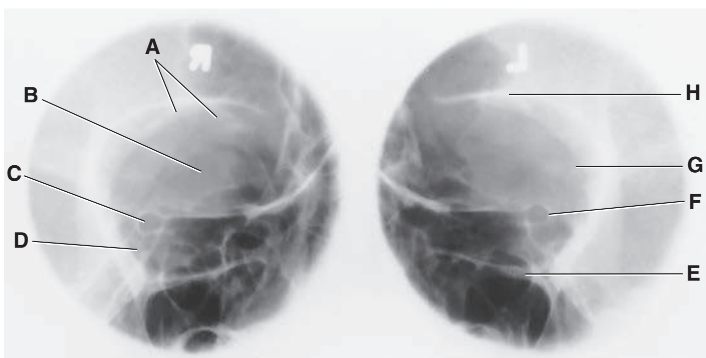


Fig. 11.75 Parieto-orbital oblique projection of orbits (optic foramina).

Facial Bones—Lateral (Figs. 11.76 and 11.77)

A. Zygomatic arch
B. Right zygomatic bone
C. Right nasal bone
D. Frontal process of right maxilla
E. Anterior nasal spine
F. Alveolar process of maxilla
G. Alveolar process of mandible
H. Mentum or mental protuberance
I. Mental foramen
J. Body of mandible
K. Angle (gonion)
L. Ramus of mandible
M. Coronoid process
N. Mandibular notch
O. Neck of mandibular condyle
P. Condyle or head of mandible
Q. EAM
R. TM fossa of temporal bone
S. Greater wings of sphenoid
T. Lesser wings of sphenoid with anterior clinoid processes
U. Ethmoid sinuses between orbits
V. Body of maxilla containing maxillary sinuses

Facial Bones—Parietoacanthial (Waters)

The photograph (Fig. 11.78) and the radiograph (Fig. 11.79) represent the skull in a parietoacanthial projection (Waters method), with the head tilted back. This is one of the more common projections used to visualize the facial bones, as follows:

A. Zygomatic prominence
B. Body of maxilla (contains maxillary sinuses)
C. Bony nasal septum (perpendicular plate of ethmoid and vomer bone)
D. Anterior nasal spine
E. Zygomatic arch
F. Coronoid process of mandible (Fig. 11.78 only)
G. Condyle (head) of mandible
H. Mastoid process of temporal bone
I. Angle of mandible
J. Foramen magnum (Fig. 11.79, which demonstrates the dens or odontoid process within the foramen magnum)

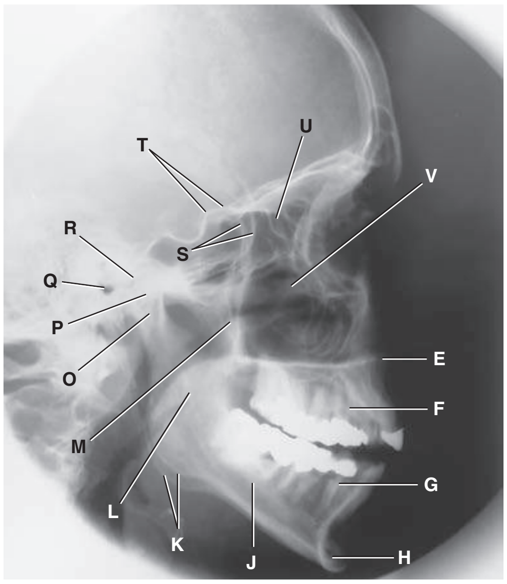


Fig. 11.77 Facial bones—lateral projection.

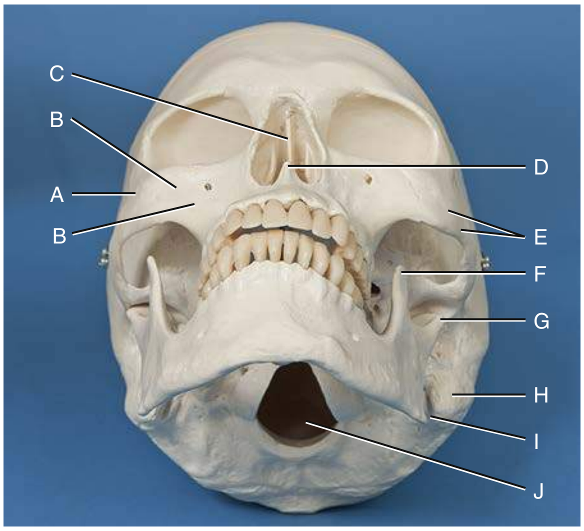


Fig. 11.78 Facial bones—parietoacanthial projection (Waters method).

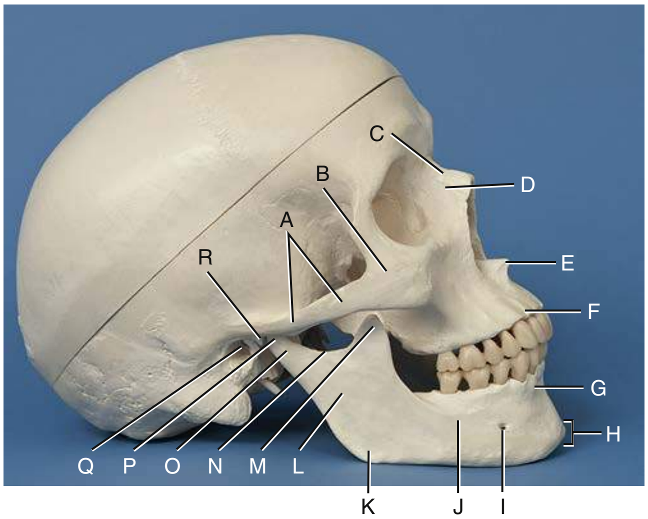


Fig. 11.76 Facial bones—lateral.

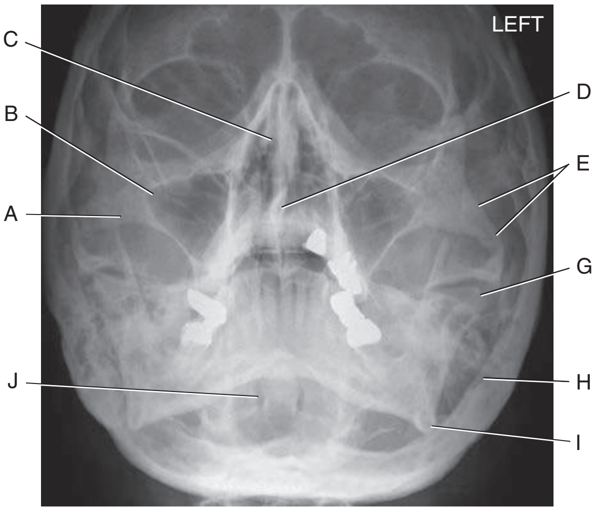


Fig. 11.79 Facial bones—parietoacanthial projection (Waters method).

Facial Bones—SMV (Inferior View)

Fig. 11.80 shows an inferior view of the dry skull with the mandible removed. The SMV projection radiograph in Fig. 11.81 demonstrates positioning whereby the top of the head (vertex) is placed against the image receptor (IR), and the CR enters under the chin (mentum).

Skull (Fig. 11.80)

A. Zygomatic arch
B. Palatine process of maxilla
C. Horizontal process of palatine bone
D. Pterygoid hamulus of sphenoid

Radiograph (Fig. 11.81)

E. Foramen ovale of sphenoid
F. Foramen spinosum of sphenoid
G. Foramen magnum
H. Petrous pyramid of temporal bone
I. Mastoid portion of temporal bone
J. Sphenoid sinus in body of sphenoid
K. Condyle (head) of mandible
L. Posterior border (vertical portion) of palatine bone
M. Vomer or bony nasal septum
N. Right maxillary sinuses
O. Ethmoid sinuses

Facial Bones—Frontal View (Fig. 11.82)

A. Left nasal bone
B. Frontal process of left maxilla
C. Optic foramen
D. Superior orbital fissure
E. Inferior orbital fissure
F. Superior and middle nasal conchae of ethmoid bone
G. Vomer bone (lower portion of bony nasal septum)
H. Left inferior nasal conchae
I. Anterior nasal spine
J. Alveolar process of left maxilla
K. Alveolar process of left mandible
L. Mental foramen
M. Mentum or mental protuberance
N. Body of right mandible
O. Angle (gonion) of right mandible
P. Ramus of right mandible
Q. Body of right maxilla (contains maxillary sinuses)
R. Zygomatic prominence of right zygomatic bone
S. Outer orbit portion of right zygomatic bone
T. Sphenoid bone (cranial bone)

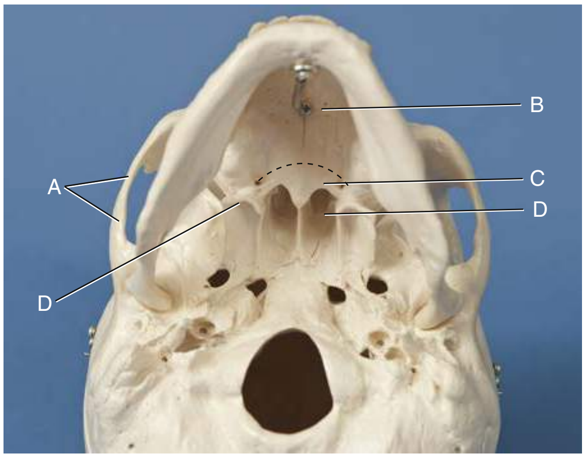


Fig. 11.80 Facial bones—inferior view.

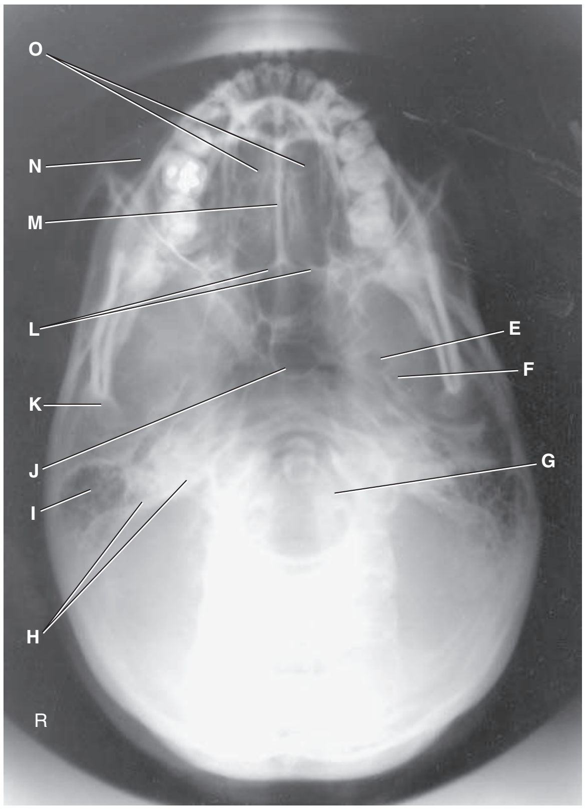


Fig. 11.81 SMV projection.

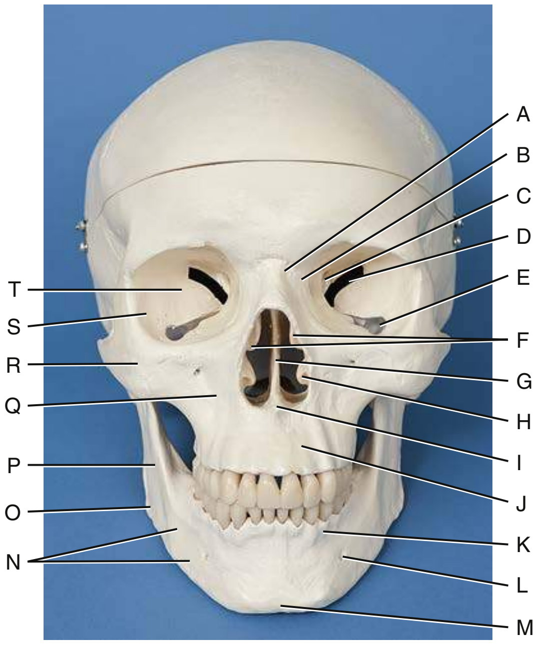


Fig. 11.82 Facial bones—frontal view.

Clinical Indications: Cranium

SKULL AND CRANIAL PATHOLOGY

Indications for skull and cranial radiographic procedures have markedly decreased because CT or MRI is increasingly available. However, smaller hospitals, clinics, and rural centers may still perform these procedures.

Skull fractures are disruptions in the continuity of **bones** of the skull.

NOTE: Although radiographic images of the skull provide excellent spatial resolution of bone, the presence or absence of a fracture is in no way an indication of underlying brain injury. Additional imaging procedures (i.e., CT or MRI) must be performed if brain tissue is to be fully assessed.

- **Linear fractures** are fractures of the skull that may appear as jagged or irregular lucent lines that lie at right angles to the axis of the bone.
- **Depressed fractures** are sometimes called *ping-pong fractures.* A fragment of bone that is separated and depressed into the cranial cavity can occur. A tangential view may be used to determine the degree of depression if CT is unavailable.
- **Basal skull fractures** are fractures through the dense inner structures of the temporal bone. These fractures are very difficult to visualize because of the complexity of the anatomy in this area. If bleeding occurs, radiographic images may reveal an air-fluid level in the sphenoid sinus if a horizontal ray is used for the lateral skull projection. CT is the modality of choice to differentiate between epidural and subdural hemorrhage.

Gunshot wounds can be visualized by radiographic images typically performed to localize bullets in gunshot victims in an antemortem or postmortem examination. The bullet is easily recognizable because of lead content.

Neoplasms are new and abnormal growths.

- **Metastases** are primary malignant neoplasms that spread to distant sites via blood and the lymphatic system. The skull is a common site of metastatic lesions, which may be characterized and visualized on the image as follows:
  - **Osteolytic** lesions are destructive lesions with irregular margins.
  - **Osteoblastic** lesions are proliferative bony lesions of increased density (brightness).
  - **Combination osteolytic and osteoblastic** lesions have a "moth-eaten" appearance of bone because of the mix of destructive and blastic lesions.

Multiple myeloma is a condition in which one or more bone tumors originate in the bone marrow. The skull is a commonly affected site.

Pituitary adenomas are investigated primarily by CT or MRI. Radiographic images may demonstrate enlargement of the sella turcica and erosion of the dorsum sellae, often as an incidental finding.

Paget disease (osteitis deformans) is a disease of unknown origin that begins as a stage of bony destruction followed by bony repair. It involves many bony sites, including the skull, pelvis, spine, and lower limbs. Radiographically, areas of lucency demonstrate the destructive stage, and a "cotton-wool" appearance with irregular areas of increased density (sclerosis) shows the reparative stage. Often these regions of bone become fragile and deformed. Nuclear medicine scans can demonstrate both regions of no (cold) and increased (hot) uptake of the radionuclide based on the stage of the disease.

TEMPORAL BONE PATHOLOGY

Common pathologic indications for temporal bone radiographic procedures include the following.

Acute mastoiditis *(mas″-toid-i′-tis)* is a bacterial infection of the mastoid process that can destroy the inner part of the mastoid process. It often results from middle ear infections. Bacteria in the middle ear can migrate to the mastoids. Mastoid air cells are replaced with a fluid-filled abscess, which can lead to progressive hearing loss. A CT scan demonstrates a fluid-filled abscess that replaces air-filled mastoid air cells.

Neoplasms are new and abnormal growths (tumors).

- **Acoustic neuroma** (vestibular schwannoma) refers to a benign, usually slow-growing tumor of the auditory nerve sheath that originates in the internal auditory canal. Symptoms include hearing loss, dizziness, and loss of balance. It typically is diagnosed with the use of CT or MRI (preferred modality), but it may be visualized on radiographic images in advanced cases with expansion and asymmetry of the affected internal acoustic canal.
- **Cholesteatoma** *(ko″-le-ste″-a-to′-ma)* is a benign, cystic mass or tumor that is most common in the middle ear. It occurs due to a congenital defect or chronic otitis media. It may destroy surrounding bone, which can lead to serious complications, including hearing loss. Surgery is required to remove a cholesteatoma.[4]

A **polyp** is a growth that arises from a mucous membrane and projects into a cavity (sinus). It may cause chronic sinusitis.

Otosclerosis *(o″-to-skle-ro′-sis)* is a hereditary disease that involves irregular ossification of the auditory ossicles of the middle ear. One common finding is fixation of the stapes to the oval window (eardrum). This leads to an impediment of sound transmission. It is the most common cause of hearing loss in adults without eardrum damage. Symptoms first become evident between the ages of 11 and 30 years. Otosclerosis is more common in women.[5] It is best demonstrated on CT imaging. See Table 11.2 for a summary of clinical indications related to the cranium.

11

TABLE 11.2 **SUMMARY OF CLINICAL INDICATIONS: CRANIUM**[a]

CONDITION OR DISEASE	MOST COMMON RADIOGRAPHIC EXAMINATION	POSSIBLE RADIOGRAPHIC APPEARANCE	EXPOSURE FACTOR ADJUSTMENT[b]
Fractures	CT, routine skull series		None
Linear	Routine skull series, CT	Jagged or irregular lucent line with sharp borders	None
Depressed	Tangential projection sometimes helpful, CT	Bone fragment depressed into cranial cavity	None
Basal	Horizontal beam lateral for potential air-fluid level in sphenoid sinuses and SMV projection if patient's condition allows, CT	Fracture visualized in dense inner structures of temporal bone	None
Gunshot wound	Routine skull series, CT	High-density object in cranial cavity if bullet has not exited; skull fracture also present because of entrance of projectile	None
Metastases	Routine skull series, bone scan	Depends on lesion type: destructive lesions with decreased density or osteoblastic lesions with increased density or a combination with a moth-eaten appearance	(+) or (−) depending on type of lesion and stage of pathology
Multiple myeloma	Routine skull series, MRI	Osteolytic (radiolucent) areas scattered throughout skull	(−) or none depending on severity
Pituitary adenoma	CT, MRI, collimated AP axial (Towne), and lateral	Enlarged, eroded aspects of sella turcica	(+) because of decreased field size
Paget disease (osteitis deformans)	Routine skull series, nuclear medicine scan	Depends on stage of disease; mixed areas of sclerotic (radiodense) and lytic (radiolucent); cotton-wool appearance; "cold" and "hot" regions of skull on nuclear medicine scan	(+) if in advanced sclerotic stage
Mastoiditis	CT, MRI	Increased densities (fluid-filled) replace mastoid air cells	None
Neoplasia			
Acoustic neuroma	MRI, CT	Widened internal auditory canal	None
Cholesteatoma	CT, MRI	Bone destruction involving middle ear	None
Polyp	Routine radiographic sinus views, CT, MRI	Increased density in affected sinus, typically with rounded borders	None
Otosclerosis	CT, MRI	Excessive bone formation involving middle and inner ear	None

[a]For the purpose of this table, a routine skull series is considered to be PA axial (Caldwell), AP axial (Towne), and lateral projections.
[b]Depends on stage or severity of disease or condition.

Clinical Indications: Facial Bones and Paranasal Sinuses

In addition to CT or MRI procedures, conventional radiographic examinations for the facial bones and paranasal sinuses still are commonly performed in smaller hospitals and clinics. For paranasal sinuses, radiographs are performed to demonstrate pathologies such as mucosal thickening, air-fluid levels, or erosion of bony margins of the sinuses.

Common clinical indications for various types of radiographic examinations for the facial bones and sinuses include the following:

Fracture is a break in the structure of a bone caused by a direct or indirect force. Examples of specific fractures involving the facial bones include the following:

- **Blowout fracture** is a fracture of the floor of the orbit caused by an object striking the eyes straight on (Fig. 11.83). As the floor of the orbit ruptures, the inferior rectus muscle is forced through the fracture into the maxillary sinus, causing entrapment and diplopia (perception of two images). A blowout fracture may also involve the medial walls of the orbit. CT is an effective imaging modality in demonstrating blowout fractures (Fig. 11.84).
- **Tripod fracture** is caused by a blow to the cheek, resulting in fracture of the zygoma in three places—orbital process, maxillary process, and arch. The result is a "free-floating" zygomatic bone, or a tripod fracture (Fig. 11.85).
- **Le Fort fractures** are severe bilateral horizontal fractures of the maxillae that may result in an unstable detached fragment.
- **Contrecoup fracture** is a fracture to one side of a structure that is caused by an impact on the opposite side. For example, a blow to one side of the mandible results in a fracture on the opposite side.

Foreign body of the eye refers to metal or other types of fragments in the eye, a relatively common industrial mishap. Radiographic images are taken to detect the presence of a metallic foreign object but are limited in their ability to demonstrate damage to tissues caused by these objects.

The patient interview before an MRI procedure includes questions regarding the history of a foreign object in the eye. Because the magnetic field causes the metal, ferrous fragments to move, injury occurs to the soft tissues (even blindness may occur if the optic nerve is damaged). Radiographic images may be obtained before MRI to confirm the presence of a foreign object.

Neoplasm describes a new and abnormal growth (tumor) that may occur in the skeletal structures of the face.

Osteomyelitis is a localized infection of bone or bone marrow. This infection may be caused by bacteria from a penetrating trauma or postoperative or fracture complications. It also may be spread by blood from a distant site.

Sinusitis *(si-nu-si'-tis)* is an infection of the sinus mucosa that may be acute or chronic. The patient complains of headache, pain, swelling over the affected sinus, and possibly a low-grade fever.

Secondary osteomyelitis, an infection of the bone and marrow secondary to sinusitis, results in erosion of the bony margins of the sinus.

TMJ syndrome describes a set of symptoms, which may include pain and clicking, that indicate dysfunction of the TMJ. This condition may be caused by malocclusion, stress, muscle spasm, or inflammation.

See Table 11.3 for a summary of clinical indications related to facial bones and sinuses.

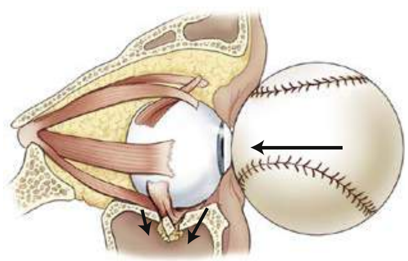


Fig. 11.83 Blowout fracture.

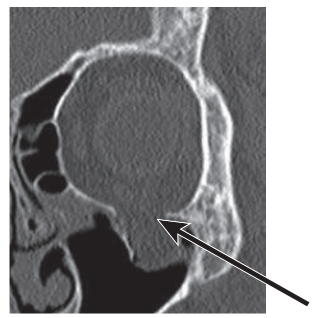


Fig. 11.84 Blowout fracture of the orbit.

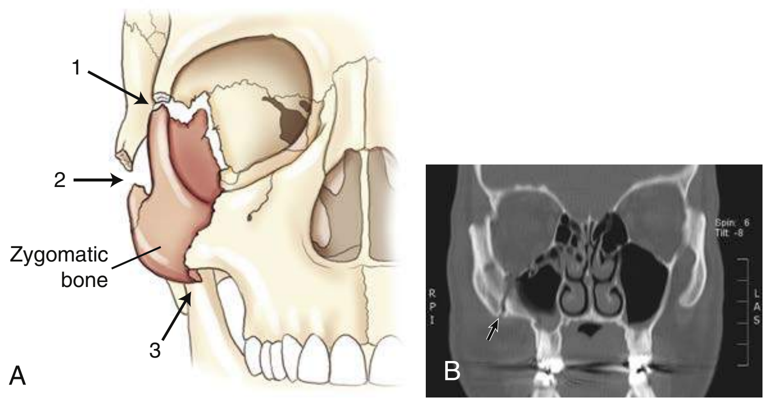


Fig. 11.85 A, Tripod fracture. B, Coronal CT of tripod fracture involving right lateral orbital wall and zygoma.

TABLE 11.3 SUMMARY OF CLINICAL INDICATIONS: FACIAL BONES AND SINUSES

CONDITION OR DISEASE	MOST COMMON RADIOGRAPHIC EXAMINATION	POSSIBLE RADIOGRAPHIC APPEARANCE	EXPOSURE FACTOR ADJUSTMENT
Fractures	Routine radiographic projections of affected area, CT	Disruption of bony cortex	None
Foreign body of eye	Routine facial bone (orbits) projections, including modified parietoacanthial projection	Increased density if foreign body is metallic	None
Neoplasms	Routine radiographic projections of affected area, CT/MRI	Possible increase or decrease in density, depending on lesion type	None
Osteomyelitis	Nuclear medicine bone scan, routine radiographic projections of affected area	Soft tissue swelling; loss of cortical margins	None
Sinusitis	Routine radiographic sinus projections, CT, MRI	Sinus mucosal thickening, air-fluid levels, opacified sinus	None
Secondary osteomyelitis	Routine radiographic sinus projections, CT	Erosion of bony margins of sinus	None
TMJ syndrome	Axiolateral projections of TMJ (open- and closed-mouth position), CT/MRI	Abnormal relationship or range of motion between condyle and TM fossa	None

RADIOGRAPHIC POSITIONING CONSIDERATIONS: CRANIUM

Traditionally, the cranium has been one of the most difficult and challenging parts of the body to image. A good understanding of the anatomy and relationships of bones and structures of the skull as described in this chapter is essential before a study of radiographic positioning of the cranium is begun. Conventional radiography of certain parts of the cranium, such as the denser temporal bone regions, is less common today because of advances in other imaging modalities such as CT and MRI. However, these imaging modalities may be unavailable in remote areas, and every technologist should be able to perform conventional skull radiography as described in this chapter.

Skull Morphology (Classifications by Shape and Size)

MESOCEPHALIC SKULL

The shape of the average head is termed **mesocephalic** *(mes″-o-se-fal′-ik).* The average caliper measurements of the adult skull are 6 inches (15 cm) between the parietal eminences (lateral), 7½ inches (19 cm) from frontal eminence to external occipital protuberance (anteroposterior [AP] or PA), and 9 inches (23 cm) from vertex to beneath the chin (SMV projection) (Fig. 11.86). Although most adults have a skull of average size and shape, exceptions to this rule exist.

A general rule for describing skull type is to compare the width of the skull at the parietal eminence with the length measured from the frontal eminence to the external occipital protuberance. For an average mesocephalic skull, the **width is 75% to 80% of the length.**[6]

BRACHYCEPHALIC AND DOLICHOCEPHALIC SKULLS

Variations of the average-shaped or mesocephalic skull include **brachycephalic** *(brak″-e-se-fal′-ik)* and **dolichocephalic** *(dol″-i-ko-se-fal′-ik)* designations. A short, broad head is termed *brachycephalic,* and a long, narrow head is called *dolichocephalic.*

The width of the brachycephalic type is **80% or greater** than the length. The width of the long, narrow dolichocephalic type is **less than 75%** of the length.[6]

A second variation is the **angle difference** between the petrous pyramids and the MSP. In the average-shaped, mesocephalic head, the petrous pyramids form an angle of **47°**. In the brachycephalic skull, the angle is **greater than 47°** (approximately 54°), and in the dolichocephalic skull, the angle is **less than 47°** (approximately 40°) (Fig. 11.87).

POSITIONING CONSIDERATIONS RELATED TO SKULL MORPHOLOGY

The positioning descriptions, including CR angles and head rotations, as described in this text are based on the average-shaped mesocephalic skull. For example, the axiolateral oblique projection (Law method) for TMJs requires 15° of head rotation. A long, narrow, dolichocephalic head requires slightly more than 15° of rotation, and a short, broad, brachycephalic type requires less than 15°.

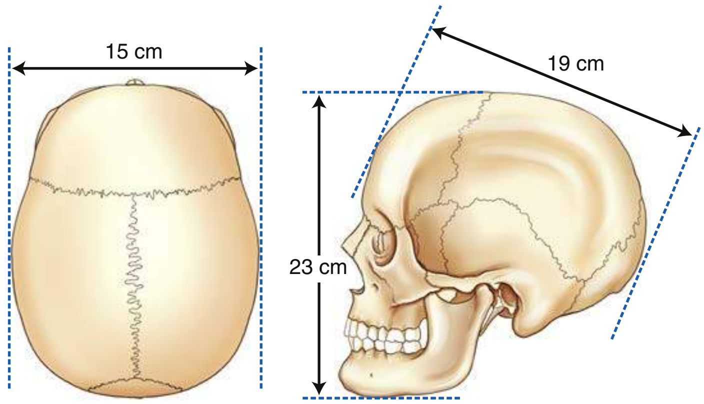


Fig. 11.86 Average skull (mesocephalic).

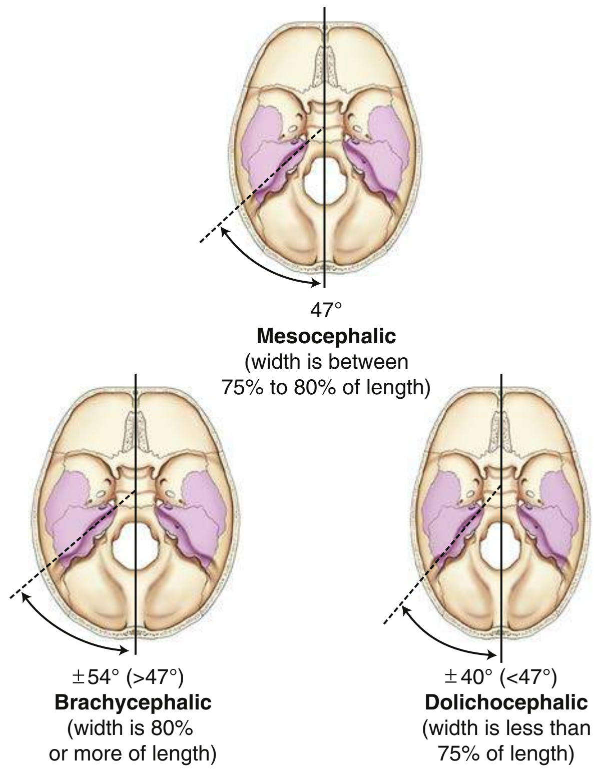


Fig. 11.87 Variable skull morphologies.

Cranial Topography (Surface Landmarks)

Certain surface landmarks and localizing lines must be used for accurate positioning of the cranium. Each of the following topographic structures can be seen or palpated.

BODY PLANES (FIG. 11.88)

The **midsagittal,** or **median, plane (MSP)** divides the body into left and right halves. This plane is important for accurate positioning of the cranium because for every AP and PA or lateral projection, the MSP is perpendicular to or parallel to the plane of the IR.

ANTERIOR AND LATERAL VIEW LANDMARKS (FIGS. 11.89 AND 11.90)

The **superciliary ridge (arch)** is the ridge or arch of bone that extends across the forehead directly above each eye. Slightly above this ridge is a slight groove or depression, the **SOG** (see Fig. 11.88).

NOTE: The SOG is important because it corresponds to the highest level of the facial bone mass, which is also the level of the **floor of the anterior fossa** of the cranial vault.

The **glabella** *(glah-bel′-ah)* is the smooth, slightly raised triangular area between and slightly superior to the eyebrows and above the bridge of the nose.

The **nasion** *(na′-ze-on)* is located at the junction of the two nasal bones and the frontal bone.

The **acanthion** *(ah-kan′-the-on)* ("little thorn") is the midline point at the junction of the upper lip and the nasal septum. This is the point where the nose and the upper lip meet.

The **angle,** or **gonion** *(go′-ne-on)*, refers to the lower posterior angle on each side of the jaw or mandible.

A flat triangular area projects forward as the **chin,** or **mentum,** in humans. Imagine the base of a triangle formed between the two mental protuberances and the two sides extending between the two innermost medial incisors to form the apex. The midpoint of this triangular area of the chin as it appears from the front is termed the **mental point.**

Ear

Parts of the ear that may be used as positioning landmarks are the **auricle,** or **pinna** (external portion of ear), the large flap of ear made of cartilage, and the **tragus,** the small cartilaginous flap that covers the opening of the ear. **TEA** refers to the superior attachment of the auricle, or the part where the side frames of eyeglasses rest. This is an important landmark because it corresponds to the highest level of the **petrous ridge** on each side.

Eye

The junctions of the upper and lower eyelids are termed **canthi** *(kan′-thi)*. The **inner canthus** *(kan′-thus)* is where the eyelids meet near the nose; the more lateral junction of the eyelids is termed the **outer canthus.**

The superior rim of the bony orbit of the eye is the **SOM**, and the inferior rim is the **IOM**. Another landmark is the **midlateral orbital margin,** which is the portion of the lateral rim that is near the outer canthus of the eye. These three landmarks contribute to the base of the orbit.

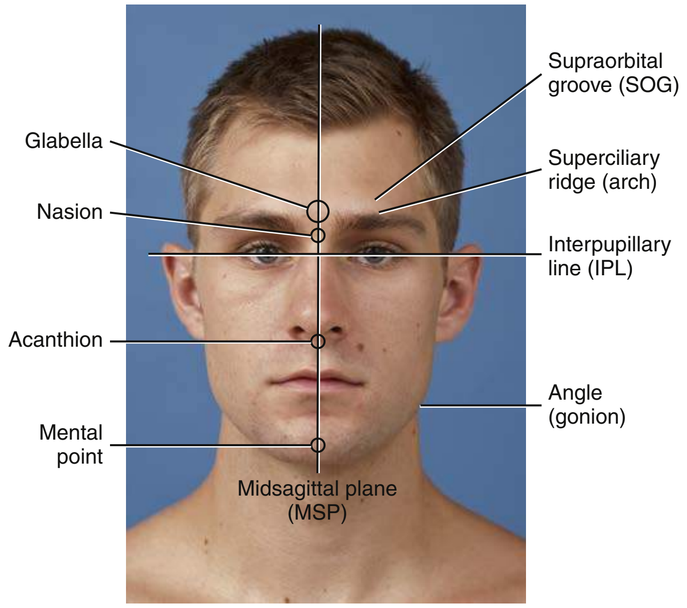


Fig. 11.88 Body planes and landmarks.

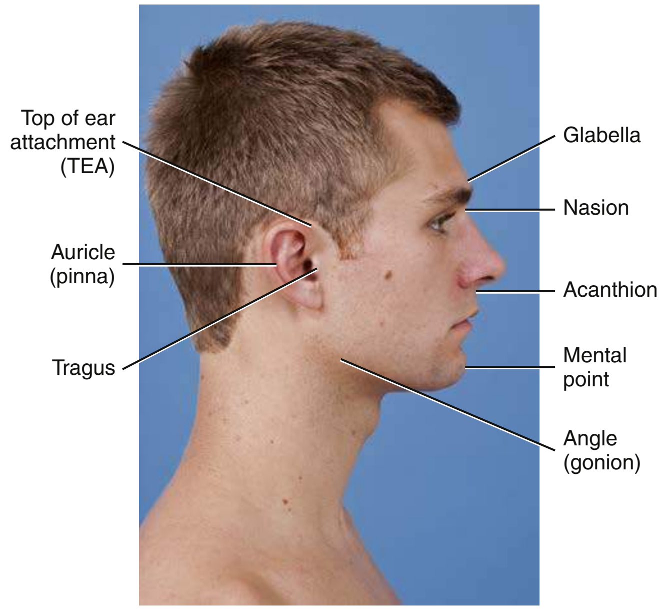


Fig. 11.89 Surface landmarks.

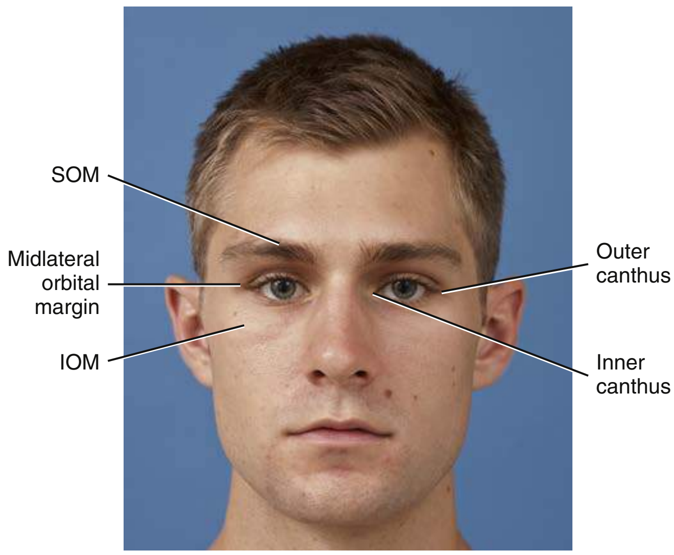


Fig. 11.90 Orbital landmarks.

CRANIAL POSITIONING LINES (FIG. 11.91)

Certain positioning lines are important in cranial radiography. These lines are formed by connecting certain facial landmarks to the midpoint of the **EAM.** The EAM is the opening of the external ear canal. The center point of this opening is called the **auricular point.**

The most superior of these positioning lines is the **glabellomeatal line (GML)**, which is not as precise as the other lines because the glabella is an area and not a specific point. The GML refers to a line between the glabella and the EAM.

The **orbitomeatal line (OML)** is a frequently used positioning line that is located between the outer canthus (midlateral orbital margin) and EAM.

The **infraorbitomeatal line (IOML)** is formed by connecting the IOM to the EAM. Two older terms identify this same line as **Reid's base line** or the **anthropologic base line.** An average difference of **7° to 8°** exists between the angles of the OML and IOML. There is also an approximate 7° to 8° average angle difference between the OML and GML. Knowing the angle differences between these three lines is helpful in making positioning adjustments for specific projections of the cranium and facial bones.

The **acanthiomeatal line (AML)** and **mentomeatal line (MML)** are important in radiography of the facial bones. Connecting the acanthion (for the AML) or the mental point (for the MML) to the EAM forms the AML or MML.

A line from the junction of the lips to the EAM, called the **lips-meatal line (LML),** is a positioning line used in this textbook to position for a specific projection of the facial bones called a modified parietoacanthial (modified Waters) projection.

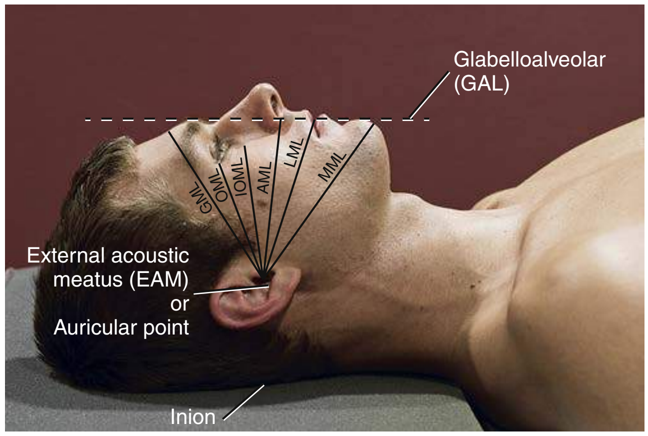
The **glabelloalveolar line (GAL)** connects the glabella to a point at the anterior aspect of the alveolar process of the maxilla. This line is used for positioning a tangential projection for the nasal bones and the lateral position of the cranium.

The **interpupillary,** or **interorbital, line (IPL)** is a line that connects the pupils or the outer canthi of the patient's eyes. When the head is placed in a **true lateral** position, the PL must be exactly perpendicular to the plane of IR (see Fig. 11.88).

The **inion** *(in′-e-on)* is the most prominent point of the external occipital protuberance. It corresponds to the highest "nuchal" line of the occipital bone and allows insertion of the occipitofrontalis muscle. A posterior extension of the IOML approximates the location of the inion.

SKULL POSITIONING AIDS

Two simple devices may be used to ensure accurate placement of a cranial positioning line. A straightedge can be used to illustrate that a cranial line is perpendicular (Fig. 11.92); an angle ruler can also be used and is often the preferred tool. The angle ruler or goniometer (Fig. 11.93) has the advantage of allowing the technologist to determine how many degrees the cranial line is from perpendicular or horizontal so that the patient position or CR angle may be accurately adjusted.



- Glabellomeatal line (GML)
- Acanthiomeatal line (AML)
- Orbitomeatal line (OML)
- Lips-meatal line (LML)
- Infraorbitomeatal line (IOML) (Reid's base line)
- Mentomeatal line (MML)

Fig. 11.91 Positioning lines.

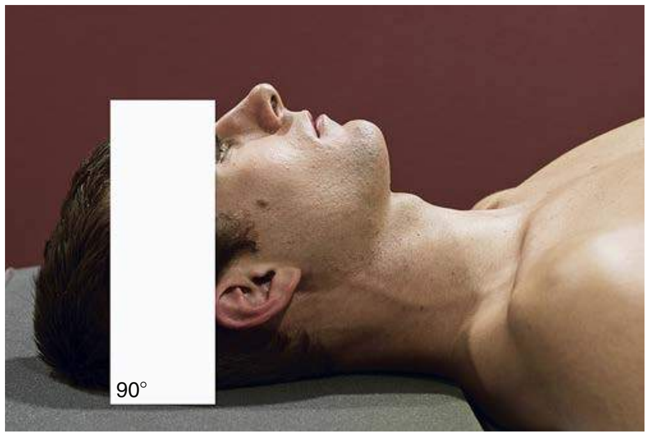


Fig. 11.92 Positioning aid—90° straightedge.

Fig. 11.93 Positioning aid—angle ruler demonstrating degrees of IOML to IR.

Positioning Considerations

ERECT VERSUS RECUMBENT

Projections of the cranium may be taken with the patient in the recumbent or erect position, depending on the patient's condition (Figs. 11.94 and 11.95). Images can be obtained in the erect position with the use of a standard x-ray table in the vertical position or an upright imaging device. The erect position allows the patient to be positioned quickly and easily and permits the use of a horizontal x-ray beam. A horizontal beam is necessary to visualize any existing air-fluid levels within the cranial or sinus cavities.

When positioning for facial bone projections, an **erect position is preferred** if the patient's condition allows for it. This positioning can be done with an erect table or an upright imaging device (see Fig. 11.94). Moving the patient's entire body in the erect position (to adjust the various planes and positioning lines) often is easier for accurate skull positioning; this is especially true with hypersthenic patients. In addition, air-fluid levels in the sinuses or other cranial cavities may indicate certain pathologic conditions that are visible only in the erect position or with the use of horizontal beam radiography.

PATIENT COMFORT

Patient motion almost always results in an unsatisfactory image. During cranial, facial bone, and sinus radiography, the patient's head must be placed in precise positions and held motionless long enough for an exposure to be obtained. Always remember a patient is attached to the skull that is being manipulated. Every effort should be made to make the patient as comfortable as possible and positioning aids such as sponges, sandbags, and pillows should be used if needed.

Except in cases of severe trauma, respiration should be suspended during the exposure to help prevent blurring of the image caused by breathing movements of the thorax. Suspending respiration is especially important when the patient is in a prone position.

Hygiene

Cranial, facial bone and sinus radiography may require the patient's face to be in direct contact with the technologist's hands and the table/upright imaging surface. In the case of infectious diseases or skin conditions, the technologist must wear gloves while positioning. It is important that proper handwashing techniques and surface disinfectants be used before and after the examination.

EXPOSURE FACTORS

Principal exposure factors for radiography of the cranium and facial bones include the following:

- Medium kVp (75 to 95)
- Small focal spot for improved sharpness (if equipment allows)
- Short exposure time

Paranasal Sinuses

EXPOSURE FACTORS

- A medium kVp range of 75 to 90 is commonly used to provide sufficient contrast of the air-filled paranasal sinuses.
- Optimum exposure as controlled by the mAs is especially important for sinus radiography to visualize pathology within the sinus cavities.
- A small focal spot should be used for improved sharpness.
- As with cranial and facial bone imaging, shielding of radiosensitive organs is recommended.
- Close collimation and elimination of unnecessary repeats are the best measures for reducing radiation dose in sinus radiography.

SOURCE–IMAGE RECEPTOR DISTANCE

The minimum source–image receptor distance (SID) with the image receptor (IR) in the table or upright imaging device is 40 inches (100 cm).

RADIATION PROTECTION

The best techniques for minimizing radiation exposure to the patient in cranial, facial bone, and paranasal sinus radiography are to (1) use **close collimation;** (2) immobilize the head when necessary, **minimizing repeats;** and (3) **center properly.**

PATIENT SHIELDING

Along with close collimation, shielding of radiosensitive organs is recommended unless it interferes with the radiographic study.

Fig. 11.94 Erect—upright imaging device.

Fig. 11.95 Recumbent—table-imaging device.

CAUSES OF POSITIONING ERRORS

When positioning a patient's head, look at various facial features and palpate anatomic landmarks to place the appropriate body plane precisely in relation to the plane of the IR. Although the human body is expected to be symmetric bilaterally (i.e., the right half is anticipated to be identical to the left half), this is not always true. The ears, nose, and jaw are often asymmetric. The nose frequently deviates to one side of the MSP, and the ears are not necessarily in the same place or of the same size on each side.

The lower jaw or mandible is also often asymmetric. Bony parts, such as the mastoid tips and the orbital margins, are safer landmarks to use. Although the patient's eyes are often used as landmarks during positioning, the nose, which may not be straight, should not be used. As a rule of thumb, use the relationship of the eye to the EAM when deciding on the use of the OML or IOML for certain positions of the skull.

FIVE COMMON POSITIONING ERRORS

Five potential positioning errors related to cranial, facial bone, and paranasal sinus positioning are as follows:

1. Rotation
2. Tilt
3. Excessive neck flexion
4. Excessive neck extension
5. Incorrect CR angle

Rotation and **tilt** are two very common positioning errors, as demonstrated by the drawings on the right. Rotation of the skull almost always results in a retake; therefore, the body planes should be correctly aligned (e.g., MSP is parallel to the IR in a lateral position) (Fig. 11.96).

Tilt is a tipping or slanting of the MSP laterally, even though rotation may not be present (Fig. 11.97).

Incorrect flexion or **extension** of the cervical spine, along with an **incorrect CR angle,** must be avoided (Fig. 11.98).

Fig. 11.96 Rotation—MSP is rotated, not parallel to tabletop and IR.

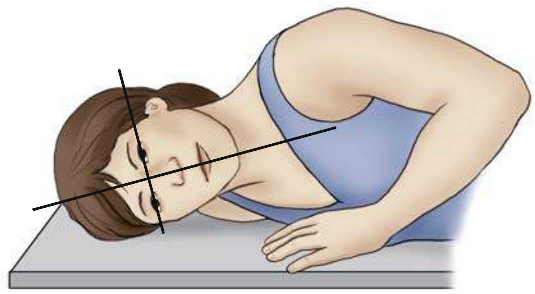


Fig. 11.97 Tilt—MSP is tipped or slanted, not parallel to tabletop or IR.

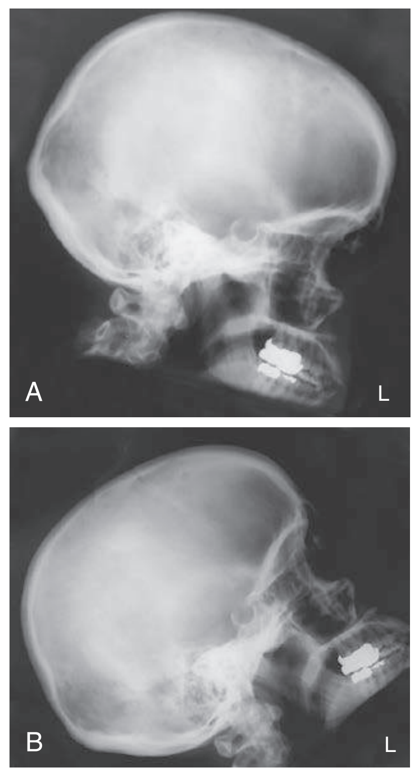


Fig. 11.98 A, Excessive flexion. B, Excessive extension.

RADIOGRAPHIC POSITIONING CONSIDERATIONS: FACIAL BONES AND PARANASAL SINUSES

Special Projections and Anatomic Relationships

Unobstructed radiographic images of various aspects of the facial bones and paranasal sinuses may be difficult to obtain because of the overall shape and structures of the skull. For example, dense internal bony structures of the skull superimpose the delicate facial bones on a routine AP or PA projection. Therefore, very specific CR angles and head positions are required, as described and illustrated subsequently.

PA Skull Projection

The PA skull projection in Fig. 11.99 was obtained with no tube angulation and with the OML (dotted line in Fig. 11.100) perpendicular to the plane of the IR. The CR is parallel to the OML. This position causes the **petrous pyramids to be projected directly into the orbits.** Drawn on both images (Figs. 11.99 and 11.100) is a line through the roof of the orbits and through the petrous ridges. With the orbits superimposed by the petrous pyramids, very little facial bone detail can be demonstrated radiographically. With the head in this position, the PA projection with a perpendicular CR has limited value for visualizing the facial bones.

Parietoacanthial (Waters Method) Projection

To visualize the facial bone mass with conventional radiography, the petrous pyramids must be removed from the facial bone area of interest. This can be done either by tube angulation or by extension of the neck. The radiographs in Figs. 11.101 and 11.102 demonstrate the result. The neck is extended by raising the chin so that the **petrous pyramids are projected just below the maxillary sinuses.** The CR is parallel to the MML. The radiograph on the right (Fig. 11.102; Waters method), if done correctly as described later in this chapter, demonstrates the petrous ridges (see arrows) projected below the maxillae and the maxillary sinuses. Except for the mandible, the **facial bones are now projected superior to** the dense petrous pyramids and are **not superimposed by** them. As stated previously, erect projections are preferred for facial bones and sinuses to demonstrate any possible air-fluid levels (Fig. 11.103). In the case of possible cervical spine injury, cranial and facial bone studies must be performed recumbent to prevent further injury (Fig. 11.104).

Fig. 11.99 Skull—PA projection.

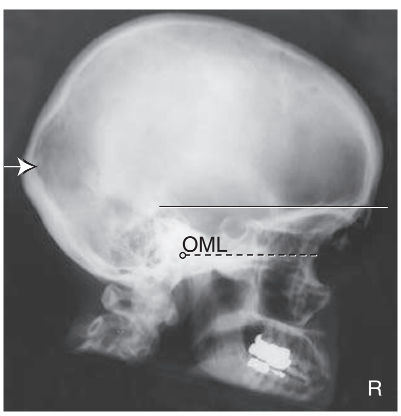


Fig. 11.100 Lateral skull for comparison of bony relationships—CR parallel to OML.

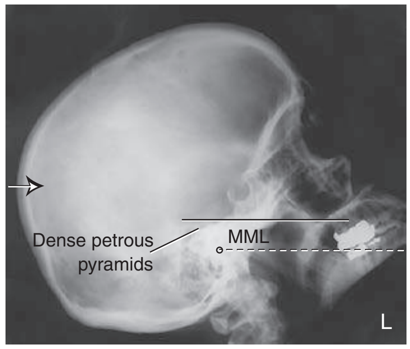


Fig. 11.101 Lateral skull for comparison of bony relationships—CR parallel to MML.

Fig. 11.102 Facial bones—parietoacanthial (Waters method) projection.

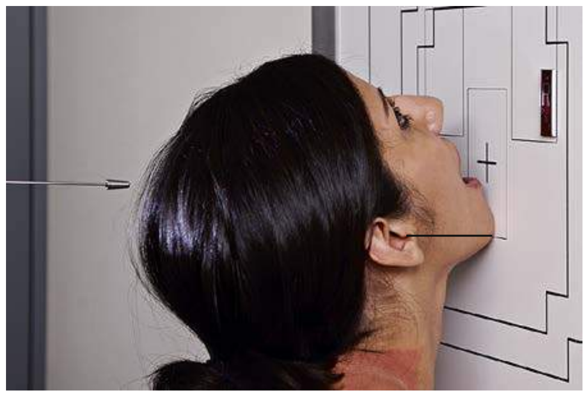


Fig. 11.103 Parietoacanthial (Waters method) erect—upright imaging device.

Fig. 11.104 AP, supine—trauma patient.

Special Patient Considerations

PEDIATRIC APPLICATIONS

Communication

A clear explanation of the procedure is required to obtain the trust and cooperation of the patient and guardian. Distraction techniques using toys, stuffed animals, and other items are also effective in maintaining patient cooperation.

Immobilization

Pediatric patients (depending on age and condition) are often unable to maintain the required positions. Use of immobilization devices to support the patient is recommended to reduce the need for the patient to be held, reducing radiation exposure. (Chapter 16 provides a detailed description of such devices.) If it is necessary for the guardian to hold the patient, the technologist must provide a lead apron or gloves or both. If the guardian is female, the technologist must ensure that no possibility of pregnancy exists.

Exposure Factors

Exposure factors vary because of various patient sizes and pathologies. Use of short exposure times (associated with the use of high mA) is recommended to reduce the risk of patient motion.

GERIATRIC APPLICATIONS

Communication and Comfort

Sensory loss (e.g., poor eyesight, hearing) associated with aging may result in the need for additional assistance, time, and patience in obtaining the required positions for cranial, facial bone, and paranasal sinus radiography in geriatric patients.

If the examination is performed with the patient in the recumbent position, decreased position awareness may cause the patient to fear falling off the radiography table. A radiolucent mattress or pad placed on the examination table provides comfort, and extra blankets may be required for warmth. Reassurance and attention from the technologist help the patient feel secure and comfortable.

If the patient is able, attaining the required positions in the erect position (sitting) at an upright imaging device may be more comfortable (Fig. 11.105), especially if he or she has an increased kyphosis. Lateral images obtained with a horizontal ray often are indicated for elderly patients who have limited movement.

Exposure Factors

Because of the high incidence of osteoporosis in geriatric patients, the kVp may require a 15% decrease.

Older patients may have tremors or signs of unsteadiness; use of short exposure times (associated with the use of high mA) is recommended to reduce the risk for motion.

BARIATRIC PATIENT CONSIDERATIONS

Positioning for cranial and facial bone projections on the bariatric patient is far more comfortable and easier to assume when performed erect. Unless the patient's health and safety prevents it, perform skull and facial bone positions erect.

If you are unable to place the OML perpendicular to the plane of the IR because of the thickness of the shoulders and restricted flexion of the neck, most cranial projections allow the technologist to use the IOML to perform the relatively same position. Please remember there is approximately 7° to 8° difference between the OML and IOML. In the case of the AP axial projection for the cranium, the technologist would increase the CR angle from 30° to 37°.

Alternative Modalities

COMPUTED TOMOGRAPHY (CT)

CT is the most commonly performed neuroimaging procedure. CT provides sectional images of the brain and bones of the cranium in axial, sagittal, or coronal planes, whereas analog and digital images provide a two-dimensional image of the bony skull only.

Because injury and pathology of the head often involve the brain and associated soft tissues, CT is a vital tool in the full evaluation of the patient. It can distinguish between blood clots, white and gray matter, cerebrospinal fluid, cerebral edema, and neoplasms.

CT provides sectional images of the facial bones, orbits, mandible, and TMJs in axial, sagittal, or coronal planes. CT assists in the full evaluation of these structures because both skeletal detail and associated soft tissues may be visualized with CT.

CT sinus studies may be performed in the prone position, which allows for coronal scans to be created. Coronal CT scans demonstrate any air-fluid levels that are present. CT also allows visualization of the soft tissue planes of the sinuses and evaluation of related bony structures. If the patient cannot be examined prone, the study can be performed supine. In the supine position, axial images are obtained in which sagittal and coronal reconstructed images can be created (Fig. 11.106). These images assist the radiologist in determining the presence of any pathology.

Intravenous contrast medium is not given for most CT sinus examinations. The most common clinical indications for a CT sinus study are sinusitis and possible masses within the sinuses. Three-dimensional reconstruction of CT scans often is useful when facial reconstructive surgery is required.

Fig. 11.105 Modified PA projection—**horizontal CR**, OML tilted 15° from perpendicular (**routine** projection for sinuses).

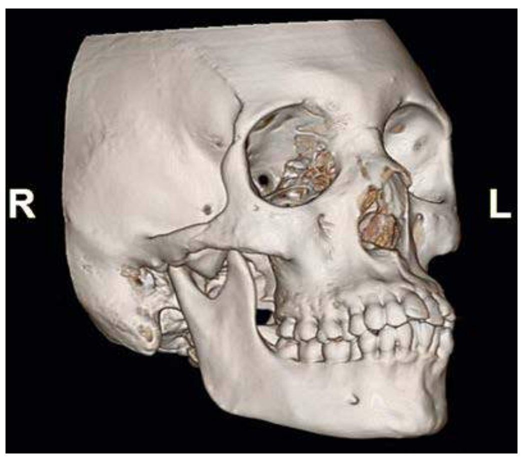


Fig. 11.106 3D CT reconstruction of skull and facial bones.

MAGNETIC RESONANCE IMAGING (MRI)

MRI also provides images of the brain in axial, sagittal, and coronal planes. MRI provides increased sensitivity in detecting differences between normal and abnormal tissues in the brain and associated soft tissues. MRI has limited usefulness in evaluation of bone; however, it is superior to other methods in evaluation of soft tissues.

The magnetic fields used in MRI are thought to be harmless, which means that the patient is spared exposure to ionizing radiation. MRI is useful for evaluating TMJ syndrome to diagnose possible damage to the articular disk of the glenoid cavity of the TM fossa.

SONOGRAPHY

Sonography of the brain of the neonate (through the fontanels) is an integral part of treatment in the intensive care unit. It allows for rapid evaluation and screening of premature infants for intracranial hemorrhage. It is preferred over CT and MRI for this purpose because it is highly portable and less expensive, requires no sedation of the patient, and provides no ionizing radiation.

Sonography can also be valuable in the investigation and follow-up of hydrocephalus. Cranial sutures also may be evaluated, assisting in the diagnosis of premature suture closure (craniosynostosis).

Research is ongoing regarding the use of sonography as a screening tool for maxillary sinusitis. Because it does not involve ionizing radiation exposure, this method would be advantageous for pediatric and pregnant patients.

NUCLEAR MEDICINE (NM)

NM provides a sensitive screening procedure (radionuclide bone scan) for detection of skeletal metastases, of which the cranium is a common site. A bone scan is ordered frequently for patients who are at risk or symptomatic for metastases. Any focal abnormality on the bone scan is investigated radiographically to examine the pathology further. Patients with a history of multiple myeloma are often exceptions to this protocol.

Brain tissue also may be studied with the use of nuclear medicine technology. New radiopharmaceuticals allow perfusion studies of the brain to be performed, typically on patients with Alzheimer disease, seizure disorders, or dementia. Tumor response to treatment also may be assessed with this modality.

Radionuclide bone scan is a sensitive diagnostic procedure for detection of osteomyelitis and occult fractures that may not be demonstrated on radiographic images.

Routine and Special Projections

Projections or positions for the cranium (skull series), facial bones, and paranasal sinuses are demonstrated and described on the following pages as suggested standard routine and special departmental procedures.

AP AXIAL PROJECTION—SKULL SERIES

TOWNE METHOD

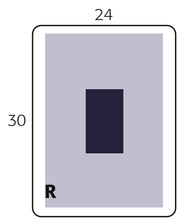
Clinical Indications

- Skull fractures (medial and lateral displacement), neoplastic processes, and Paget disease

Skull Series
ROUTINE
- AP axial (Towne method)
- Lateral
- PA axial 15° (Caldwell method) or PA axial 25° to 30°
- PA

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–90



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metal, plastic, or other removable objects from the patient's head. Take radiograph with the patient in the erect or supine position.

Part Position

- Depress chin, bringing **OML perpendicular** to IR. For patients unable to flex the neck to this extent, align **IOML** perpendicular to IR. Add radiolucent support under the head if needed (see NOTE).
- Align MSP to CR and to midline of the grid or the table/imaging device surface.
- Ensure that no head rotation or tilt exists.
- Ensure that the vertex of the skull is within collimation field.

CR

- Angle CR 30° caudad to OML or 37° caudad to IOM (Fig. 11.107) (see NOTE).
- Center at MSP 2½ inches (6.5 cm) above the glabella to pass through the foramen magnum at the level of the base of the occiput.
- Center IR to projected CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: If patient is unable to depress the chin sufficiently to bring **OML** perpendicular to IR even with a small sponge under the head, **IOML** can be placed perpendicular instead and the CR angle increased to **37°** caudad. This maintains the **30° angle between OML and CR** and demonstrates the same anatomic relationships. (A 7° to 8° difference exists between OML and IOML.)

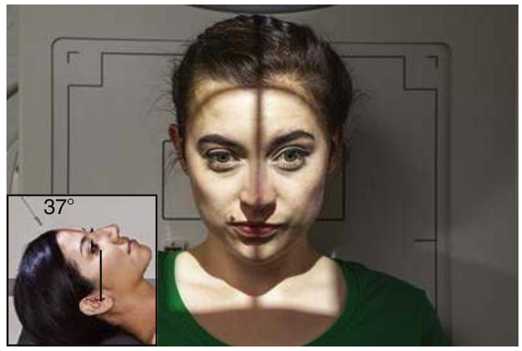


Fig. 11.107 Erect and supine (*inset*)—AP axial.

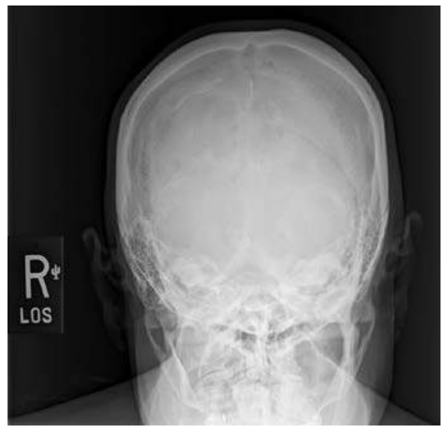


Fig. 11.108 AP axial.

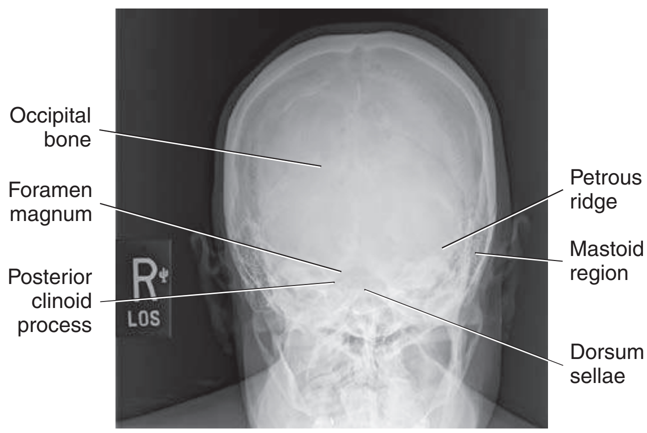


Fig. 11.109 AP axial.

Evaluation Criteria

Anatomy Demonstrated: • Occipital bone, petrous pyramids, and foramen magnum are demonstrated with the dorsum sellae and posterior clinoid processes visualized in the shadow of the foramen magnum (Figs. 11.108 and 11.109).

Position: • Petrous ridges should be symmetric, indicating **no rotation** (petrous ridge will appear narrowed in the direction of rotation). • Dorsum sellae and posterior clinoid processes visualized in the foramen magnum indicate **correct CR angle and proper neck flexion/extension.** • **Underangulation** of CR or insufficient flexion of neck projects the dorsum sellae superior to **the foramen magnum. Overangulation** of CR or excessive neck flexion superimposes the **posterior arch of C1 over the dorsum sellae** within the foramen magnum and produces foreshortening of the dorsum sellae. • Shifting of the anterior or posterior clinoid processes laterally within the foramen magnum indicates tilt.[7] • Collimation to area of interest.

Exposure: • Density (brightness) and contrast are sufficient to visualize occipital bone and sellar structures within foramen magnum. • Sharp bony margins indicate **no motion.**

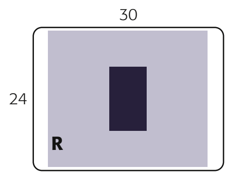
LATERAL POSITION: RIGHT OR LEFT LATERAL—SKULL SERIES

Skull Series
ROUTINE
- AP axial (Towne method)
- Lateral
- PA axial 15° (Caldwell method) or PA axial 25° to 30°
- PA

Clinical Indications

- Skull fractures, neoplastic processes, and Paget's disease

Trauma Routine A horizontal beam projection is required to obtain a lateral perspective for trauma patients. This may demonstrate air-fluid levels in the sphenoid sinus—a sign of a basal skull fracture if intracranial bleeding occurs. See Chapter 15 for details on trauma skull projections.



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), landscape
- Grid
- kVp range: 70–85

Fig. 11.110 Lateral skull—erect and recumbent (*inset*).

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metal, plastic, or other removable objects from patient's head. Take radiograph with patient in the erect or recumbent semiprone position.

Part Position

- Place the head in a **true lateral position,** with the side of interest closest to IR and the patient's body in a semiprone or erect position as needed for comfort. Align **MSP parallel** to IR, ensuring **no rotation or tilt.**
- Align **IPL perpendicular** to IR, ensuring no tilt of head (Fig. 11.110) (see NOTE).
- Adjust neck flexion to align **IOML perpendicular** to front edge of IR. (GAL is parallel to front edge of IR.)

CR

- Align CR **perpendicular** to IR.
- Center to a point 2 inches (5 cm) superior to EAM or halfway between the glabella and the inion for other types of skull morphologies.
- Center IR to CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: For patients in the recumbent position, a radiolucent support placed under the chin helps in maintaining a true lateral position. A patient with a broad chest may require a radiolucent sponge under the entire head to prevent tilt, and a thin patient may require support under the upper thorax.

Fig. 11.111 Lateral.

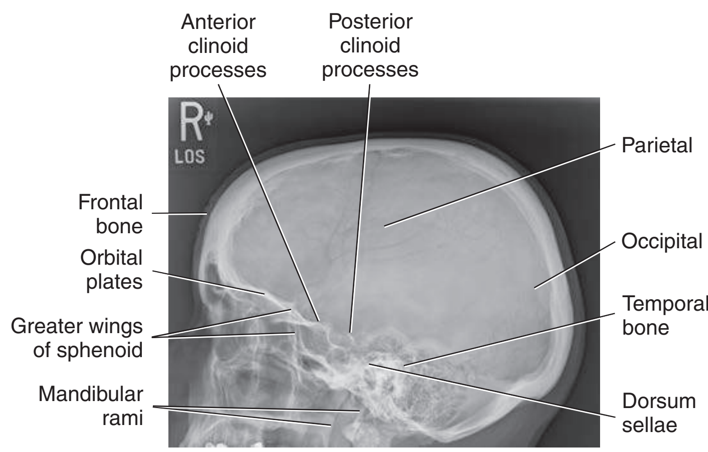


Fig. 11.112 Lateral.

Evaluation Criteria

Anatomy Demonstrated: • Entire cranium visualized and superimposed parietal bones of cranium. • The entire sella turcica, including anterior and posterior clinoid processes and dorsum sellae, is also demonstrated. • The sella turcica and clivus are demonstrated in profile (Figs. 11.111 and 11.112).

Position: • **No rotation or tilt** of the cranium is evident. • **Rotation** is evident by **anterior and posterior separation** of symmetric vertical bilateral structures such as the mandibular rami, and greater wings of the sphenoid. • **Tilt** is evident by **superior and inferior separation** of symmetric horizontal structures such as the orbital plates **and greater wings of sphenoid bone.** • Collimation to area of interest.

Exposure: • Density (brightness) and contrast are sufficient to visualize bony detail of bony structures and surrounding skull. • Sharp bony margins indicate **no motion.**

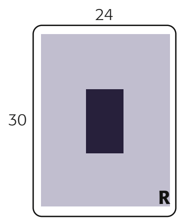
PA AXIAL PROJECTION—SKULL SERIES

15° CR (CALDWELL METHOD) OR 25° TO 30° CR

Clinical Indications
- Skull fractures, neoplastic processes, and Paget disease

Skull Series
ROUTINE
- AP axial (Towne method)
- Lateral
- PA axial 15° (Caldwell method) or PA axial 25° to 30°
- PA

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from the patient's head and neck. Take radiograph with patient in the erect or prone position.

Part Position
- Rest patient's nose and forehead against table/imaging device surface.
- Flex neck as needed to align **OML perpendicular** to IR.
- Align the **MSP perpendicular** to the IR to **prevent rotation and/or tilt.**
- Center IR to CR.

CR
- Angle CR 15° caudad, and center to exit at nasion (Fig. 11.113).
- Alternative with CR 25° to 30° caudad, and center to exit at nasion.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

Alternative 25° to 30° An alternative projection is a **25° to 30° caudad** tube angle (Fig. 11.114) that allows better visualization of the superior orbital fissures (black arrows), the foramen rotundum (small white arrows) (Fig. 11.114), and the inferior orbital rim region. CR exits at level of mid orbit.

NOTE: Decreased caudal angulation of the CR to 15° and/or increased neck flexion (chin down) will result in projection of the petrous pyramids to the lower third of the orbits.

Alternative AP Axial Projection For patients who are unable to be positioned for a PA projection (e.g., trauma patients), an AP axial projection may be obtained with the use of a 15° cephalic angle, with OML positioned perpendicular to IR (see Chapter 15).

Fig. 11.113 PA axial—CR 15° caudad, OML perpendicular; *inset - (solid arrow),* and alternative CR 30° caudad *(dotted arrow).*

Fig. 11.114 Alternative PA axial—30° caudad.

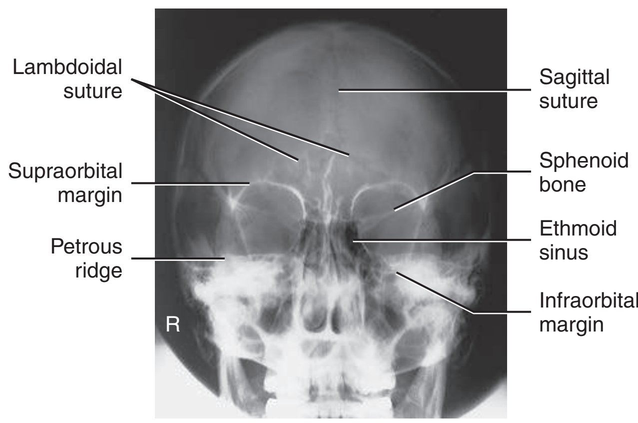


Fig. 11.115 PA axial—15° caudad (Caldwell method).

Evaluation Criteria

Anatomy Demonstrated: • Frontal bone, greater and lesser sphenoid wings, superior orbital fissures, frontal and anterior ethmoid sinuses, supraorbital margins, and crista galli are demonstrated (Fig. 11.115).

PA Axial 25° to 30° Caudad Angle • In addition to the structures mentioned previously, the foramen rotundum adjacent to each IOM is visualized, and the superior orbital fissures (see Fig. 11.114, white and black arrows) are visualized within the orbits.

Position: • **No rotation** as assessed by equal distance from the midlateral orbital margins to the lateral cortex of the cranium on each side and the superior orbital fissures symmetric within the orbits, correct extension of neck (OML alignment). • **Example:** If the distance between the right lateral orbit and lateral cranial cortex is greater than the left side, the face is rotated toward the left side. • **No tilt with the MSP perpendicular to IR.**

PA Axial 15° Caudad Angle: • Petrous pyramids are projected into the lower one-third of the orbits. • Supraorbital margin is visualized without superimposition. CR angle and OML alignment will impact location of petrous ridges within the orbits.

PA Axial 25° to 30° Caudad Angle: • Petrous pyramids are projected at or just below the IOM to allow visualization of the entire orbital base. CR angle and OML alignment will impact location of petrous ridges within the orbits. • Collimation to area of interest.

Exposure: • Density (brightness) and contrast are sufficient to visualize the frontal bone and sellar structures without overexposure to perimeter regions of skull. • Sharp bony margins indicate **no motion.**

PA PROJECTION—SKULL SERIES

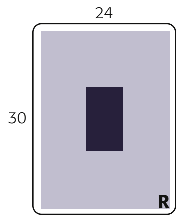
Clinical Indications

- Skull fractures (medial and lateral displacement), neoplastic processes, and Paget disease. This projection is intended to demonstrate the frontal bone with minimal distortion.

Skull Series
ROUTINE
- AP axial (Towne method)
- Lateral
- PA axial 15° (Caldwell method) or PA axial 25° to 30°
- PA

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from patient's head and neck. Exposure is taken with patient in the erect or prone position.

Part Position

- Rest patient's nose and forehead against table/imaging surface.
- Flex neck, aligning **OML perpendicular** to IR.
- Align **MSP perpendicular** to midline of table/imaging device to prevent head rotation or tilt (EAM same distance from table/imaging device surface).
- Center IR to CR.

CR

- CR is perpendicular to IR (parallel to OML) and is centered to exit at glabella (Fig. 11.116).

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

Evaluation Criteria

Anatomy Demonstrated: • Frontal bone, crista galli, internal auditory canals, frontal and anterior ethmoid sinuses, petrous ridges, greater and lesser wings of sphenoid, and dorsum sellae are demonstrated (Figs. 11.117 and 11.118).

Position: • **No rotation** is evident, as indicated by equal distance bilaterally from lateral orbital margin to lateral cortex of skull. • Petrous portion of temporal bone fills the orbits with the petrous ridges at the level of the supraorbital margin. • Posterior and anterior clinoid processes are visualized just superior to ethmoid sinuses. • Collimation to area of interest.

Exposure: • Density (brightness) and contrast are sufficient to visualize frontal bone and surrounding bony structures. • Sharp bony margins indicate **no motion.**

Fig. 11.116 PA skull - erect and prone (*inset*).

Fig. 11.117 PA skull.

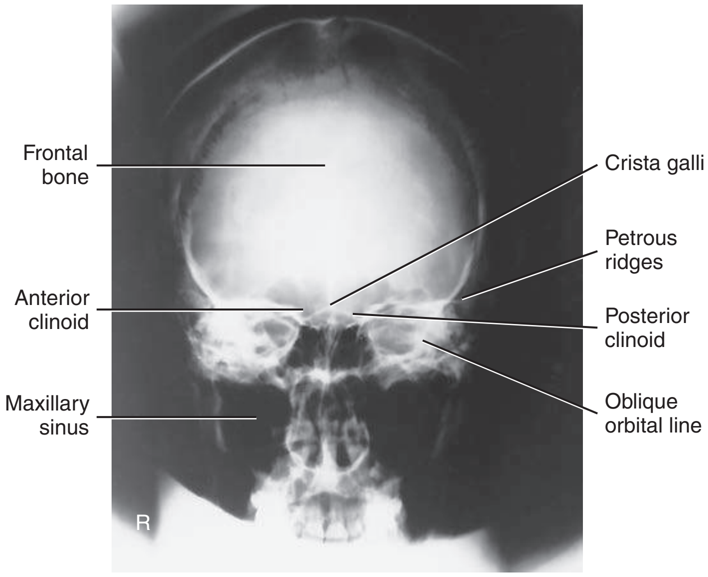


Fig. 11.118 PA skull.

SUBMENTOVERTICAL (SMV) PROJECTION—SKULL SERIES

WARNING: Rule out cervical spine fracture or subluxation on trauma patient before attempting this projection.

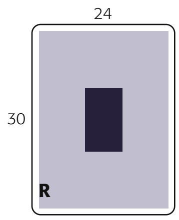
Skull Series
SPECIAL
- **SMV**

Clinical Indications

- Advanced bony pathology of the inner temporal bone structures (skull base)
- Possible basal skull fracture

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metal, plastic, and other removable objects from patient's head. Take radiograph with patient in an erect or supine position.

The erect position is recommended using an erect table or an upright imaging device (Fig. 11.119, inset). A wheelchair can also be used. A wheelchair offers support for the back and provides greater stability in maintaining the position. (Ensure wheels are locked before positioning patient.)

Part Position

- Raise patient's chin and hyperextend the neck if possible until **IOML is parallel to IR** (see NOTE).
- Rest patient's head on vertex.
- Align **MSP perpendicular** to the midline of the grid or table/imaging device surface, **avoiding tilt or rotation.**

Supine With patient in the supine position, extend patient's head over end of table and support grid cassette and head as shown in Fig. 11.119, keeping **IOML parallel to IR** and **perpendicular to CR.** Place a positioning sponge/pillow under the patient's back to support neck extension.

Erect If patient is unable to extend the neck sufficiently, compensate by angling CR to remain **perpendicular to IOML.** Depending on the equipment used, IR also may be angled to maintain the perpendicular relationship with CR (e.g., with an adjustable upright imaging device).

NOTE: This position is very uncomfortable for patients in the erect or the supine position; perform it as quickly as possible.

CR

- CR is perpendicular to infraorbitomeatal line.
- Center 1½ inch (4 cm) inferior to mandibular symphysis, or midway between the gonions (approximately ¾ inch [2 cm] anterior to level of EAM).
- Center IR to CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

Fig. 11.119 SMV tabletop with grid cassette (*inset* demonstrates use of upright imaging device). CR perpendicular to IOML.

Fig. 11.120 SMV.

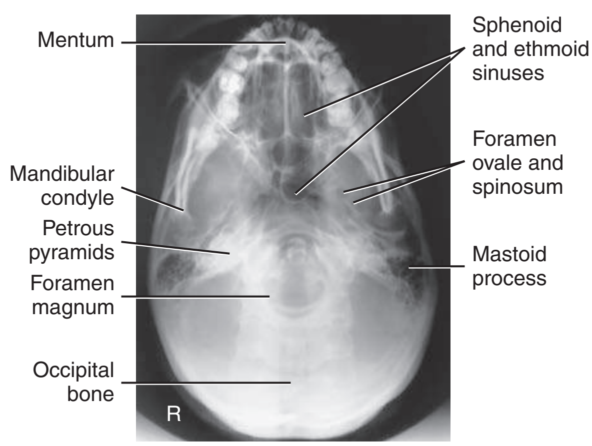


Fig. 11.121 SMV.

Evaluation Criteria

Anatomy Demonstrated: • Foramen ovale and spinosum, mandible, sphenoid and posterior ethmoid sinuses, mastoid processes, petrous ridges, hard palate, foramen magnum, and occipital bone are demonstrated (Figs. 11.120 and 11.121).

Position: • Correct extension of neck and relationship between IOML and CR as indicated by mandibular mentum anterior to the ethmoid sinuses. • **No rotation** evidenced by the MSP parallel to edge of IR. • **No tilt** evidenced by equal distance between mandibular ramus and lateral cranial cortex. • **Example:** If the distance on the left side between the ramus and lateral cranium is greater on the left than the right, the cranial vertex is tilted to the left. • Collimation to area of interest.

Exposure: • Density (brightness) and contrast are sufficient to visualize clearly outline of ethmoid and sphenoid sinuses and cranial foramen. • Sharp bony margins indicate **no motion.**

PA AXIAL PROJECTION—SKULL SERIES

HAAS METHOD

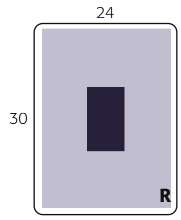
Clinical Indications

- Skull fractures (medial and lateral displacement), neoplastic processes, and Paget disease

This is an **alternative projection** for patients who cannot flex the neck sufficiently for AP axial (Towne). It can also be performed for hypersthenic and bariatric patients with limited range of cervical motion. It results in magnification of the occipital area but in lower doses to facial structures and the thyroid gland.

This projection is not recommended when the occipital bone is the area of interest because of excessive magnification.

Skull Series
SPECIAL
- SMV
- PA axial (Haas method)



Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–90

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from patient's head and neck. Take radiograph with patient in the erect or prone position.

Part Position

- Rest patient's nose and forehead against the table/imaging device surface.
- Flex neck, bringing **OML perpendicular** to IR (Fig. 11.122).
- Align MSP to CR and to the midline of the grid or table/imaging device surface.
- Ensure that **no rotation or tilt** exists (MSP perpendicular to IR).

CR

- Angle CR 25° cephalad to OML.
- Center CR to MSP and 1½ inches (4 cm) inferior to the inion and exit 1½ inches (4 cm) superior to nasion.
- Center IR to projected CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

Evaluation Criteria

Anatomy Demonstrated: • Occipital bone, petrous pyramids, and foramen magnum are demonstrated, with the dorsum sellae and posterior clinoid processes visualized in the shadow of the foramen magnum (Figs. 11.123 and 11.124).
Position: • **No rotation** is evident, as indicated by bilateral symmetric petrous ridges. • Dorsum sellae and posterior clinoid processes are visualized in the foramen magnum, which indicates correct CR angle and proper neck flexion and extension. • **No tilt** as evidenced by correct placement of anterior clinoid processes within the middle of the foramen magnum • Collimation to area of interest.
Exposure: • Density (brightness) and contrast are sufficient to visualize occipital bone and sellar structures within foramen magnum. • Sharp bony margins indicate **no motion.**

Fig. 11.122 PA axial—CR 25° cephalad to OML, erect and prone (*inset*).

Fig. 11.123 PA axial.

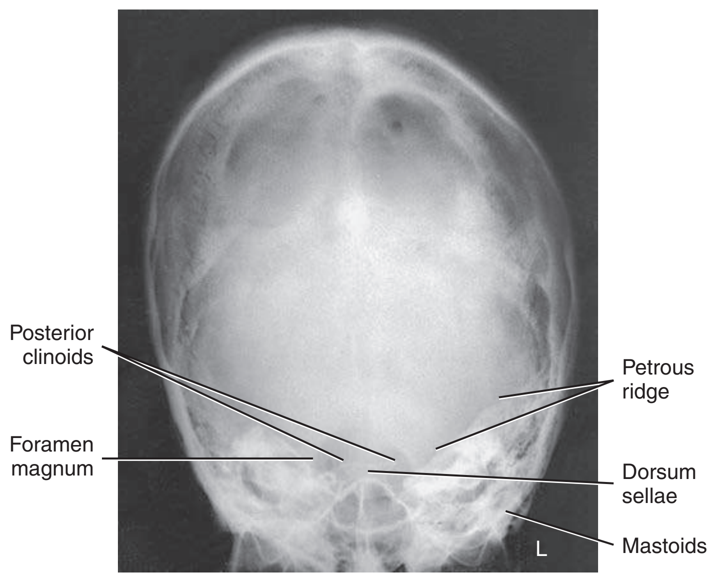


Fig. 11.124 PA axial.

LATERAL POSITION: RIGHT OR LEFT LATERAL—FACIAL BONES

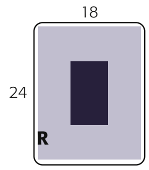
Clinical Indications

- Fractures and neoplastic or inflammatory processes of the facial bones, orbits, and mandible

Facial Bones
ROUTINE
- Lateral
- Parietoacanthial (Waters method)
- PA axial (Caldwell method)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Grid
- kVp range: 70–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or recumbent semiprone.

Part Position

- Rest lateral aspect of head against table or upright imaging device surface, **with side of interest closest to IR.**
- Adjust head into a **true lateral position** and oblique body as needed for patient's comfort. (Palpate external occipital protuberance posteriorly and nasion or glabella anteriorly to ensure that these two points are equidistant from IR [Fig. 11.125].)
- Align **MSP parallel** to IR.
- Align IPL perpendicular to IR.
- Adjust chin to bring **IOML perpendicular** to front edge of IR.

CR

- Align CR **perpendicular** to IR.
- Center CR to **zygoma (prominence of the cheek)**, midway between outer canthus and EAM (Fig. 11.126).
- Center IR to CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: Use radiolucent support under the head if needed to bring IPL perpendicular to tabletop on patient with a large chest.

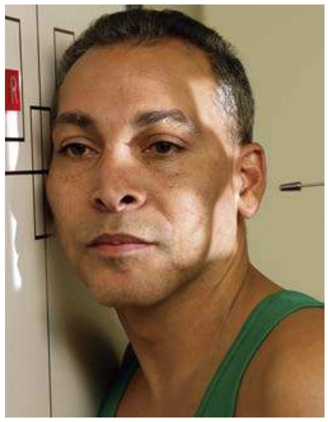


Fig. 11.125 Right lateral—erect.

Fig. 11.126 Right lateral—recumbent semiprone.

Fig. 11.127 Lateral facial bones.

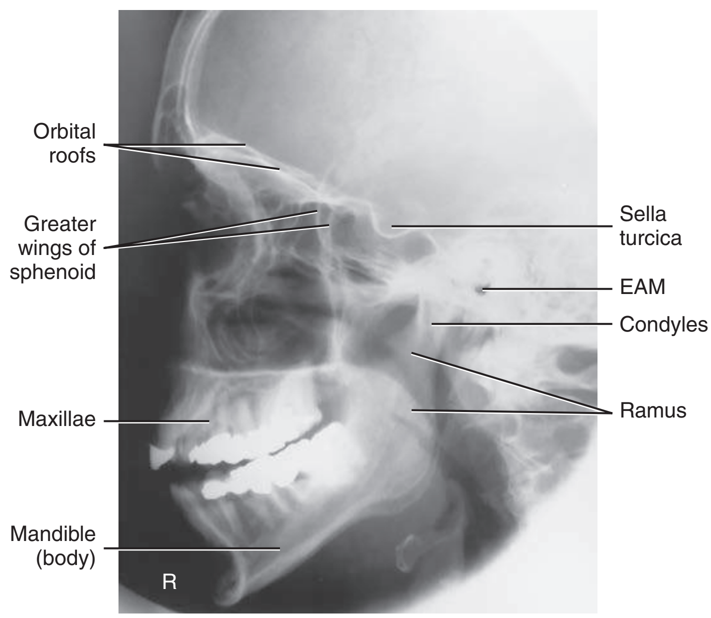


Fig. 11.128 Lateral facial bones.

Evaluation Criteria

Anatomy Demonstrated: • Superimposed facial bones, greater wings of the sphenoid, orbital plates, sella turcica, zygoma, and mandible are demonstrated (Figs. 11.127 and 11.128).

Position: • An accurately positioned lateral image of the facial bones demonstrates no rotation or tilt. • **Rotation** is evident by **anterior and posterior separation** of symmetric vertical bilateral structures such as the mandibular rami and greater wings of the sphenoid. • **Tilt** is evident by **superior and inferior separation** of the orbital plates. • Collimation to area of interest.

Exposure: • Contrast and density (brightness) are sufficient to visualize the maxillary region. • Sharp bony margins indicate **no motion.**

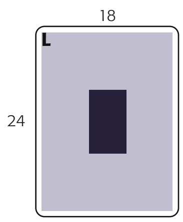
PARIETOACANTHIAL PROJECTION—FACIAL BONES

WATERS METHOD

Facial Bones
ROUTINE
- Lateral
- **Parietoacanthial (Waters method)**
- PA axial (Caldwell method)

Clinical Indications
- Fractures (particularly tripod and Le Fort fractures) and neoplastic or inflammatory processes
- Foreign bodies in the eye

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 70–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or prone (erect is preferred if patient's condition allows).

Part Position
- Extend neck, resting chin against table/upright imaging device surface.
- Adjust head until **MML is perpendicular to plane of IR.** OML forms a 37° angle with the table/upright imaging device surface (Fig. 11.129).
- Position **MSP perpendicular** to midline of grid or table/imaging device surface, preventing rotation or tilting of head. (One way to check for rotation is to palpate the mastoid processes on each side and the lateral orbital margins with the thumb and fingertips to ensure that these lines are equidistant from the IR.)

CR
- Align CR perpendicular to IR, to exit at acanthion.
- Center IR to CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

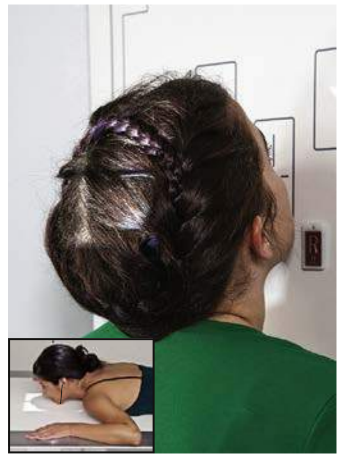


Fig. 11.129 Parietocanthial (Waters - MML perpendicular (OML 37° to IR) - erect and recumbent (*inset*).

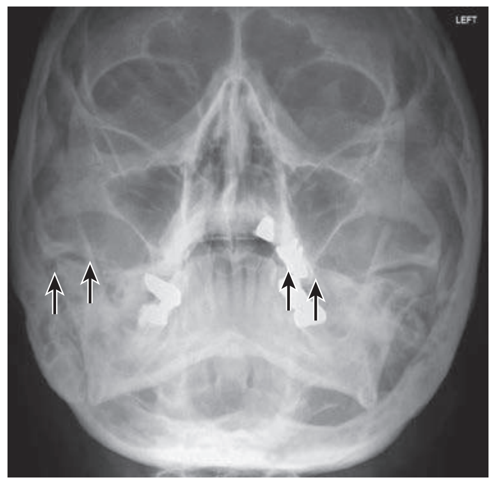


Fig. 11.130 Parietoacanthial (Waters) projection.

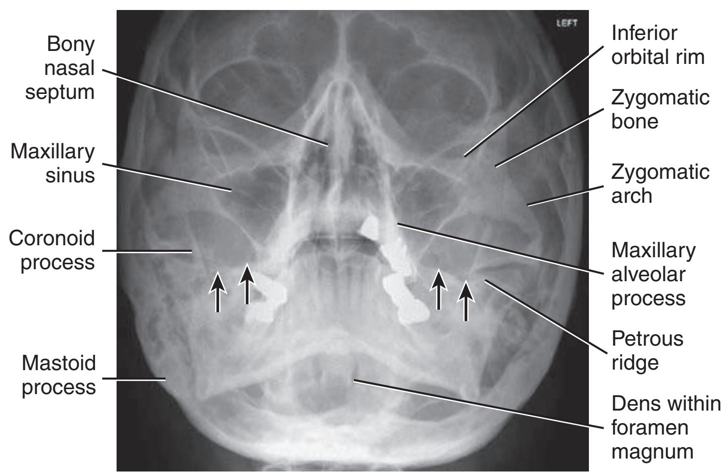


Fig. 11.131 Parietoacanthial (Waters) projection.

Evaluation Criteria
Anatomy Demonstrated: • IOMs, maxillae, nasal septum, zygomatic bones, zygomatic arches, and anterior nasal spine.
Position: • Correct neck extension demonstrates petrous ridges (black arrows) just inferior to the maxillary sinuses (Figs. 11.130 and 11.131). • **No** patient **rotation** exists, as indicated by equal distance from the midlateral orbital margin to the lateral cortex of cranium on each side. • Collimation to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize maxillary region. • Sharp bony margins indicate **no motion.**

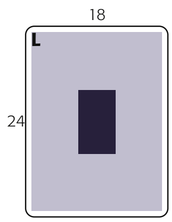
PA AXIAL PROJECTION—FACIAL BONES

CALDWELL METHOD

Facial Bones
ROUTINE
- Lateral
- Parietoacanthial (Waters method)
- PA axial (Caldwell method)

Clinical Indications
- Fractures and neoplastic or inflammatory processes of the facial bones

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 70–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or prone (erect is preferred if patient's condition permits it).

Part Position
- Rest patient's nose and forehead against the imaging device.
- Tuck chin, bringing **OML perpendicular** to IR.
- Align **MSP perpendicular** to midline of grid or table/imaging device surface. Ensure **no rotation or tilt** of head (Fig. 11.132).

CR
- Angle CR 15° caudad, to exit at nasion (see NOTE).
- Center CR to IR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: If area of interest is the orbital margins, use a **30°** caudad angle to project the petrous ridges below the IOM. CR will exit level of midorbits.

Fig. 11.132 PA axial Caldwell—OML perpendicular, CR 15° caudad, erect and prone (*inset*).

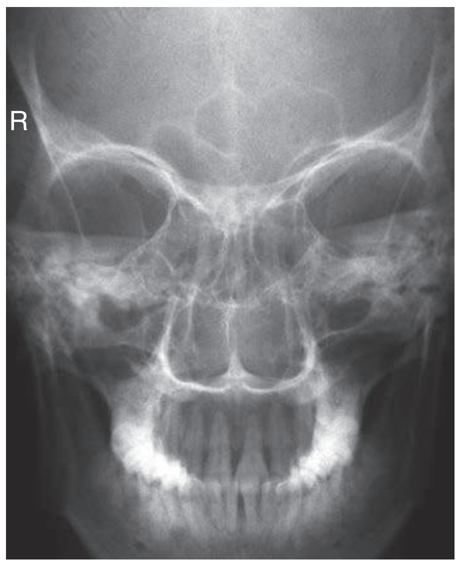


Fig. 11.133 PA axial Caldwell—CR 15°.

Evaluation Criteria
Anatomy Demonstrated: • Superior orbital margins, maxillae, nasal septum, zygomatic bones, and anterior nasal spine (Figs. 11.133 and 11.134).
Position: • Correct patient position/CR angulation is indicated by petrous ridges projected into the **lower one-third of orbits** with 15° caudad CR. If the inferior orbital margins are the area of interest, 30° caudad angle projects the petrous ridges below the IOMs. • **No rotation** of cranium is indicated by equal distance from midlateral orbital margin to the lateral cortex of the cranium (a narrower distance would indicate rotation towards the IR); superior orbital fissures are symmetric. • Collimation to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize maxillary region and superior orbital margins. • Sharp bony margins indicate **no motion.**

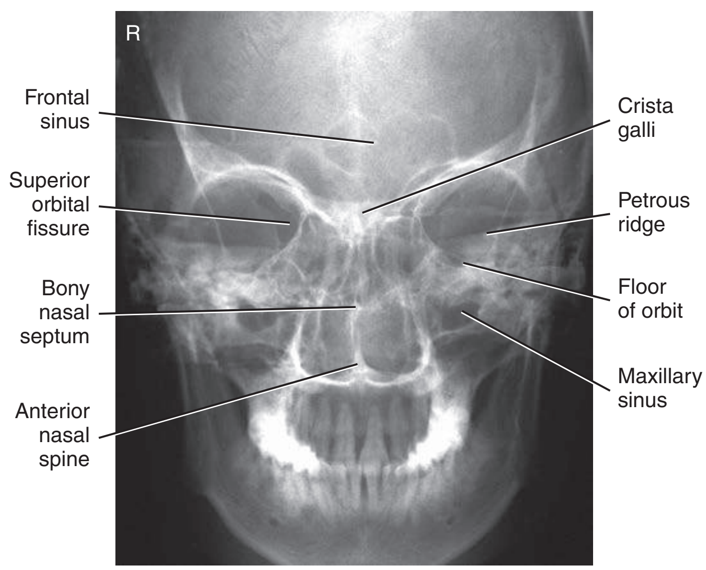


Fig. 11.134 PA axial Caldwell—CR 15°.

MODIFIED PARIETOACANTHIAL PROJECTION—FACIAL BONES

MODIFIED WATERS METHOD

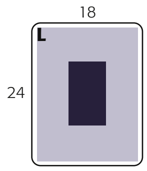
Clinical Indications

- Orbital fractures (e.g., blowout) and neoplastic or inflammatory processes
- Foreign bodies in the eye

Facial Bones
SPECIAL
- Modified parietoacanthial (modified Waters method)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 70–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from the head and neck. Patient position is erect or prone (erect is preferred if patient's condition allows).

Part Position

- Extend neck, resting chin and nose against table/upright imaging device surface.
- Adjust head until **LML is perpendicular;** OML forms a **55°** angle with IR (Fig. 11.135).
- Position **MSP perpendicular** to midline of grid or table/upright imaging device surface. Ensure **no rotation or tilt** of head.

CR

- Align CR perpendicular, centered to exit at acanthion.
- Center IR to CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

Evaluation Criteria

Anatomy Demonstrated: • Inferior orbital margins are perpendicular to IR, which also provides a less distorted view of the orbital base than a parietoacanthial (Waters) projection (Figs. 11.136 and 11.137).
Position: • Correct position/CR angulation is indicated by petrous ridges projected into the lower half of the maxillary sinuses, below the IOMs. • **No rotation** of the cranium is indicated by equal distance from the midlateral orbital margin to the lateral cortex of the cranium. • Collimation to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize the inferior orbital margins. • Sharp bony margins indicate **no motion.**

Fig. 11.135 Modified parietoacanthial (Waters)—**LML perpendicular (OML 55°).** (Prone shown in *inset.*)

Fig. 11.136 Modified parietoacanthial (Waters).

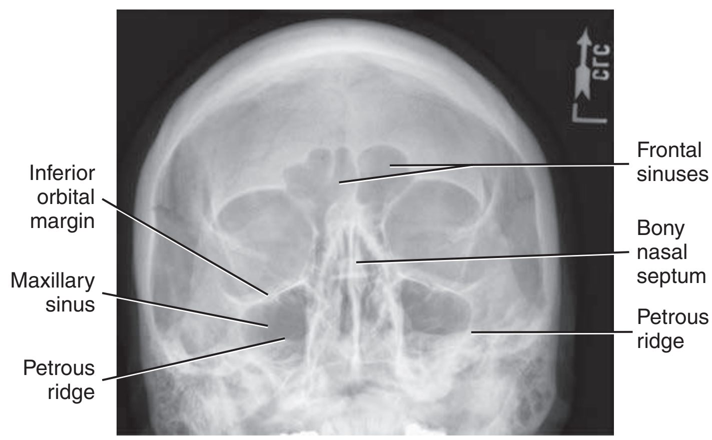


Fig. 11.137 Modified parietoacanthial (Waters).

LATERAL POSITION—NASAL BONES

Clinical Indications

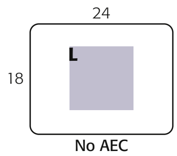
- Nasal bone fractures

Both sides should be examined for comparison, with side closest to IR best demonstrated.

Nasal Bones
ROUTINE
- Lateral
- Parietoacanthial (Waters method)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Nongrid
- kVp range: 65–80



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is recumbent semiprone or erect.

Part Position

- Rest lateral aspect of head against the table/upright imaging device surface, with side of interest closest to IR.
- Position nasal bones to center of IR.
- Adjust head into a **true lateral position** and oblique body as needed for patient's comfort (place sponge block under chin if needed) (Fig. 11.138).
- Align **MSP parallel** with a table/upright imaging device surface.
- Align **IPL perpendicular** to table/upright imaging device surface.
- Position **IOML perpendicular** to front edge of IR.

CR

- Align CR perpendicular to IR.
- Center CR to ½ inch (1.25 cm) inferior to nasion.

Recommended Collimation Collimate on all sides to within 2 inches (5 cm) of nasal bone.

Respiration Suspend respiration.

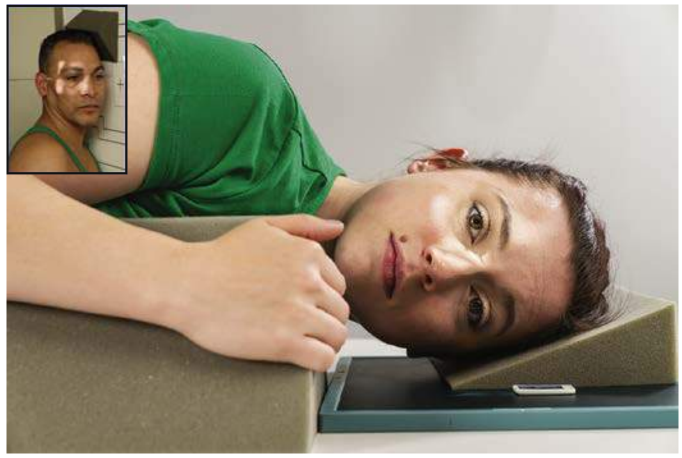


Fig. 11.138 Left lateral nasal bones—recumbent semiprone and erect (*inset*).

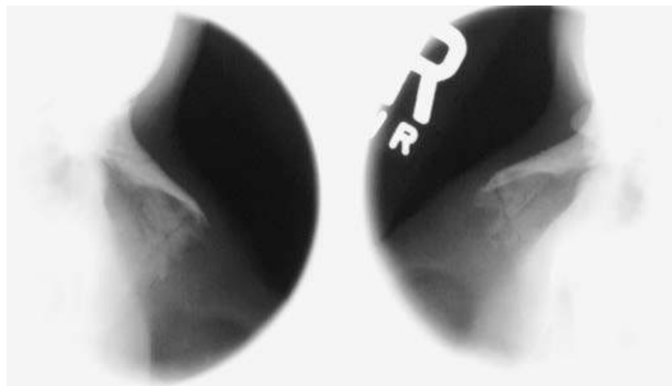


Fig. 11.139 Lateral (L and R).

Evaluation Criteria

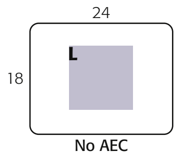
Anatomy Demonstrated: • Nasal bones with soft tissue nasal structures, the region of the frontonasal suture, and the anterior nasal spine are demonstrated (Fig. 11.139).
Position: • Nasal bones are demonstrated **without rotation.** • Collimation to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize nasal bone and soft tissue structures. • Sharp bony structures indicate **no motion.**

SUPEROINFERIOR TANGENTIAL (AXIAL) PROJECTION—NASAL BONES

Nasal Bones
SPECIAL
- Superoinferior tangential (axial)

Clinical Indications
- Fractures of the nasal bones (medial-lateral displacement)

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Nongrid
- kVp range: 65–80



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Patient is seated erect in a chair at end of table or in the prone position on table.

Part Position (insert part position icon here)
- Extend and rest chin on IR. Place angled support under IR, as demonstrated, to **place IR perpendicular to GAL** (Fig. 11.140).
- Align **MSP** perpendicular to CR and to IR midline.

CR
- Center CR to nasion and angle as needed to ensure that it is **parallel to GAL.** (CR must just skim glabella and anterior upper front teeth.)

Recommended Collimation Collimate on all sides to nasal bones.

Respiration Suspend respiration.

Evaluation Criteria
Anatomy Demonstrated: • Tangential projection of midnasal and distal nasal bones (with minimal superimposition of the glabella or alveolar ridge) and nasal soft tissue (Figs. 11.141 and 11.142). Petrous ridges are inferior to maxillary sinuses.
Position: • **No** patient **rotation** is evident, as indicated by equal distance from anterior nasal spine to outer soft tissue borders on each side. • Incorrect neck position is indicated by visualization of alveolar ridge (excessive extension) or visualization of too much glabella (excessive flexion).
Exposure: • Contrast and density (brightness) are sufficient to visualize nasal bones and nasal soft tissue. • Sharp bony margins indicate **no motion.**

Fig. 11.140 Superoinferior tangential (axial) projection.

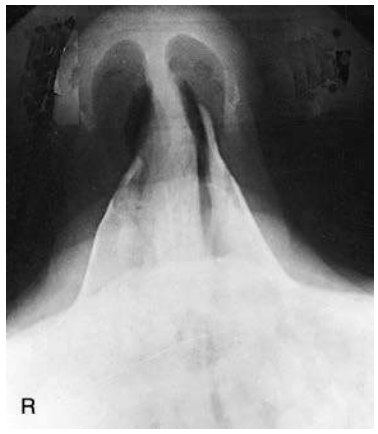


Fig. 11.141 Superoinferior tangential (axial) projection.

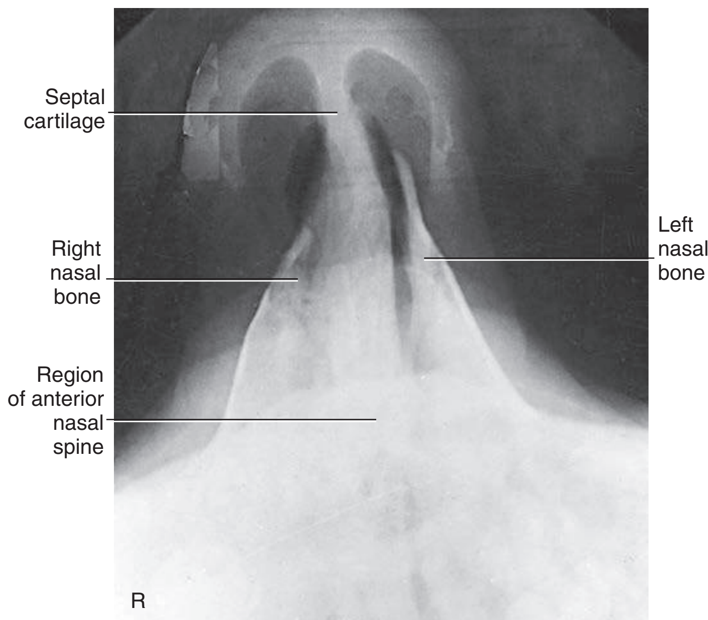


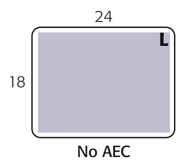
Fig. 11.142 Superoinferior tangential (axial) projection.

SUBMENTOVERTICAL (SMV) PROJECTION—ZYGOMATIC ARCHES

Zygomatic Arches
ROUTINE
- SMV
- Oblique inferosuperior (tangential)
- AP axial (modified Towne method)

Clinical Indications
- Fractures of zygomatic arch
- Neoplastic or inflammatory processes

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10inches (18 × 24 cm), landscape
- Grid or nongrid
- kVp range: 75–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. This projection may be taken with the patient erect or supine.

Part Position
- Raise chin, hyperextend neck until **IOML is parallel** to IR (see NOTE 1).
- Rest head on vertex of skull.
- Align **MSP perpendicular** to midline of the grid or the table/ upright imaging device surface, **avoiding all tilt or rotation.**

CR
- Align CR **perpendicular** to IR (see NOTE 2).
- Center CR **midway between zygomatic arches**, at a **level** 1½ **inches (4 cm) inferior to mandibular symphysis.**
- Center IR to CR, with plane of IR parallel to IOML.

Recommended Collimation Collimate to outer margins of zygomatic arches.

Respiration Suspend respiration.

NOTE 1: This position is very uncomfortable for patients; complete the projection as quickly as possible.

NOTE 2: If patient is unable to extend neck adequately, angle CR **perpendicular to IOML.** If equipment allows, IR should be angled to maintain CR/ IR perpendicular relationship (Fig. 11.143, inset).

Evaluation Criteria
Anatomy Demonstrated: • Zygomatic arches are demonstrated laterally from each mandibular ramus (Figs. 11.144 and 11.145).
Position: • Correct IOML/CR relationship, as indicated by superimposition of mandibular symphysis on frontal bone. • **No** patient **rotation**, as indicated by zygomatic arches visualized symmetrically. • Collimation to area of interest.
Exposure: • Sufficient contrast and density (brightness) to visualize zygomatic arches. • Sharp bony margins indicate **no motion.**

Fig. 11.143 SMV projection, erect and supine (*inset*)—IOML parallel to IR; CR perpendicular to IOML.

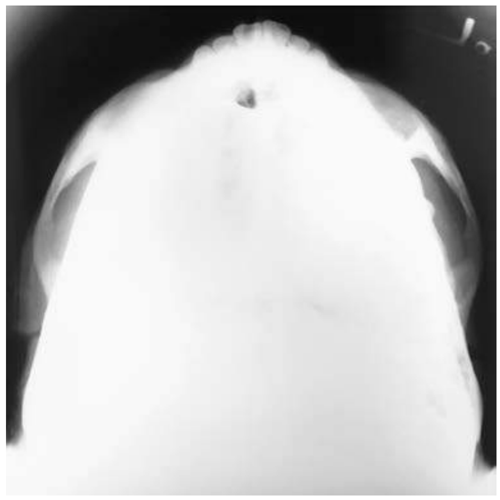


Fig. 11.144 SMV projection.

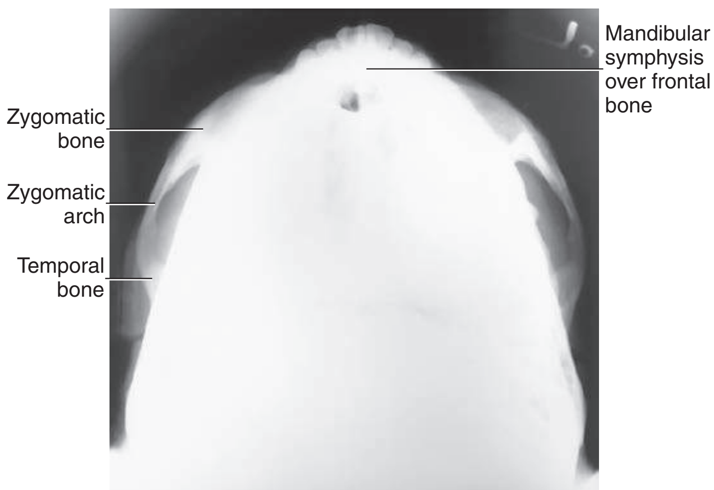


Fig. 11.145 SMV projection.

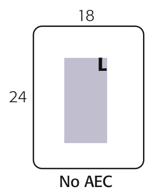
OBLIQUE INFEROSUPERIOR (TANGENTIAL) PROJECTION—ZYGOMATIC ARCHES

Zygomatic Arches
ROUTINE
- SMV
- **Oblique inferosuperior (tangential)**
- AP axial (modified Towne method)

Clinical Indications
- Fractures of zygomatic arch
- Especially useful for depressed zygomatic arches caused by trauma or skull morphology

Radiographs of both sides generally are taken for comparison.

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Grid or nongrid
- kVp range: 70–85
- AEC not recommended



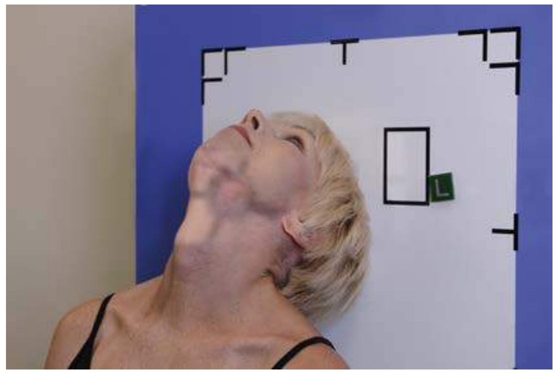


Fig. 11.146 Oblique inferosuperior (tangential), upright imaging device (15° tilt, 15° rotation, CR perpendicular to IOML).

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or supine. Erect, which is easier for the patient, may be done with erect table or upright imaging device.

Part Position
- Raise chin, hyperextending neck until **IOML is parallel** to IR (see NOTE 1).
- Rest head on vertex of skull.
- Rotate head 15° toward side to be examined; also tilt chin 15° toward side of interest (Fig. 11.146).

CR
- Align CR perpendicular to IR and IOML (see NOTE 2).
- Center CR to zygomatic arch of interest (CR skims mandibular ramus, passes through arch, and skims parietal eminence on the downside).
- Adjust IR so it is parallel to IOML and perpendicular to CR.

Recommended Collimation Collimate closely to zygomatic bone and arch.

Respiration Suspend respiration.

NOTE 1: This position is very uncomfortable for the patient; complete the projection as quickly as possible.

NOTE 2: If patient is unable to extend neck sufficiently, angle CR **perpendicular to IOML**. If equipment allows, IR should be angled to maintain CR/IR perpendicular relationship.

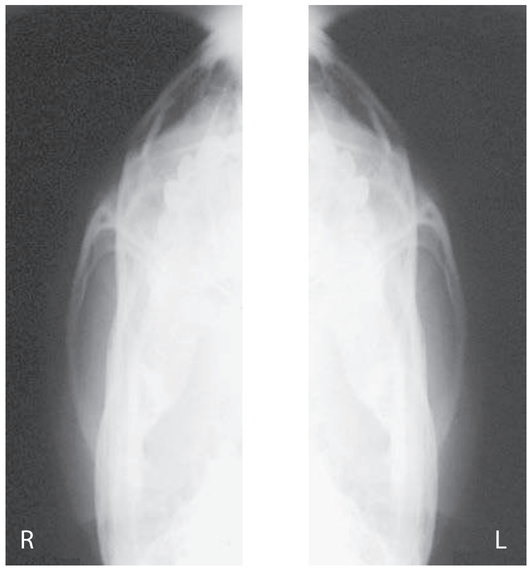


Fig. 11.147 Oblique inferosuperior (tangential).

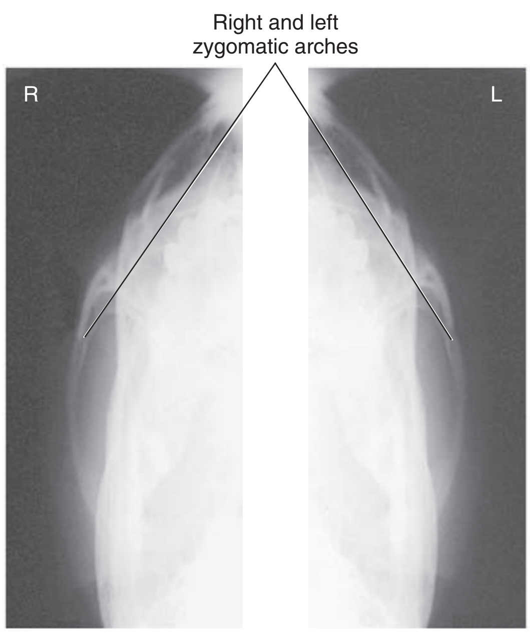


Fig. 11.148 Oblique inferosuperior (tangential).

Evaluation Criteria
Anatomy Demonstrated: • Single zygomatic arch, free of superimposition, is demonstrated (Figs. 11.147 and 11.148).
Position: • Correct patient position provides for demonstration of zygomatic arch without superimposition of parietal bone or mandible. • Collimation to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize zygomatic arch. • Sharp bony margins indicate **no motion.**

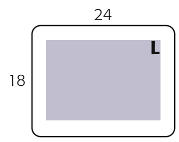
AP AXIAL PROJECTION—ZYGOMATIC ARCHES

MODIFIED TOWNE METHOD

Zygomatic Arches
ROUTINE
- SMV
- Oblique inferosuperior (tangential)
- AP axial (modified Towne method)

Clinical Indications
- Fractures and neoplastic or inflammatory processes of zygomatic arch

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Grid
- kVp range: 70–85
- AEC not recommended



Shielding Shield radiosensitive tissues outside region of interest

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or supine.

Part Position
- Rest patient's posterior skull against table/upright imaging device surface.
- Tuck chin, bringing **OML (or IOML) perpendicular** to IR (see NOTE).
- Align **MSP perpendicular** to midline of grid or table/upright imaging device surface to **prevent head rotation or tilt** (Fig. 11.149).

CR
- Angle CR 30° caudad to OML or 37° to IOML (see NOTE).
- Center CR to 1 inch (2.5 cm) superior to nasion (to pass through midarches) at level of the gonion.
- Center IR to projected CR.

Recommended Collimation Collimate to outer margins of zygomatic arches.

Respiration Suspend respiration.

NOTE: If patient is unable to depress the chin sufficiently to bring OML perpendicular to IR, **IOML** can be placed perpendicular instead and CR angle increased to 37° caudad. This positioning maintains the 30° angle between OML and CR and demonstrates the same anatomic relationships. (A 7° to 8° difference is noted between OML and IOML.)

Evaluation Criteria

Anatomy Demonstrated: • Bilateral zygomatic arches, free of superimposition, are demonstrated (Figs. 11.150 and 11.151).
Position: • Zygomatic arches are visualized **without** patient **rotation** as indicated by symmetric appearance of arches bilaterally. • Collimation to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize zygomatic arches. • Sharp bony margins indicate **no motion.**

Fig. 11.149 AP axial—zygomatic arches—**CR 30° to OML** (37° to IOML), erect and supine (*inset*).

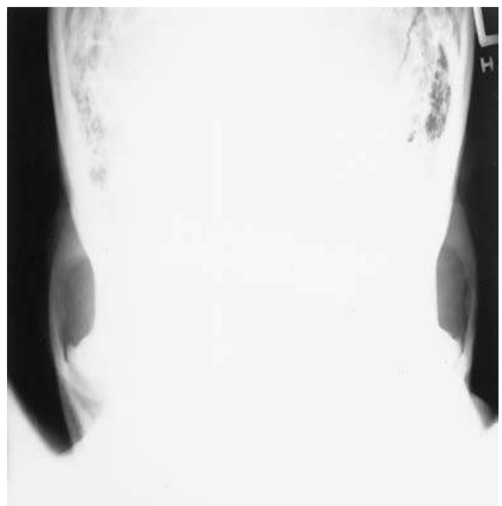


Fig. 11.150 AP axial.

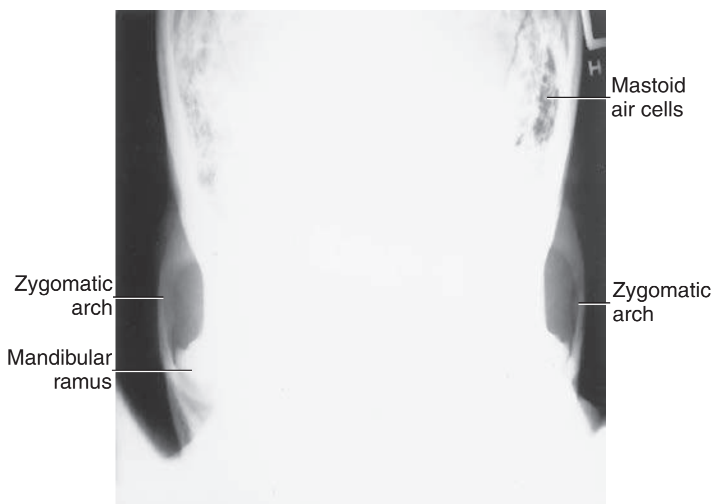


Fig. 11.151 AP axial.

PARIETO-ORBITAL OBLIQUE PROJECTION—OPTIC FORAMINA

RHESE METHOD

Clinical Indications

- Bony abnormalities of the optic foramen
- Demonstrate lateral margins of orbits and foreign bodies within eye

CT is the preferred modality for a detailed investigation of the optic foramina. Radiographs of both sides generally are taken for comparison. This projection can also provide an excellent image of the mid-lateral and inferior orbital margins.

Optic Foramina

ROUTINE

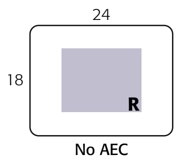
- Parieto-orbital oblique (Rhese method)
- Parietoacanthial (Waters method)

SPECIAL

- Modified parietoacanthial (modified Waters method)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Grid
- kVp range: 70–85
- AEC not recommended



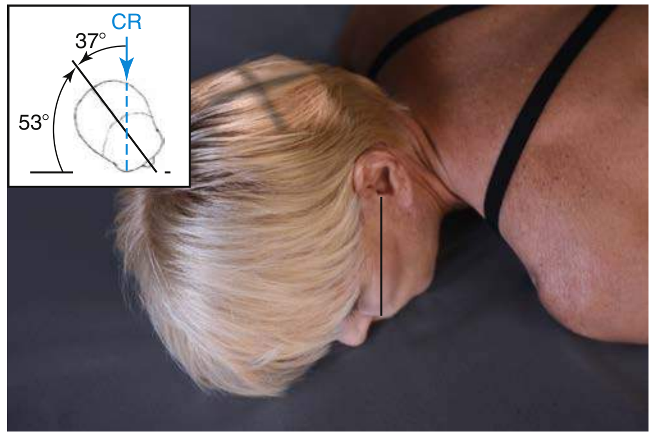


Fig. 11.152 Parieto-orbital oblique projection—53° rotation; AML perpendicular; CR perpendicular.

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Position patient erect or supine.

Part Position

- As a starting reference, position patient's head in a prone position with MSP perpendicular to IR. Adjust flexion and extension so that AML is perpendicular to IR. Adjust the patient's head so that the chin, cheek, and nose touch the table/upright imaging device surface (this position is historically known as the "3 point landing").
- Rotate the head 37° toward the affected side. The angle formed between MSP and plane of IR measures 53° (Fig. 11.152). (An angle indicator should be used to obtain an accurate angle of 37° from CR to MSP.)

CR

- Align CR **perpendicular** to IR at the midportion of the **downside orbit.**

Recommended Collimation Collimate on all sides to yield a field size of approximately 3 inches (7.5 cm) square.

Respiration Suspend respiration during exposure.

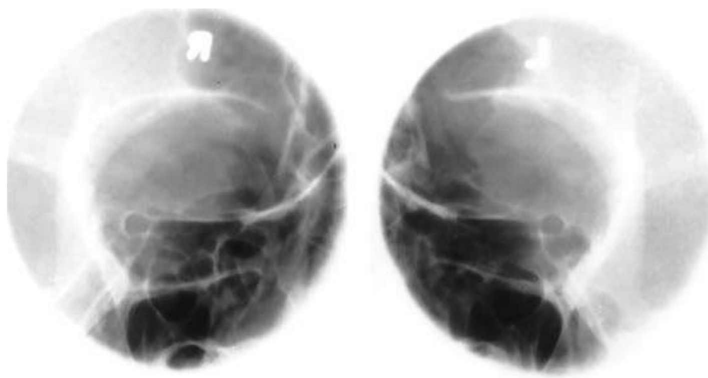


Fig. 11.153 Bilateral parieto-orbital oblique projection.

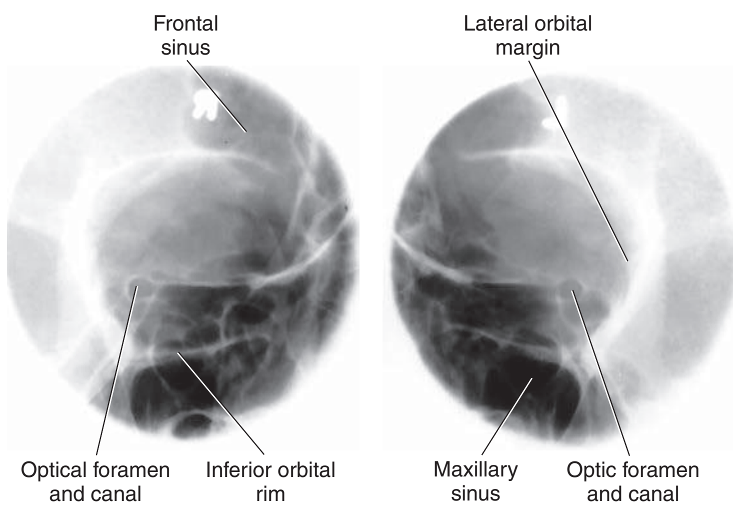


Fig. 11.154 Bilateral parieto-orbital oblique projection.

Evaluation Criteria

Anatomy Demonstrated: • Bilateral, nondistorted view of the optic foramen. • Lateral orbital margins are demonstrated (Figs. 11.153 and 11.154).

Position: • Accurate positioning projects the optic foramen into the lower outer quadrant of the orbit. • Proper positioning results when AML is correctly placed perpendicular to IR and correct rotation of skull. • Collimation to area of interest.

Exposure: • Contrast and density (brightness) are sufficient to visualize the optic foramen. • Sharp bony margins indicate **no motion.**

AXIOLATERAL OR AXIOLATERAL OBLIQUE PROJECTION—MANDIBLE

Clinical Indications

- Fractures and neoplastic or inflammatory processes of mandible

  Both sides of mandible are examined for comparison.

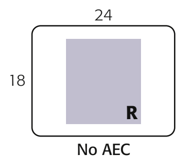
Mandible

ROUTINE

- Axiolateral or axiolateral oblique
- PA (or PA axial)
- AP axial (Towne method)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm) landscape
- Grid (often performed nongrid)
- kVp range: 70–85
- AEC not used



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or recumbent. If performed recumbent, place IR on wedge sponge to minimize OID (Fig. 11.155). For erect position, place region of interest against wall bucky and parallel to IR (Fig. 11.156). For horizontal beam trauma position, place IR (and grid if used) parallel to mandible (Fig. 11.157).

Part Position

- Place head in a true lateral position, with side of interest against IR.
- If possible, have patient close mouth and bring teeth together.
- Extend neck slightly to prevent superimposition of the gonion over the cervical spine.
- Rotate head toward IR (for axiolateral oblique) to place the mandibular area of interest parallel to IR. The degree of rotation/obliquity depends on which section of the mandible is of interest.
- Head in **true lateral** position best demonstrates **ramus.**
- **10° to 15°** rotation best provides a **general survey** of the mandible.
- **30°** rotation toward IR best demonstrates **body.**
- **45°** rotation best demonstrates **mentum.**

CR

- Three methods are suggested for demonstrating the specific region of the mandible of interest (side closest to IR) without superimposing the opposite side:
  1. Angle CR 25° cephalad from IPL; for horizontal beam trauma position.
  2. Use a combination of head tilt and CR angle that does not exceed 25° cephalad (e.g., angle the tube 10° cephalad and add 15° of head tilt toward the IR).
  3. Use 25° of head tilt toward IR and use perpendicular CR.
- Direct CR to exit mandibular region of interest.
- Center IR to projected CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

Fig. 11.155 Semisupine—15° wedge sponge and 10° cephalad CR angle.

Fig. 11.156 Erect 10°–15° head rotation toward IR and 10° cephalad CR angle.

Fig. 11.157 Horizontal beam trauma projection—**25° cephalad CR angle;** left lateral.

Fig. 11.158 Axiolateral oblique (general survey).

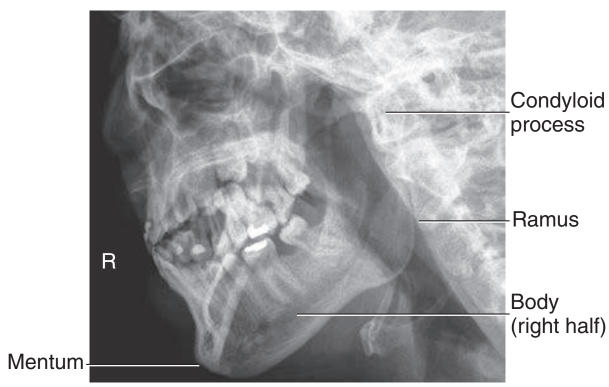


Fig. 11.159 Axiolateral oblique (general survey).

Evaluation Criteria

Anatomy Demonstrated: • Ramus, condyloid, and coronoid processes, body, and mentum of mandible nearest the IR are demonstrated (Figs. 11.158 and 11.159).

Position: • The appearance of the image/position of the patient depends on the structures under examination. • For the ramus and body, the ramus of interest is demonstrated with **no superimposition** from the opposite mandible (indicating correct CR angulation). • **No superimposition** of the cervical spine by the ramus should occur (indicating sufficient extension of neck). • The ramus and body should be demonstrated without foreshortening (indicating correct rotation of head). • The area of interest is demonstrated with minimal superimposition and minimal foreshortening. • **Collimation** to area of interest.

Exposure: • Contrast and density (brightness) are sufficient to visualize the mandibular area of interest. • Sharp bony margins indicate **no motion.**

PA OR PA AXIAL PROJECTION—MANDIBLE

Clinical Indications

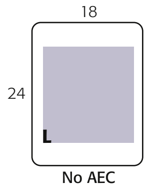
- Fractures
- Neoplastic or inflammatory processes of mandible

Optional PA axial best demonstrates proximal rami and elongated view of condyloid processes.

Mandible
ROUTINE
- Axiolateral oblique
- PA (or PA axial)
- AP axial (Towne method)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm) portrait
- Grid
- kVp range: 75–90



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or prone.

Part Position

- Rest patient's forehead and nose against table/upright imaging device surface (Fig. 11.160).
- Tuck chin, bringing **OML perpendicular** to IR (see NOTE).
- Align **MSP perpendicular** to midline of grid or table/imaging device surface (ensuring **no rotation or tilt** of head).
- Center IR to projected CR (to junction of lips).

CR

- PA: Align CR perpendicular to IR, centered to exit at junction of lips. For trauma patients, this position is best performed supine.
- Optional PA axial: Angle CR 20° to 25° cephalad, centered to exit at acanthion.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: For a true PA projection of the body (if this is area of interest), raise chin to bring AML perpendicular to IR.

Evaluation Criteria

Anatomy Demonstrated: • **PA:** Mandibular rami and lateral portion of body are visible (Fig. 11.161). • **Optional PA axial:** TMJ region and heads of condyles are visible through mastoid processes; condyloid processes are well visualized (slightly elongated) (Fig. 11.162).
Position: • **No** patient **rotation** exists, as indicated by mandibular rami visualized symmetrically, lateral to the cervical spine. • Midbody and mentum are faintly visualized, superimposed on cervical spine. • Collimation to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize mandibular body and rami. • Sharp bony margins indicate **no motion.**

Fig. 11.160 PA—CR perpendicular, exit at junction of lips. *Inset,* Optional PA axial—CR 20° to 25° cephalad, exit at acanthion.

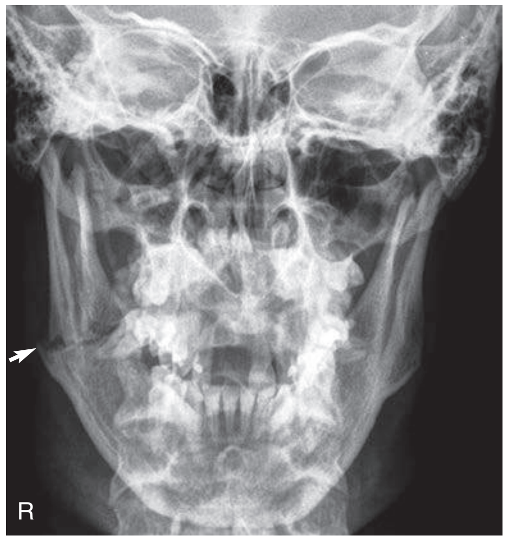


Fig. 11.161 PA; fracture through left ramus.

Fig. 11.162 Optional PA axial—CR 20° to 25° cephalad.

AP AXIAL PROJECTION—MANDIBLE

TOWNE METHOD

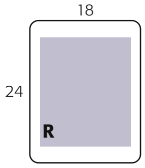
Clinical Indications

- Fractures
- Neoplastic or inflammatory processes of condyloid processes of mandible

Mandible
SPECIAL
- SMV
- Orthopantomography (mandible or TMJs or both)

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–90



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or supine.

Part Position

- Rest patient's posterior skull against table/upright imaging device surface.
- Tuck chin, aligning **OML perpendicular** to IR or place IOML perpendicular and adjust the CR angle accordingly (see NOTE).
- Align **MSP perpendicular** to midline of grid or table/upright imaging device surface to prevent head rotation or tilt.

CR

- Angle CR 35° caudad if OML is perpendicular to IR, or 42° caudad if IOML is perpendicular to IR (see NOTE).
- Center CR 1 inch (2.5 cm) superior to glabella.
- Center IR to CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: If patient is unable to bring OML perpendicular to IR, align IOML perpendicular and increase the 35° CR angle by 7° to 42° caudad (Fig. 11.163). If area of interest is the TM fossae, increase CR angle to 40° to OML to reduce superimposition of TM fossae and mastoid portions of the temporal bone.

Evaluation Criteria

Anatomy Demonstrated: • Condyloid processes of mandible and TM fossae.
Position: • A correctly positioned image with **no rotation** demonstrates the following: condyloid processes visualized symmetrically, lateral to the cervical spine; clear visualization of condyle/TM fossae relationship, with minimal superimposition of the TM fossae and mastoid portions (Figs. 11.164 and 11.165). • Collimation to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize condyloid process and TM fossa. • **No motion** exists, as indicated by sharp bony margins.

Fig. 11.163 AP axial—CR 35° to 40° to OML, erect and supine (*inset*).

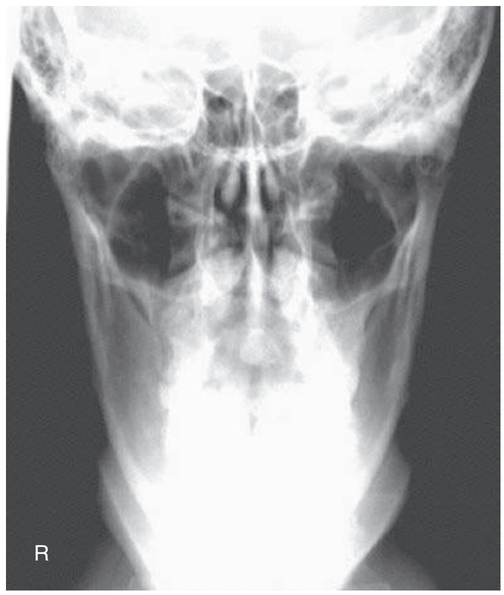


Fig. 11.164 AP axial—mandible.

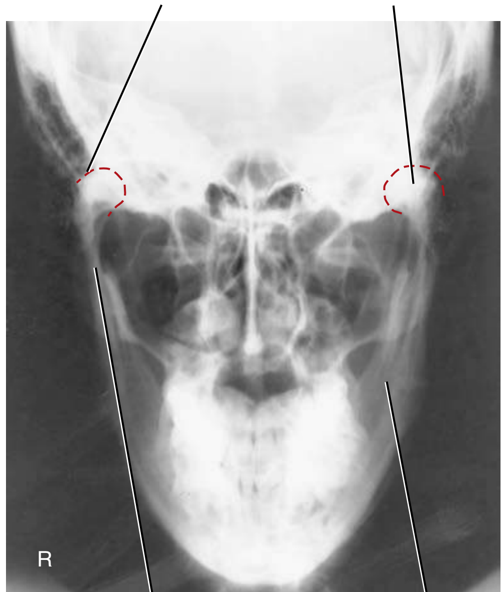


Fig. 11.165 AP axial—mandible.

SUBMENTOVERTICAL (SMV) PROJECTION—MANDIBLE

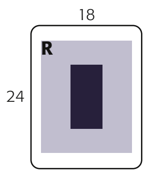
Mandible
SPECIAL
- SMV
- Orthopantomography (mandible or TMJs or both)

Clinical Indications

- Fractures and neoplastic or inflammatory process of mandible

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 80–95



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Patient position is erect or supine (erect preferred, if patient's condition allows). Erect may be done with an upright imaging device (Fig. 11.166).

Part Position

- Hyperextend neck until **IOML is parallel** to IR.
- Rest head on vertex of skull.
- Align **MSP perpendicular** to midline of grid or table/upright imaging device surface to prevent head rotation or tilt.

CR

- Align CR perpendicular to IR or IOML (see NOTE).
- Center CR to a point midway between angles of mandible or at a level 1 ½ inches (4 cm) inferior to mandibular symphysis.
- Center IR to projected CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: If patient is unable to extend the neck sufficiently, angle tube to align CR **perpendicular to IOML.** This position is very uncomfortable for the patient; complete the projection as quickly as possible.

Evaluation Criteria

Anatomy Demonstrated: • Entire mandible and coronoid and condyloid processes are demonstrated (Figs. 11.167 and 11.168).

Position: • Correct neck extension is indicated by the following: mandibular symphysis superimposing frontal bone; mandibular condyles projected anterior to petrous ridges. • **No** patient **rotation or tilt** is indicated by the following: no tilt as evidenced by equal distance from mandible to lateral border of skull; no rotation as evidenced by symmetric mandibular condyles. • **Collimation** to area of interest.

Exposure: • Contrast and density (brightness) are sufficient to visualize the mandible superimposed on the skull. • Sharp bony margins indicate **no motion.**

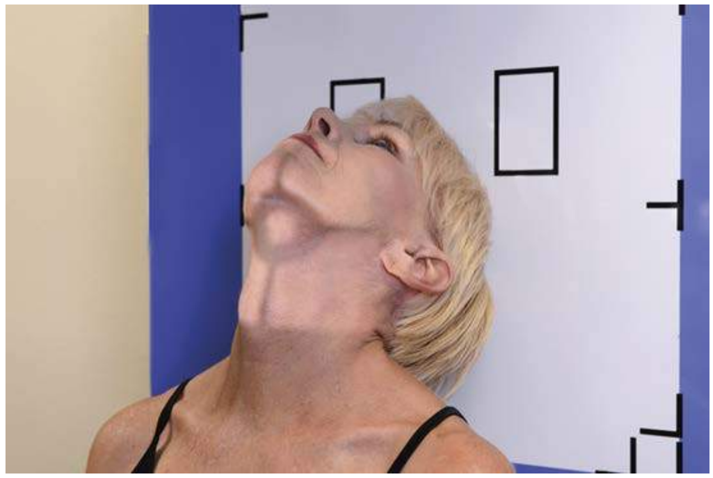


Fig. 11.166 SMV—mandible.

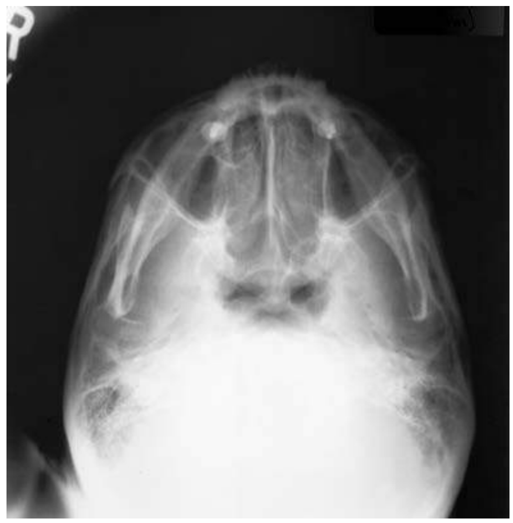


Fig. 11.167 SMV—mandible.

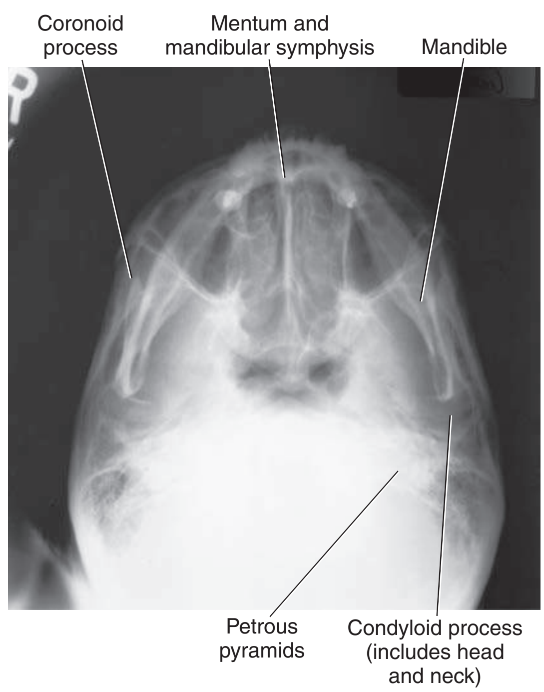


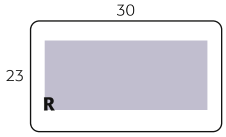
Fig. 11.168 SMV—mandible.

ORTHOPANTOMOGRAPHY: PANORAMIC TOMOGRAPHY—MANDIBLE

Mandible
ROUTINE
- Axiolateral oblique
- PA (or PA axial)
- AP axial (Towne method)

Clinical Indications
- Fractures or infectious processes of mandible
- Pre-surgical assessment before bone marrow transplants

Technical Factors (Conventional Radiographic Systems)
- IR size—9 × 12 inches (23 × 30 cm), landscape
- Curved nongrid cassette or digital detector
- kVp range: 70–85



Unit Preparation
- Attach IR to panoramic unit.
- Position tube and IR at starting position.
- Raise chin rest to approximately same level as patient's chin.

Shielding Wrap vest-type lead apron around patient.

Patient Position
- Remove all metal, plastic, and other removable objects from head and neck.
- Explain to patient how tube and IR rotate and the time span needed for exposure.
- Guide patient into unit, resting patient's chin on bite-block (Fig. 11.169).
- Position patient's body, head, and neck as demonstrated in Figs. 11.170 and 11.171. Do not allow head and neck to stretch forward (Fig. 11.172); have patient stand with spine straight and hips forward.

Part Position
- Adjust height of chin rest until **IOML is aligned parallel with floor.** The occlusal plane (plane of biting surface of teeth) declines 10° from posterior to anterior.
- Align **MSP** with vertical center line of chin rest.
- Position bite-block between patient's front teeth (see NOTE).
- Instruct patient to place lips together and position tongue on roof of mouth.

CR
- CR is fixed and directed slightly cephalic to project anatomic structures, positioned at the same height, on top of one another.
- Fixed SID, per panoramic unit.

Recommended Collimation A narrow, vertical-slit diaphragm is attached to tube, providing collimation.

NOTE: When TMJs are of interest, a second panoramic image is taken with the mouth open. This requires placement of a larger bite-block between the patient's teeth.

Digital Orthopantomography The first digital **orthopantomography** system was developed in 1995. Since 1997, digital orthopantomography systems have been replacing the analog systems. These systems do not require a cassette or chemical processing of images. They use a digital detector or a photostimulable phosphor to convert the analog signal into a digitized image. A key advantage of digital orthopantomography over film-based systems is increased exposure latitude and fewer repeat studies. This leads to reduced costs and patient exposure (see Figs. 11.169 and 11.171).

Advantages of Orthopantomography Compared with Conventional Mandible Positioning
- More comprehensive image of the mandible, TMJs, surrounding facial bones, and teeth
- Low patient radiation dose (slit collimation reduces exposure to eyes and thyroid gland)
- Convenience of examination for patient (one position provides the panoramic view of entire mandible)
- Ability to image the teeth in a patient who cannot open the mouth or when the oral cavity is restricted
- Shorter examination time

Fig. 11.169 Digital orthopantomography head correctly positioned.

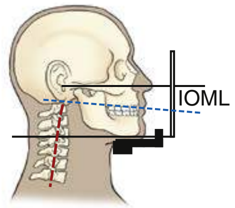


Fig. 11.170 Correct position.

Fig. 11.171 Digital orthopantomography—correct body position.

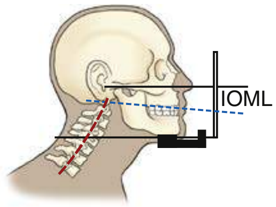


Fig. 11.172 Incorrect position.

Fig. 11.173 Orthopantomogram.

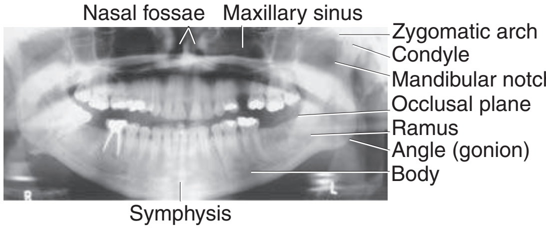


Fig. 11.174 Orthopantomogram.

Evaluation Criteria
Anatomy Demonstrated: • A single image of the teeth, mandible, TMJs, nasal fossae, maxillary sinus, zygomatic arches, and maxillae is demonstrated (Figs. 11.173 and 11.174). • A portion of the cervical spine is visualized.
Position: • The mandible visualized **without rotation or tilting** is indicated by the following: TMJs on the same horizontal plane in the image; rami and posterior teeth equally magnified on each side of the image; anterior and posterior teeth sharply visualized with uniform magnification. • Correct positioning of the patient is indicated by the following: mandibular symphysis projected slightly below the mandibular angles; mandible oval in shape; occlusal plane parallel with the long axis of the image; upper and lower teeth positioned slightly apart with **no superimposition**; cervical spine demonstrated with **no superimposition** of the TMJs.
Exposure: • Density (brightness) of mandible and teeth is uniform across entire image; no density loss is evident at the center. • No artifacts are superimposed on the image.

AP AXIAL PROJECTION—TEMPOROMANDIBULAR JOINTS

MODIFIED TOWNE METHOD

WARNING: Opening the mouth should not be attempted with possible fracture.

Clinical Indications

- Fractures and abnormal relationship or range of motion between condyle and TM fossa.

See NOTE 1 on open-mouth and closed-mouth comparisons.

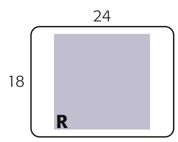
Temporomandibular Joints

ROUTINE
- AP axial (modified Towne method)

SPECIAL
- Axiolateral oblique (modified Law method)
- Axiolateral (Schuller method)
- Orthopantomography

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), landscape
- Grid
- kVp range: 75–85



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Position patient erect or supine.

Part Position

- Rest patient's posterior skull against table/upright imaging device surface.
- Tuck chin, bringing **OML perpendicular** to table/imaging device surface or bringing IOML perpendicular and increasing CR angle by 7° (Fig. 11.175).
- Align **MSP perpendicular** to midline of the grid or the table/upright imaging device surface to prevent head rotation or tilt.

CR

- Angle CR 35° caudad from OML or 42° from IOML (see NOTE 2).
- Direct CR 3 inches (7.5 cm) superior to the nasion. Center IR to projected CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE 1: Some departmental protocols indicate that these projections should be taken in both closed-mouth and open-mouth positions for comparison purposes when patient's condition allows.

NOTE 2: An additional 5° increase in CR may best demonstrate the TM fossae and TMJs.

Fig. 11.175 AP axial—CR 35° to OML (closed-mouth position) or 42° to the IOML *(inset).*

Fig. 11.176 AP axial (closed-mouth position).

Evaluation Criteria

Anatomy Demonstrated: • Condyloid processes of mandible and TM fossae are demonstrated (Fig. 11.176).

Position: • Correctly positioned patient, with **no rotation**, is indicated by the following: condyloid processes visualized symmetrically, lateral to the cervical spine; clear visualization of condyle and TM fossae relationship. • Collimation to area of interest.

Exposure: • Contrast and density (brightness) are sufficient to visualize condyloid process and TM fossa. • Sharp bony margins indicate **no motion.**

AXIOLATERAL OBLIQUE PROJECTION—TEMPOROMANDIBULAR JOINT

MODIFIED LAW METHOD

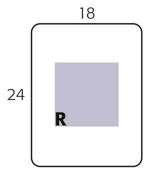
Temporomandibular Joints

ROUTINE
- AP axial (modified Towne method)

SPECIAL
- Axiolateral 15° oblique (modified Law method)
- Axiolateral (Schuller)
- Orthopantomography

Clinical Indications
- Abnormal relationship or range of motion between condyle and TM fossa

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Grid
- kVp range: 75–85
- AEC not recommended



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Patient position is erect or semiprone (erect is preferred if patient's condition allows). Rest lateral aspect of head against table/upright imaging device surface, with side of interest closest to IR.

Part Position
- Prevent tilt by maintaining **IPL perpendicular** to IR. MSP is parallel to IR to start.
- Align **IOML perpendicular** to front edge of IR (Fig. 11.177).
- From lateral position, **rotate face toward IR 15°** (with MSP of head rotated 15° from plane of IR).
- Closed- and open-mouth projections are often taken to demonstrate range of motion of the TMJ (Fig. 11.178).

CR
- Angle CR **15° caudad,** centered to 1 ½ **inches (4 cm) superior to upside EAM** (to pass through downside TMJ).
- Center IR to projected CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

Evaluation Criteria
Anatomy Demonstrated: • TMJ nearest IR is visible. • Closed-mouth image demonstrates condyle within mandibular fossa; the condyle moves to the anterior margin (articular tubercle) of the mandibular fossa in the open-mouth position (Figs. 11.179 and 11.180).
Position: • Correctly positioned images demonstrate TMJ closest to IR clearly, **without superimposition** of opposite TMJ (15° rotation prevents superimposition). • TMJ of interest is **not superimposed** by cervical spine. • **Collimation** to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize TMJ. • Sharp bony margins indicate **no motion.**

Fig. 11.177 Right TMJ—closed mouth; 15° oblique; CR 15° caudad.

Fig. 11.178 Right TMJ—open mouth; 15° oblique; CR 15° caudad.

Fig. 11.179 TMJ—closed mouth.

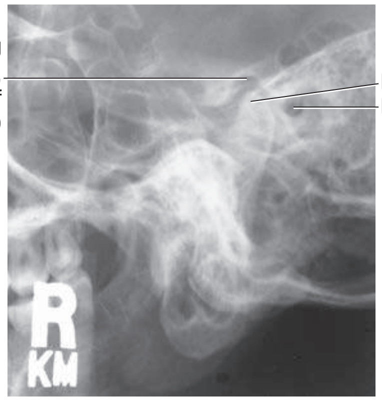


Fig. 11.180 TMJ—closed mouth.

AXIOLATERAL PROJECTION—TEMPOROMANDIBULAR JOINT

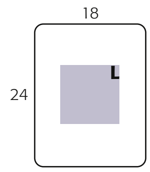
SCHULLER METHOD

Clinical Indications
- Abnormal relationship or range of motion between condyle and TM fossa

Temporomandibular Joints
ROUTINE
- AP axial (modified Towne method)

SPECIAL
- Axiolateral 15° oblique (modified Law method)
- Axiolateral (Schuller method)
- Orthopantomography

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Grid
- kVp range: 75–85
- AEC not recommended



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Position patient erect or semiprone. Place the head in a true lateral position, with side of interest nearest IR.

Part Position
- Adjust head into **true lateral position** and move patient's body in an oblique direction, as needed for patient's comfort.
- Align IPL perpendicular to IR.
- Align **MSP parallel** with table/imaging device surface.
- Position **IOML perpendicular** to front edge of IR (Fig. 11.181). Closed- and open-mouth projections are often taken to demonstrate range of motion of the TMJ (Fig. 11.182).

CR
- Angle CR **25° to 30° caudad,** centered to ½ inch (1.3 cm) anterior and 2 inches (5 cm) superior to upside EAM.
- Center IR to projected TMJ.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

Evaluation Criteria
Anatomy Demonstrated: • TMJ nearest IR is visible. • Closed-mouth image (Figs. 11.183 and 11.184) demonstrates the condyle within the mandibular fossa; the condyle moves to the anterior margin (articular tubercle) of fossa in the open-mouth position (Fig. 11.185).
Position: • TMJs are demonstrated **without rotation**, as evidenced by superimposed lateral margins. • **Collimation** to area of interest.
Exposure: • Contrast and density (brightness) are sufficient to visualize TMJ. • Sharp bony margins indicate **no motion.**

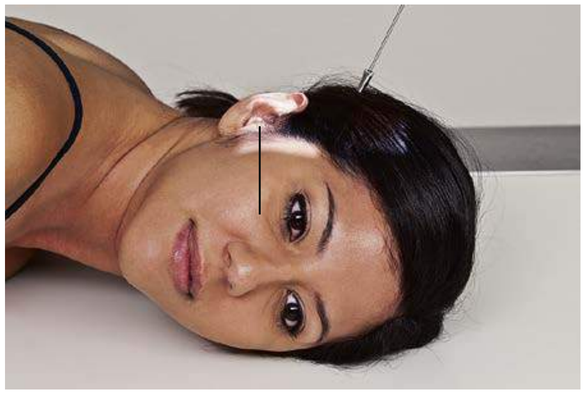


Fig. 11.181 Left TMJ—closed mouth; true lateral, CR 25° to 30° caudad angle.


Fig. 11.182 Left TMJ—open mouth; true lateral, CR 25° to 30° caudad angle.

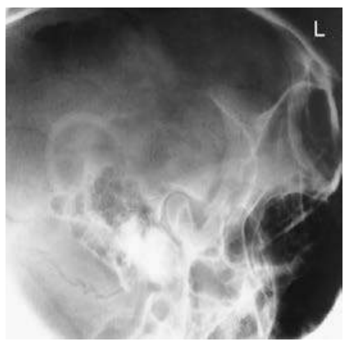


Fig. 11.183 Closed mouth.

Fig. 11.185 Open mouth.

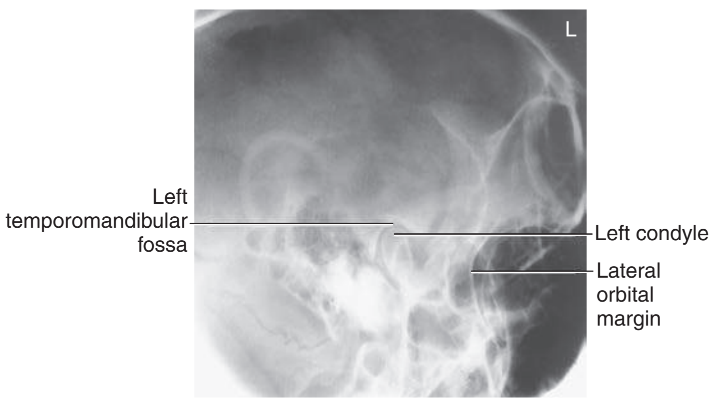


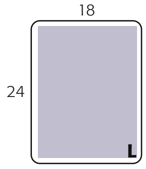
Fig. 11.184 Closed mouth.

LATERAL POSITION: RIGHT OR LEFT LATERAL—SINUSES

Sinuses
ROUTINE
- Lateral
- PA (Caldwell method)
- Parietoacanthial (Waters method)

Clinical Indications
- Inflammatory conditions (sinusitis, secondary osteomyelitis)
- Sinus polyps or cysts

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8× 10 inches (18 × 24 cm), portrait
- Grid
- kVp range: 75–85
- AEC not recommended



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metal, plastic, and other removable objects from head. Position patient **erect** (see NOTE).

Part Position
- Place lateral aspect of head against table/upright imaging device surface, with side of interest closest to IR (Fig. 11.186).
- Adjust head into **true lateral** position, moving body in an oblique direction as needed for patient's comfort (MSP parallel to IR).
- Align **IPL perpendicular to IR** (ensures no tilt).
- Adjust chin to align IOML perpendicular to front edge of IR.

CR
- Align **horizontal CR** perpendicular to IR.
- Center CR to a point midway between outer canthus and EAM.
- Center IR to CR.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: To visualize air-fluid levels, an erect position with a horizontal beam is required. Fluid within the paranasal sinus cavities is thick and gelatinous, causing it to cling to the cavity walls. To visualize this fluid, allow a short time (at least 5 minutes) for the fluid to settle after patient's position has been changed (i.e., from recumbent to erect). If patient is unable to be placed in the upright position, the image may be obtained with the use of a horizontal beam, similar to trauma lateral facial bones, as described in Chapter 15.

Evaluation Criteria
Anatomy Demonstrated: • All four paranasal sinus groups are demonstrated (Figs. 11.187 and 11.188).
Position: • Accurately positioned cranium without rotation or tilt. • **Rotation** is evident by **anterior and posterior separation** of symmetric bilateral vertical structures such as the mandibular rami and greater wings of the sphenoid. • **Tilt** is evident by **superior and inferior separation** of symmetric horizontal structures such as the orbital plates **and greater wings of sphenoid** • Collimation to area of interest.
Exposure: • Density (brightness) and contrast are sufficient to visualize the sphenoid sinuses through the cranium without overexposing the maxillary and frontal sinuses. • Sharp bony margins indicate **no motion.**

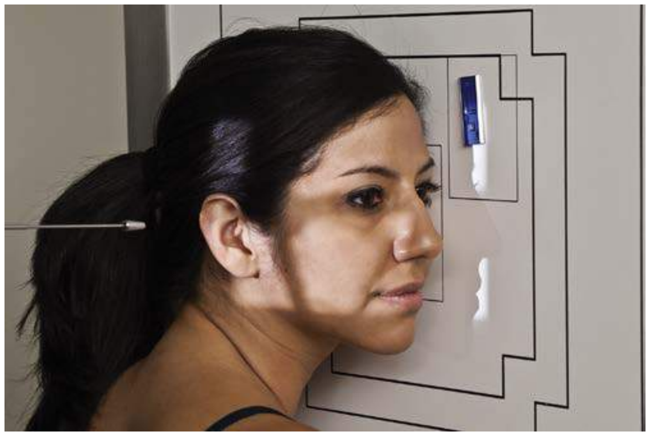


Fig. 11.186 Erect left lateral—sinuses (upright imaging device).

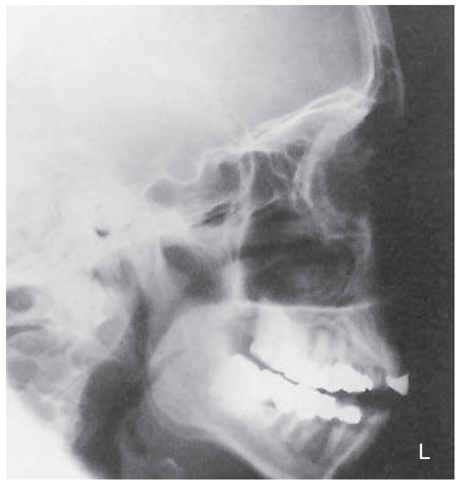


Fig. 11.187 Lateral sinuses.

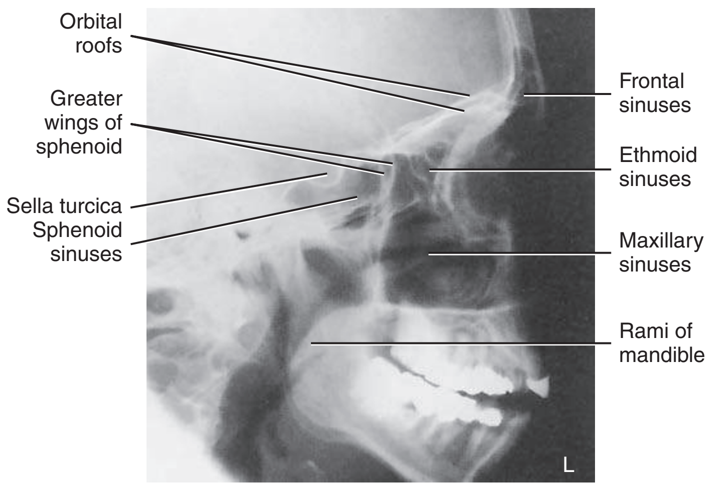


Fig. 11.188 Lateral sinuses.

PA PROJECTION—SINUSES

CALDWELL METHOD

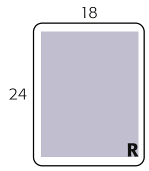
Sinuses
ROUTINE
- Lateral
- PA (Caldwell method)
- Parietoacanthial (Waters method)

Clinical Indications

- Inflammatory conditions (sinusitis, secondary osteomyelitis)
- Sinus polyps or cysts

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm), portrait
- Grid
- kVp range: 75–85
- Upright imaging device angled 15° if possible, CR horizontal (see NOTE)
- AEC not recommended



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Position patient erect (see NOTE).

Part Position

- Place patient's nose and forehead against upright imaging device or table with neck extended to elevate **OML 15° from horizontal.** A radiolucent support between forehead and upright imaging device or table may be used to maintain this position (Fig. 11.189). **CR remains horizontal.** (See alternative method if imaging device can be tilted 15°.)
- Align **MSP perpendicular to midline** of grid or upright imaging device surface.
- Center IR to CR and to nasion, ensuring **no rotation.**

CR

- Align **CR horizontal,** parallel with floor (see NOTE).
- Center CR to **exit at nasion.**

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: To assess air-fluid levels accurately, **CR must be horizontal,** and the **patient must be erect.**

Alternative Method An alternative method if the **imaging device can be tilted 15°** is shown (see Fig. 11.189, inset). The patient's forehead and nose can be supported directly against the imaging device with **OML perpendicular to imaging device surface** and **15° to horizontal CR.**

Evaluation Criteria

Anatomy Demonstrated: • Frontal sinuses projected above the frontonasal suture are demonstrated. • Anterior ethmoid air cells are visualized lateral to each nasal bone, directly below the frontal sinuses (Figs. 11.190 and 11.191).

Position: • Accurately positioned cranium with **no rotation or tilt** is indicated by the following: equal distance from the lateral margin of the orbit to the lateral cortex of the cranium on both sides; equal distance from the MSP (identified by the crista galli) to the lateral orbital margin on both sides; superior orbital fissures symmetrically visualized within the orbits. • Correct alignment of OML and CR projects petrous ridges into lower one-third of orbits (black arrows). • Collimation to area of interest.

Exposure: • Density (brightness) and contrast are sufficient to visualize the frontal and ethmoid sinuses. • Sharp bony margins indicate **no motion.**

Fig. 11.189 **CR horizontal,** OML 15° to CR (if cannot be tilted). *Inset,* If upright, imaging device can be tilted 15°.

Fig. 11.190 PA projection—sinuses.

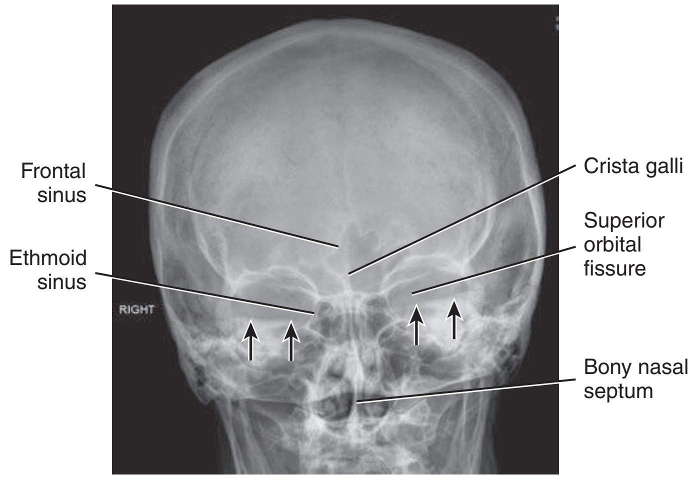


Fig. 11.191 PA projection—sinuses.

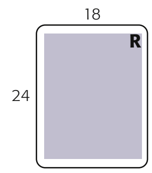
PARIETOACANTHIAL PROJECTION—SINUSES

WATERS METHOD

Clinical Indications
- Inflammatory conditions (sinusitis, secondary osteomyelitis)
- Sinus polyps and cysts

Sinuses
ROUTINE
- Lateral
- PA (Caldwell method)
- Parietoacanthial (Waters method)

Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–85
- AEC not recommended



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Position patient erect (see NOTE).

Part Position
- Extend neck, placing chin and nose against table/upright imaging device surface.
- Adjust head until **MML is perpendicular** to IR; OML forms a 37° angle with plane of IR (Fig. 11.192).
- Position **MSP perpendicular** to midline of grid.
- Ensure that **no rotation or tilt** exists.
- Center IR to CR and to acanthion.

CR
- Align horizontal CR perpendicular to IR centered to exit at acanthion.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: CR must be horizontal, and patient must be erect to demonstrate air-fluid levels within the paranasal sinus cavities.

Evaluation Criteria
Anatomy Demonstrated: • Maxillary sinuses with the inferior aspect visualized free from superimposing alveolar processes and petrous ridges, the inferior orbital margin, and an oblique view of the frontal sinuses (Figs. 11.193 and 11.194).
Position: • **No rotation** of the cranium is indicated by the following: equal distance from MSP (identified by the bony nasal septum) to lateral orbital margin on both sides; equal distance from the lateral orbital margin to the lateral cortex of the cranium on both sides. • Adequate extension of neck demonstrates petrous ridges just inferior to the maxillary sinuses. • Collimation to area of interest.
Exposure: • Density (brightness) and contrast are sufficient to visualize maxillary sinuses. • Sharp bony margins indicate **no motion.**

Fig. 11.192 Parietoacanthial projection (upright imaging device/table)—CR and MML perpendicular (OML 37° to IR).

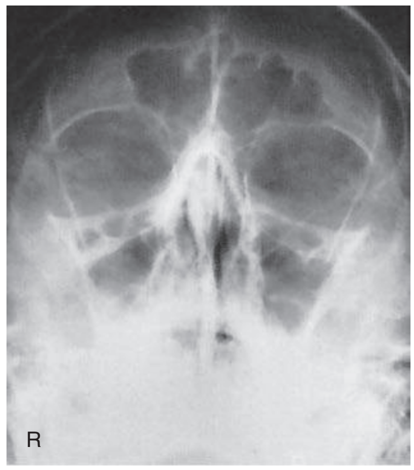


Fig. 11.193 Parietoacanthial projection—sinuses.

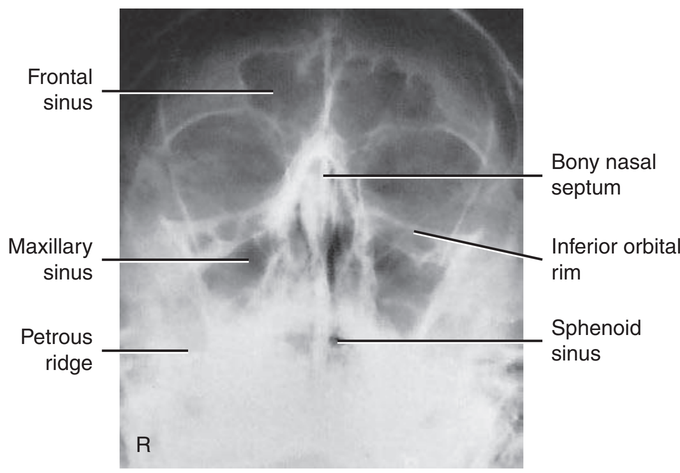


Fig. 11.194 Parietoacanthial projection—sinuses.

SUBMENTOVERTICAL (SMV) PROJECTION—SINUSES

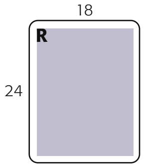
Sinuses
SPECIAL
- Submentovertical (SMV)

Clinical Indications

- Inflammatory conditions (sinusitis, secondary osteomyelitis)
- Sinus polyps and cysts

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–85
- AEC not recommended



Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Position patient erect, if possible, to show air-fluid levels.

Part Position

- Raise chin, hyperextend neck if possible until **IOML is parallel** to table/upright imaging device surface (see NOTE 1).
- Head rests on vertex of skull.
- Align **MSP perpendicular** to midline of the grid; ensure **no rotation or tilt.**

CR

- CR directed perpendicular to IOML (see NOTE 2).
- CR centered midway between angles of mandible, at a level 1½ to 2 inches (4 to 5 cm) inferior to mandibular symphysis (Fig. 11.195).
- CR centered to IR

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE 1: This position is very uncomfortable for the patient; have all factors set before positioning the patient, and complete the projection as quickly as possible.

NOTE 2: If patient is unable to extend neck sufficiently, angle the tube from horizontal as needed to align CR perpendicular to IOML.

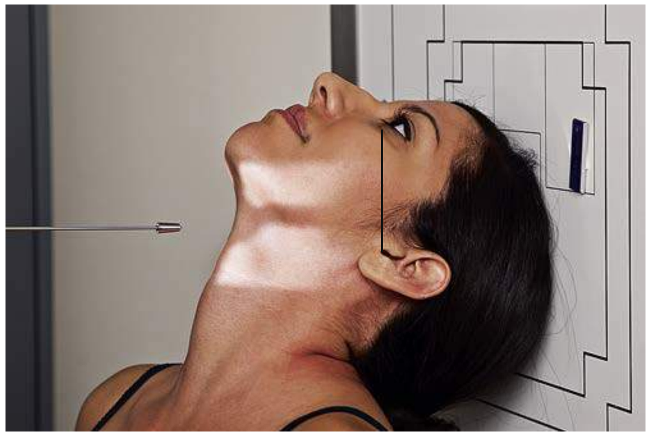


Fig. 11.195 SMV projection (upright imaging device/table).

Fig. 11.196 SMV projection—sinuses.

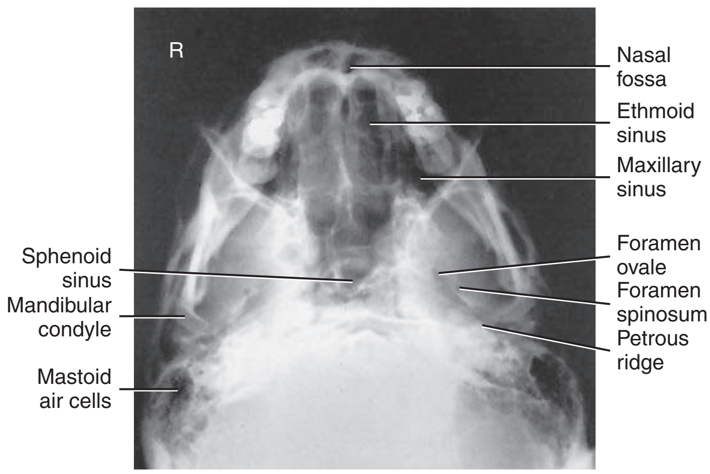


Fig. 11.197 SMV projection—sinuses.

Evaluation Criteria

Anatomy Demonstrated: • Sphenoid sinuses, ethmoid sinuses, nasal fossae, and maxillary sinuses are demonstrated (Figs. 11.196 and 11.197).

Position: • Accurate IOML and CR relationship is demonstrated by the following: correct extension of neck and relationship between IOML and CR as indicated by **mandibular mentum anterior to** ethmoid sinuses. • **No rotation** evidenced by MSP parallel to edge of IR. • **No tilt** evidenced by equal distance between mandibular ramus and lateral cranial cortex. • Collimation to area of interest.

Exposure: • Density (brightness) and contrast are sufficient to visualize sphenoid and ethmoid sinuses. • Sharp bony margins indicate **no motion.**

PARIETOACANTHIAL TRANSORAL PROJECTION—SINUSES

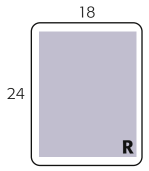
OPEN-MOUTH WATERS METHOD

Sinuses

SPECIAL
- Submentovertical (SMV)
- Parietoacanthial transoral (open-mouth Waters method)

Clinical Indications
- Inflammatory conditions (sinusitis, secondary osteomyelitis)
- Sinus polyps and cysts

NOTE: This projection is a good alternative to demonstrate the sphenoid sinuses for patients who cannot perform the submentovertex (SMV) position.



Technical Factors
- Minimum SID—40 inches (100 cm)
- IR size—8 × 10 inches (18 × 24 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range: 75–85
- AEC not recommended

Shielding Shield radiosensitive tissues outside region of interest.

Patient Position Remove all metallic or plastic objects from head and neck. Position patient **erect**.

Part Position
- Extend neck, placing chin and nose against table/upright imaging device surface.
- Adjust head until **OML forms 37° angle** with IR (**MML is perpendicular** with mouth closed) (Fig. 11.198).
- Position **MSP perpendicular** to the midline of grid; ensure **no rotation or tilt.**
- Instruct patient to open mouth (i.e., "drop your jaw without moving your head"); MML may not be perpendicular.
- Center IR to CR and to **acanthion.**

CR
- Align horizontal CR perpendicular to IR.
- Center CR to exit at acanthion.

Recommended Collimation Collimate on four sides to anatomy of interest.

Respiration Suspend respiration.

NOTE: Remember, the CR must be horizontal and the patient erect to demonstrate air-fluid levels within the paranasal sinuses.

Evaluation Criteria
Anatomy Demonstrated: • Maxillary sinuses with the inferior aspect visualized, free from superimposing alveolar processes and petrous ridges; the inferior orbital margin, an oblique view of the frontal sinuses, and the sphenoid sinuses visualized through the open mouth (Figs. 11.199 and 11.200).
Position: • **No rotation** of the cranium is indicated by the following: equal distance from the MSP (identified by the bony nasal septum) to the lateral orbital margin on both sides; equal distance from the lateral orbital margin to the lateral cortex of the cranium on both sides; accurate extension of the neck demonstrating petrous ridges just inferior to the maxillary sinuses. • Collimation to area of interest.
Exposure: • Density (brightness) and contrast are sufficient to visualize the maxillary and sphenoid sinuses. • Sharp bony margins indicate **no motion.**

Fig. 11.198 Parietoacanthial transoral projection (upright imaging device/table).

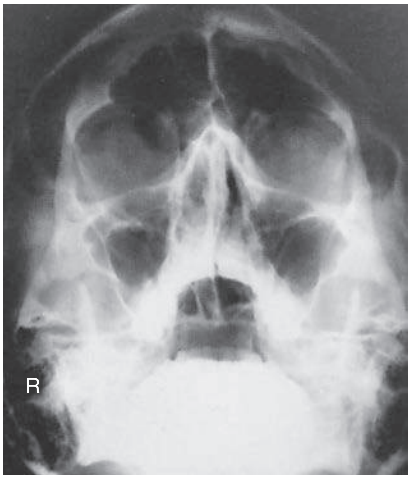


Fig. 11.199 Parietoacanthial transoral projection.

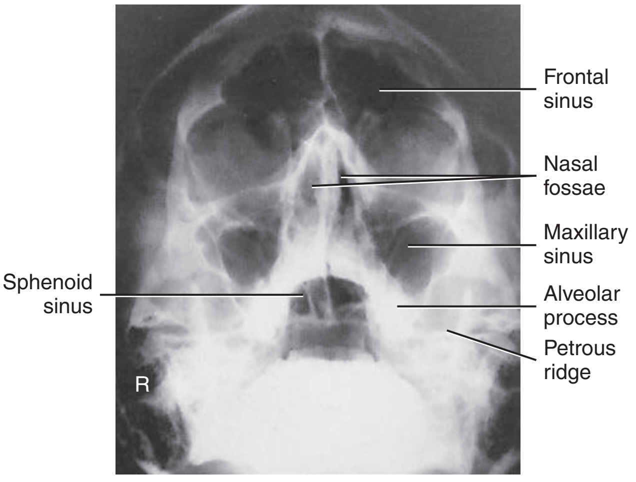


Fig. 11.200 Parietoacanthial transoral projection.

RADIOGRAPHS FOR CRITIQUE—CRANIUM

This section consists of an ideal projection (Image A) along with one or more projections that may demonstrate positioning and/or technical errors. Critique Figures C11.201 through C11.203. Compare Image A to the other projections and identify the errors. While examining each image, consider the following questions:

1. Is all essential anatomy demonstrated on the image?
2. What positioning errors are present that compromise image quality?
3. Are technical factors optimal?
4. Is there evidence of collimation and are pre-exposure anatomic side markers visible on the image?
5. Do these errors require a repeat exposure?

Feedback for each set of images is located on the faculty Evolve site.

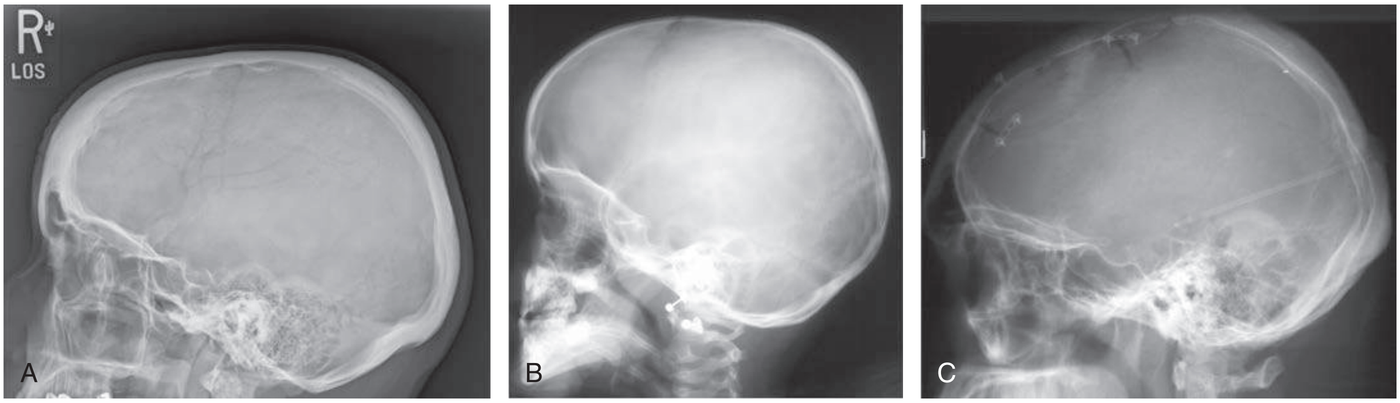


Fig. C11.201 Lateral cranium.

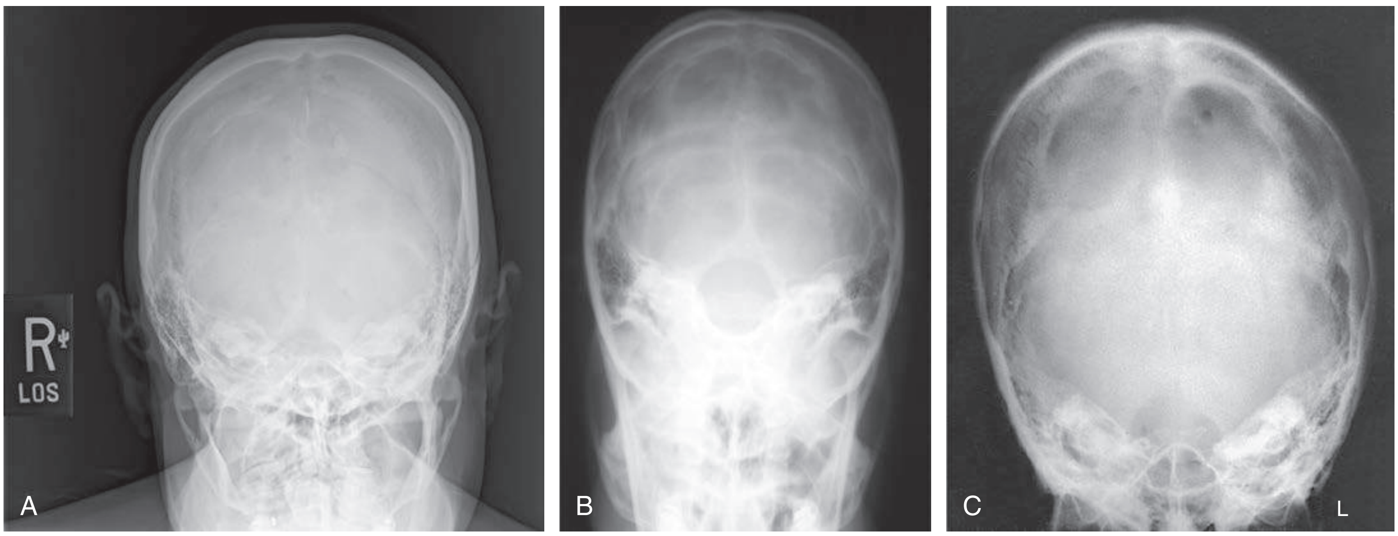


Fig. C11.202 AP axial cranium.

Fig. C11.203 PA Axial (Caldwell method) cranium.

RADIOGRAPHS FOR CRITIQUE—FACIAL BONES

This section consists of an ideal projection (Image A) along with one or more projections that may demonstrate positioning and/or technical errors. Critique Figures C11.204 through C11.206. Compare Image A to the other projections and identify the errors. While examining each image, consider the following questions:

1. Is all essential anatomy demonstrated on the image?
2. What positioning errors are present that compromise image quality?
3. Are technical factors optimal?
4. Is there evidence of collimation and are pre-exposure anatomic side markers visible on the image?
5. Do these errors require a repeat exposure?

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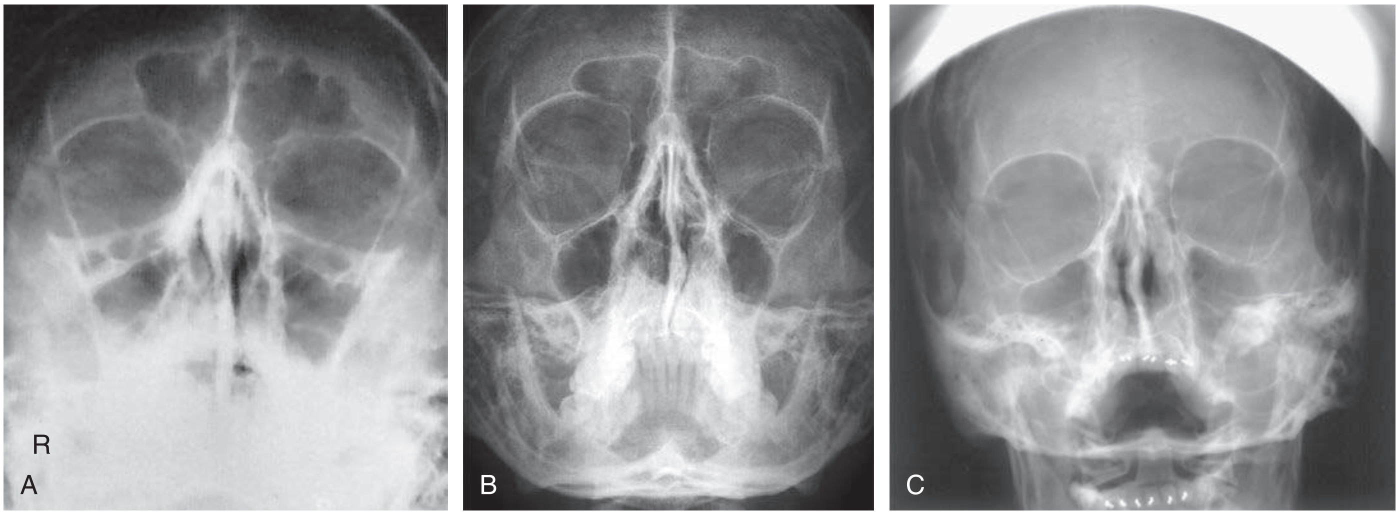

Fig. C11.204 Parietoacanthial projection—facial bones (Waters method).

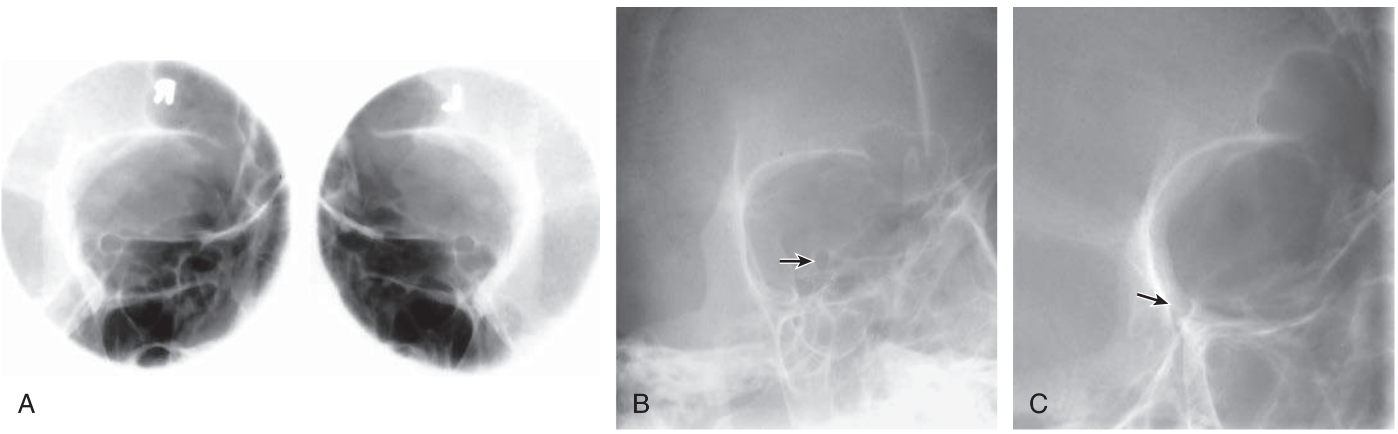

Fig. C11.205 Parieto-orbital oblique (Rhese method)—optic foramen.

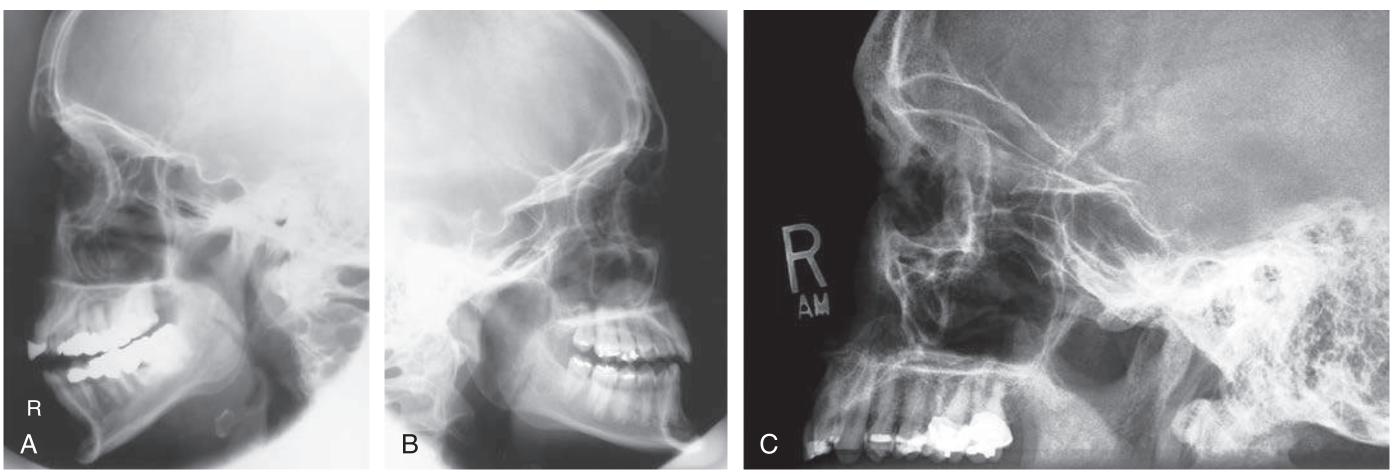

Fig. C11.206 Lateral facial bones.

RADIOGRAPHS FOR CRITIQUE—PARANASAL SINUSES

This section consists of an ideal projection (Image A) along with one or more projections that may demonstrate positioning and/or technical errors. Critique Figure C11.207. Compare Image A to the other projections and identify the errors. While examining each image, consider the following questions:

1. Is all essential anatomy demonstrated on the image?
2. What positioning errors are present that compromise image quality?
3. Are technical factors optimal?
4. Is there evidence of collimation and are pre-exposure anatomic side markers visible on the image?
5. Do these errors require a repeat exposure?

Feedback for each set of images is located on the faculty Evolve site.

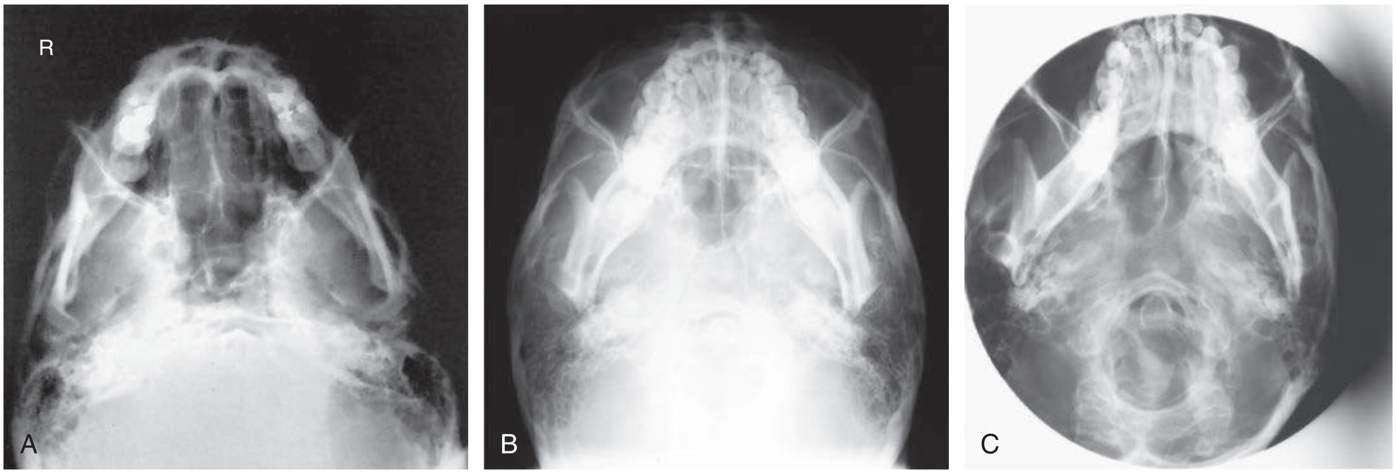

Fig. C11.207 SMV projection—paranasal sinuses.

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